Ministry of High Education
And Scientific Research
*** * ***
University of Carthage
*** * ***
National Institute of Applied Sciences and Technology

Graduation Project

In order to obtain the

National Engineering Diploma
in Applied Sciences and Technology

Major: Industrial Computing and Automation

Subject:

Embedded Software Product Quality and Handle-Non Conformity Processes Quality Improvement

Realized by: **Baha-Eddine ALAYET**

Host company:

Volvo Group Trucks

Presented on the 10/10/2017

President: Mr.Khaled ELLOUMI
Examiner: Mrs.Imen JAAFAR

Company supervisor : Mr.Denis MARTIN

INSAT supervisor : Pr.Faouzi BEN AMMAR

Academic year: 2016/2017

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Dedications

To the Omniscient, the Omnipotent, the Omnipresent... To the One beyond his creatures, thank you

Lord for giving me this chance, to fail, to resist, to love, to defy, to think, to disobey, to fall, to rise, to live; and to exist. The leap of faith comes first, the Beginning of all beginnings.

To the best Mom in the world, for the unconditional love, affection and support during all these years. To the woman who inspired me with her patience and love, who has always encouraged me to be a better person, even with rebukes. The word itself doesn't sound very kind. But it was. "Place no one above you, sweet lady and dear mama"

To my dad, for everything, the blue bicycle, the first ball, the first sports shoes, even the Sousse trip that we never had, thank you for everything and for every single sacrifice...

To the worst brothers that one can ever have, **Houssem** & **Mohamed Dhia**, I love you guys.

To a brotherhood shaped in the cave of Ordeals, in the dark. To the siblings of days, **Houcem**, **Amine**, **Karim**, **Mahmoud**, to the ones beyond the oceans, **Michael**, **Khalil**, **Hideya**. You have all played an integral part in my change over the past years. Thank you for everything.
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To all my Basket-Ball community, **AS Marsa**, **INSAT**, **Renault Trucks**, **AL Vénissieux Parilly** the best four Basket-Ball teams in the history of the game. Thank you teammates, coaches and fans.
To the new friends that made me feel like I know them since ever and were there for me this particular year, **Imene**, **Emna**, **Oumaima**. Love you guys.

To all those who matter to me …I dedicate this work to you. And I know, there is no great fulfillment here. But I'm not celebrating this very modest achievement; I'm celebrating the twisting course that led me here. Because meeting all of you on the road is and will always be my greatest achievement.

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Without forgetting to thank all the members of my working groups and specially **Capucine BOUSQUET**, **Loan BESSE, Emeric JAUSIONS, Ederson CALSAVARA** et **Philippe MAUGUIN.**

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Glossary

ARGUS: IT support tool for QJ and QSP handling and follow up.

Electronic Control Unit, ECU: complete piece of hardware including software that implements a control system

External campaign: set of organized actions to secure that the product delivered to the customer is at the right quality level by releasing campaigns to implement solutions on potential or real Customer quality issues on delivered products.

Green OK: describes the necessary steps to determine that a truck is ready to be delivered from a factory.

Implementation: the coding phase of software development implying the translation of the design proposal to source code.

Process: set of interrelated activities, which transform inputs into outputs (ISO/IEC 12207).

Product Quality Leader: Ensure the product quality performance within determined responsibility scope through investigating and prioritizing market product quality issues related to GTT / GTO to achieve committed quality and warranty targets. Secure the voice of the customer in the problem solving processes.

Software Design: umbrella term for analysis, design and implementation of a software system.

Software Green OK: The truck fulfills all functional requirements as stated in quality requirements

- It has passed the Dyno/roller test and/or test drive without any remarks (all functions OK)
- The vehicle software as specified corresponds to the order and has been installed successfully
- No active fault codes
- All calibration results/ values are according to specification
- The truck is not in factory mode when leaving the factory

Software System: umbrella term for the complete software program, typically consisting of one or more application programs, parameter settings, drivers

Abbreviations

APAC: Asia/ Pacific

CAB: Cabin
E&BM: Europe & Brazil Manufacturing

EL: Electrical

EMEA: Europe Middle East Africa

FF1: Failure Frequency 1 Month

GTA: Group Trucks Asia

GTO: Group Trucks Operations

GTS: Group Trucks Sales

GTS&M: Group Trucks Sales & Marketing

GTT: Group Trucks Technology
HNC: Handle Non-Conformity

KPI: Key Performance Indicator

LS: logistics and services

NAM: North America Manufacturing

OU: Operation Unit

PI: Performance Indicator

PQL: Product Quality Leader

PTP: Powertrain (Production)

Q & CS: quality and customer satisfaction

QJ: Quality Journal

QSP: Quick Solving Process

QV: QlikView

QULIS: Quality Information System

SW/ HW: Software/ Hardware

TEA: Truck Electronic Architecture

VMT: Vehicle Module Team

WG: Working Group

WoW: Way of Working

General Introduction

The product quality is a primordial requirement, from design to manufacturing till the moment the product is put into circulation. In a competitive and regulatory vehicle environment in a constant evolution, the quality strategy of **Volvo Group** -in order to achieve its goal to become the world leader in sustainable transport solutions- is to keep the costumer in focus in whatever it does.

However, let us not forget that competition is tough and that its place is not gained. In a context that is rapidly changing environmentally, regulatory, digitally and economically, the automotive world is evolving at a fast pace.

The truck got electrified, is now connected and is becoming automated. The number of innovations developed on vehicles is multiplying.

Continuous leverage of more intelligent vehicles can only be achieved through functionalities implemented as embedded software. There are claims that 80 to 90 per cent of automotive innovation is enabled through software.

Thus, the international deployment of engineering requires strict execution of processes within shared and coherent development logic.

Also, working on the continuous improvement of Handling Non-Conformities Processes whether detected inside the manufacturing line or at the customer is a key point to achieve the wanted place.

Knowing the importance of those two subjects, **Volvo Group** entrusted me, first of all, during my graduation project with the responsibility to initiate and lead two working groups in order to ensure the improvement of product quality from the embedded software point of view and subsequently reduce its contribution to customer claims. Secondly, to ensure the evolution of the operational excellence in Handling Non-Conformity Processes through the definition of performance indicators and the establishment of a global dashboard to monitor and control these processes.

All the activities carried out during my six months internship are illustrated by a report of three chapters.

The first chapter is a presentation of the host company, its brands and products to finish with the definition of the project specifications.

The second chapter is a diagnosis based on customer feedback on embedded software problems in order to prioritize the most critical defects, detailing the standards and approaches taken into account. The second part is an analysis of the Handling Non-Conformity processes of the group and their IT support application.

The third chapter focuses on a more detailed analysis of customer claims in order to establish an action plan and monitor its execution. As well as the definition of performance/key performance indicators, the development and implementation of a dashboard in order to monitor and continuously improve the operational excellence at all **Volvo Group** plants. The report concludes with a **General Conclusion** in which we outline both the quantitative and qualitative outcomes of our graduation project as well as its perspectives.

Chapter I

Project Context

I. Introduction

Our graduation project was carried out within the **Volvo Group Trucks**. In this first chapter we present the **Volvo AB group**, its history, its organization and its values and vision. Then we detail the specifications of the work.

II. Company presentation

1. General presentation

The Swedish group **Volvo AB** -born in 2001 of the grouping of two centennial companies (**Volvo** & **Renault VI**) - is today one of the world leaders in the design and production of transport solutions with products such as:

- Trucks (**Volvo**, **Renault**, **Mack**, **UD**, **Eicher**)
- Bus & Cars (**Volvo**, **NovaBus**, **Prevost**, **Eicher**)
- Construction machinery (**Volvo CE**, **SDLG**)
- Engines for marine & industrial applications (**Volvo Penta**)

Various entities assist the group in logistics and manufacturing development, including: **Volvo Powertrain** (Powertrain Production), **Volvo 3P**, **Volvo IT**, **Volvo Logistics** and **Volvo Parts**. The Volvo Group now employs approximately **110,000** people, has production sites in **18** countries and markets its products in more than **190** markets and has a turnover of more than **32 Billion euros** in **2016** by selling more than **190,000** truck that year.[1]

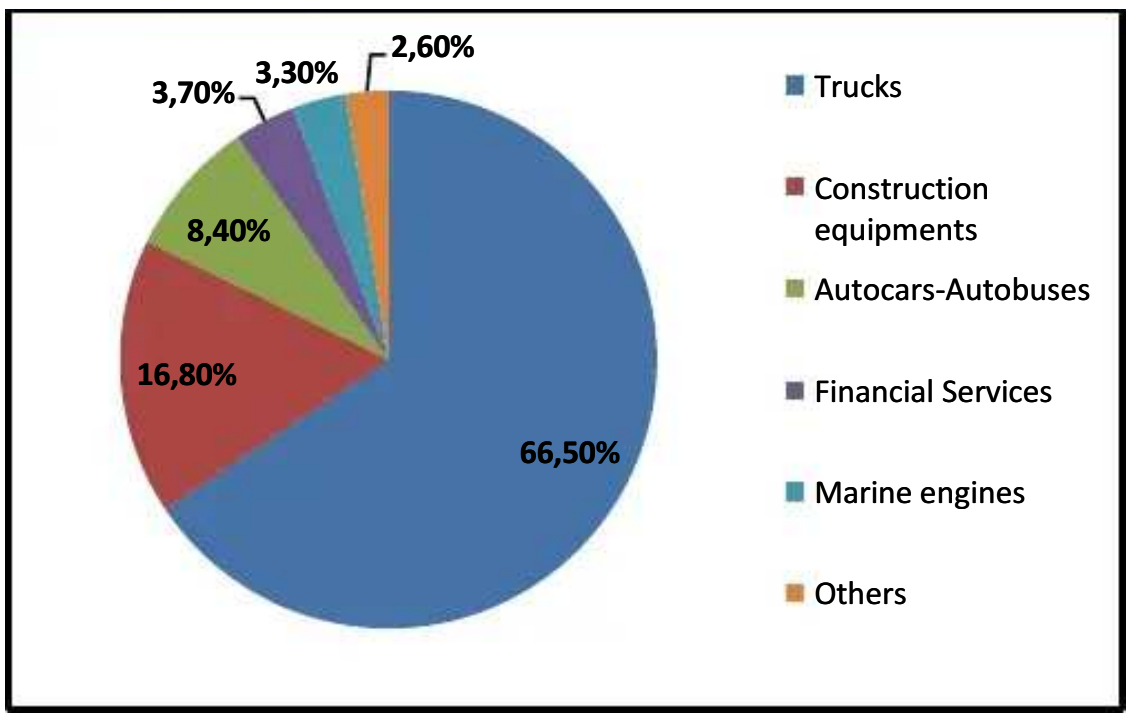


Figure 1: Contribution of products to the total turnover of Volvo AB (2016) [1]



Figure 2: Volvo Group Production Sites [1]

The group is one of the leading manufacturers of trucks, buses, construction equipment, propulsion systems for marine and industrial applications.

Volvo offers a complete range of transport and financing solutions. Volvo provides transport products and services that meet the criteria of excellence in terms of quality, safety and respect for the environment. Since its inception over 80 years ago, Volvo has built a solid reputation throughout the world. The Group's values revolve around three areas: Quality, Safety and Environment. [1]

The **Volvo Group** aims to become the world leader in sustainable transport solutions by:

- creating value for its clients in targeted sectors
- Using its expertise to create innovative products and services for transport companies
- guaranteeing quality, safe and environmentally friendly transport
- Working with energy, passion and respect for the individual

This point was resumed in the **Volvo Group** Pyramid in order to have a clear vision of the wanted place and how we should operate to get there.

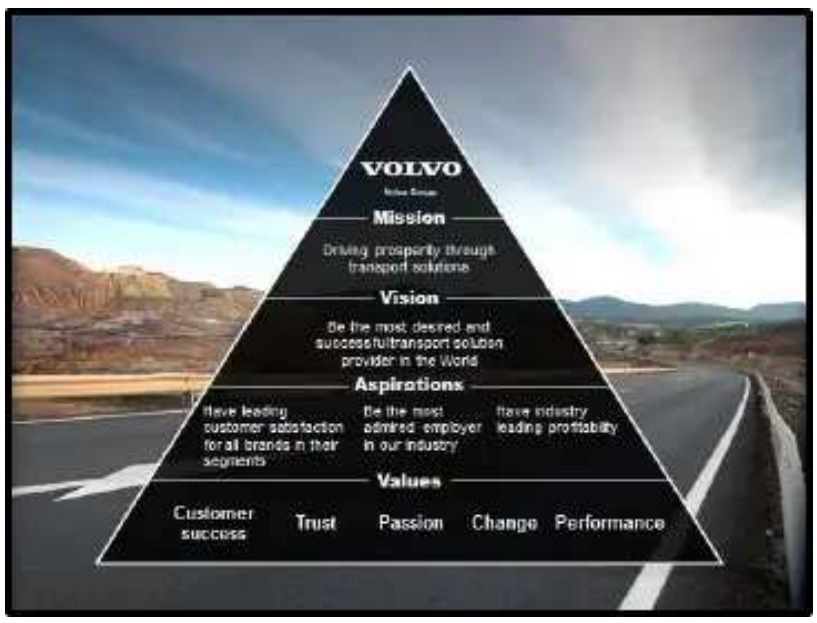


Figure 3 : The Volvo Group Pyramid [1]

2. Volvo Group History

Founded in 1927 by **Assar GABRIELSSON** and **Gustaf LARSON**, Volvo is one of the world's leading manufacturers of heavy duty trucks and diesel engines. The **Volvo Group** also offers a full range of customized solutions: financing, leasing, insurance and services, complete transportation systems for urban environments.

The 1920s were a period of progress for trucks and, more generally, for trucking. At the beginning of the decade, trucks were generally equipped with chain driven systems and rigid rubber tires. Volvo's first truck, the "Series 1", came out of the production lines in 1928. Until 1932, Volvo medium-duty pickup trucks with two service brakes and wooden spokes did not stand out by their modernity. A smaller, more modern truck became indispensable. Volvo's LV71 and LV73 series were designed to be such a success that Volvo became one of the largest exporters of trucks in the mid-1930s. The early 1940s were marked by the Second World War. Sales of civil trucks were growing at a very slow pace. The army, meanwhile, was generating strong demand, and Volvo became a leading supplier to the Swedish armed forces. During the Second World War, thousands of Round nose trucks were delivered in different models: standard, simplified design or all-wheel drive.

The fifties (1950s) were the theater of trucking. Gasoline engines and rudimentary pre-combustion diesel engines were replaced by direct-injection diesel engines; Introduced in 1951, the "Titan" was one of the most popular Volvo trucks of all time. It was in the 1960s that the truck became the king of freight transport, not only because of its flexibility, but also because the national and international road infrastructure now made it possible to transport goods quickly and safely.

Volvo then launched the Titan TIPTOP / F88 / G88. These models have played a major role in the history of Volvo trucks. In the 1990s, the transport sector was dominated by three aspects: environmental protection, the production of more profitable couplings and, by the end of the decade, the integration of information technology through such programs. As well as Volvo Dynafleet 2.0, which made it possible for transport companies and drivers to increase their efficiency and safety on a daily basis. For Volvo, it was then natural to start designing a successor to the F10 / F12 / F16 range, incorporating new generation engines ready to replace the 12-liter engine. The new millennium witnesses demand for road transport diversifying. Although the main objectives remain efficiency and cost reduction, safety, ergonomics and the environment are also top priorities. At the Amsterdam Truck Show, in February 2000, visitors were able to discover the European version of the Volvo NH12: a "high-tech" truck designed for the new millennium. Equipped with the ultra-modern Volvo D12C engine, a direct injection diesel engine and electronic management, this truck can be delivered with Volvo's Dynafleet 2.0 software tool.

The history of the group is very full of events and transitions. In 1999, Volvo Cars was sold to Ford Motor Company. In 2001, **Volvo Group** acquired **Renault Trucks** and **Mack Trucks**. 2007 was a

busy year for **Volvo**; the group acquired **Nissan Diesel** known as **UD Trucks** and in particular, **Lingong**, a Chinese manufacturer of construction vehicles. In 2008, a joint venture with **Eicher Motors** was created. And in 2012, **Volvo Aero's** aerospace components company is sold to the English engineering group **GKN**. Innovation is still part of the Volvo spirit with energy, passion and respect for the individual [1]

Table 1: Volvo Group Innovation History [1]

Year	Innovation
1954	First Turbo
1960	First Volvo Trucks Safety Cab
1969	Establishment of the accident research team
1979	Three point belt
1979	12 Liters intercooler engine
1995	Truck Environmental Concept
2005	Alcolock equipped with factory
2008	Gand, First CO2 free factory
2011	FM Methane Diesel & FE Hybrid

3. Volvo Group Brands

The Volvo group brand portfolio consists of twelve brands divided into four product categories: trucks, buses, construction equipment and marine & industrial engines.

Brands owned by the Volvo group include Volvo, Volvo Penta, UD, Terex Trucks, Prevost, Renault Trucks, Nova Bus and Mack.

Brands facilitated through strategic alliances and joint ventures include Sunwin, SDLG (Shandong Lingong Construction Machinery), Eicher and Dongfeng Trucks.

Each brand within the Volvo group brand portfolio has a unique identity and profile. [1]

3.1 Trucks



The Volvo brand was created in 1928 and has a strong global position. It is one of the most well-known and respected brands in the world within the industrial vehicle industry. It is associated with the Group's core values - quality, safety and respect for the environment.

Volvo Trucks is one of the world's largest manufacturers of medium and high range vehicles for long distance and regional transport, as well as infrastructure projects.



Figure 4: Volvo Trucks Model [2]

Renault Trucks: Constructor of heavy-duty vehicles for long distances. Since 1894, building on the legacy of the French truck industry's expertise, Renault Trucks provides strong tools and links with transport professionals. The main objective of Renault Trucks is to reduce the costs of transport operations for its customers. [2]



Figure 5: Renault Trucks Model [2]

UD Trucks Manufactures and sells a wide range of light, medium and heavy vehicles. UD trucks markets are medium and heavy trucks in Japan, Asia, Australia, Africa, Middle East and Latin America. The UD Trucks motto for its customers is "Going the Extra Mile". [2]



Figure 6: UD Trucks Model [2]

Mack Trucks The emblematic brand of the American truck is known for its longevity, excellent application and customer support. Today, Mack is one of the strongest heavy-duty brands for the North American market. [2]



Figure 7: Mack Trucks Model [2]

Eicher: Thanks to a joint venture, the Volvo Group sells trucks and buses under the Eicher brand, mainly in India. Eicher is one of the major players in the Indian market, with a strong image of energy efficiency and profitability for customers, leader in modernizing the commercial vehicle industry in India and emerging countries. [2]



Figure 8: Eicher Trucks Model [2]

Products are segmented in different feature and price levels, thus giving them a different position on the market and meeting different customer expectations.

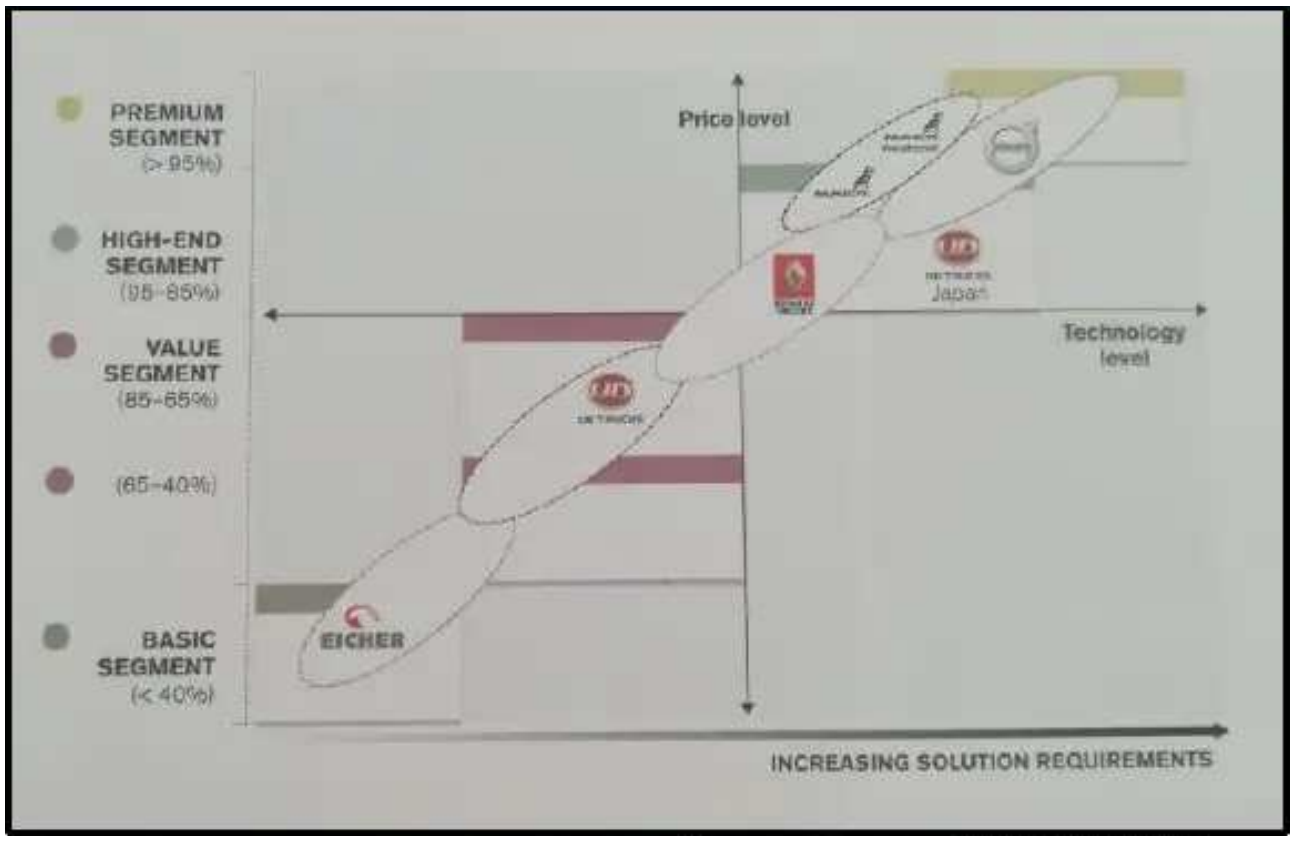


Figure 9 : Products segments [2]

3.2 Construction equipment

Volvo Construction Equipment (VCE): One of the world's leading manufacturers of construction equipment, such as dumpers, wheel loaders, excavators, compactors and pavers.



Figure 10: Volvo Construction equipment [2]

SDLG: Through a joint venture, Volvo Group sells construction equipment under the Lingong brand, SDLG, mainly in China.

3.3 Buses



Volvo Buses Volvo Buses is one of the world's largest manufacturers of buses and coaches: it also distributes chassis, transport solutions and telematics systems.



Figure 11: Volvo Buses Model [2]

UD Buses is a brand new activity for UD since 2014. The UD bus range will offer modern transport solutions developed specifically for expanding markets. It will include intercity buses and coaches.

Prevost Leading manufacturer in North America of top-of-the-range touring coaches, converting vehicles into luxury motorhomes and other special constructions.

Nova Bus One of the leading companies in North America in the design, production and marketing of urban buses.

Sunwin Bus is one of the leading Chinese bus manufacturers and one of the world's largest manufacturers of fully electric buses. The Sunwin brand is sold mainly in China. [2]

3.4 Marine & Industrial engines

Volvo Penta is the world's leading supplier of marine engines and navigation systems for recreational and professional boats as well as for industrial applications such as generators and handling equipment.

4. The Volvo Group Organization

The below organigram represents the Volvo Group organization.

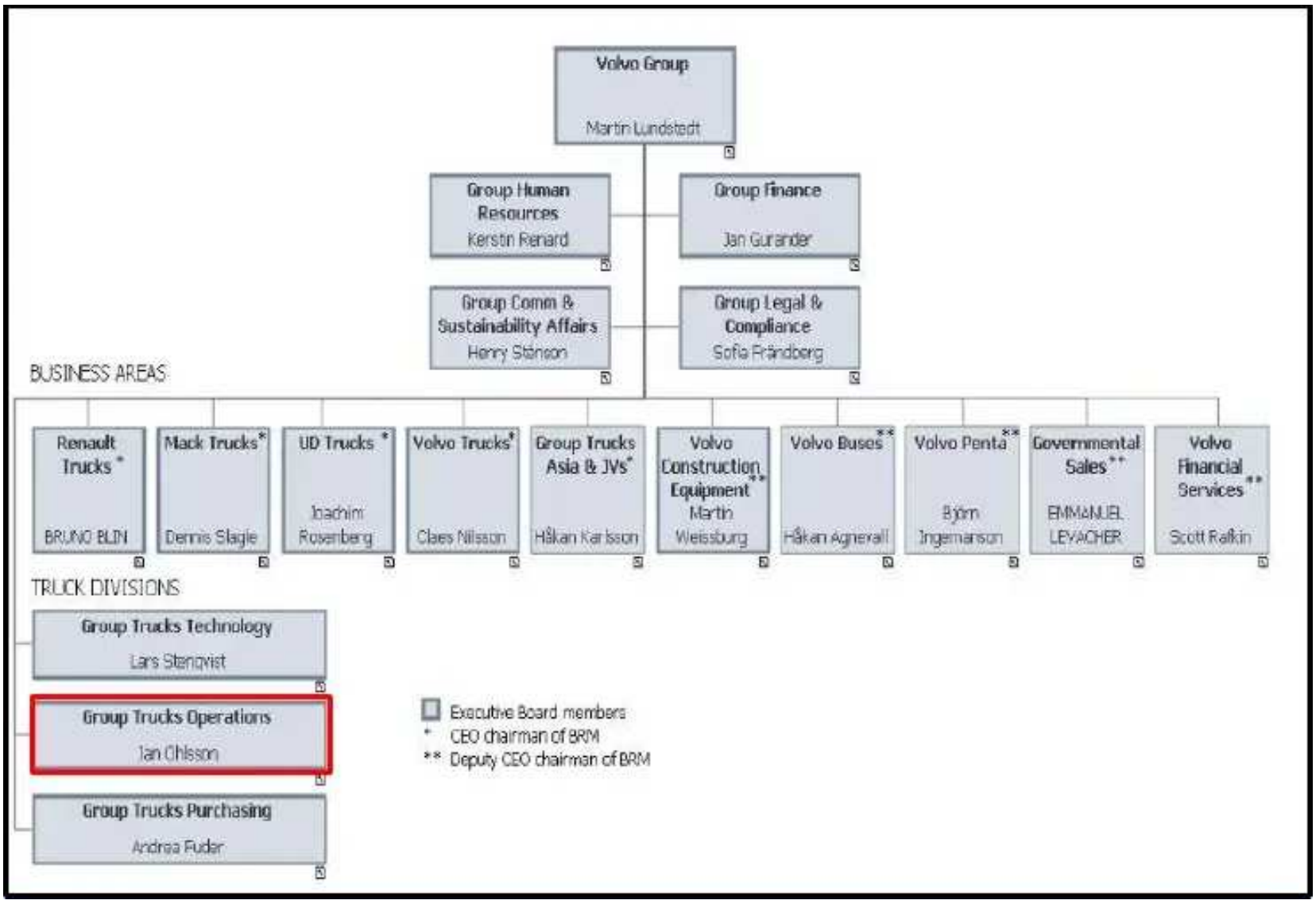


Figure 12: Volvo AB Organigram [3]

The Volvo Group consists of four divisions, each one got a well-defined functions:

Table 2 : Volvo Group Divisions and their roles

Division	Function
Group Trucks Sales	Global responsibility for sales and marketing activities for all truck brands, as well as for the joint venture between Volvo Group and Eicher Motors Limited.
Group Trucks Technology	Research and development of complete vehicles, engines and transmissions, components and ancillary services.
Group Trucks Operations	Manufacture of cabs and trucks for the Volvo, Renault Trucks, Mack and UD Trucks brands as well as the Group's engines and transmissions. Supply of spare parts for Group customers and logistics operations. Responsibility for purchases for the Volvo Group.

Group Trucks Area	Sectors: **Volvo Penta**: marine engines and navigation systems for pleasure and commercial boats and industrial applications. **Bus**: urban and interurban buses, coaches and chassis. **Government Sales**: Sales to government agencies and organizations.

Business Areas encompasses all maritime, governmental, construction site and bus activities of the Group Trucks, on the other hand, includes all of the Group's trucking activities, including product development, purchasing, manufacturing, sales and after-sales.

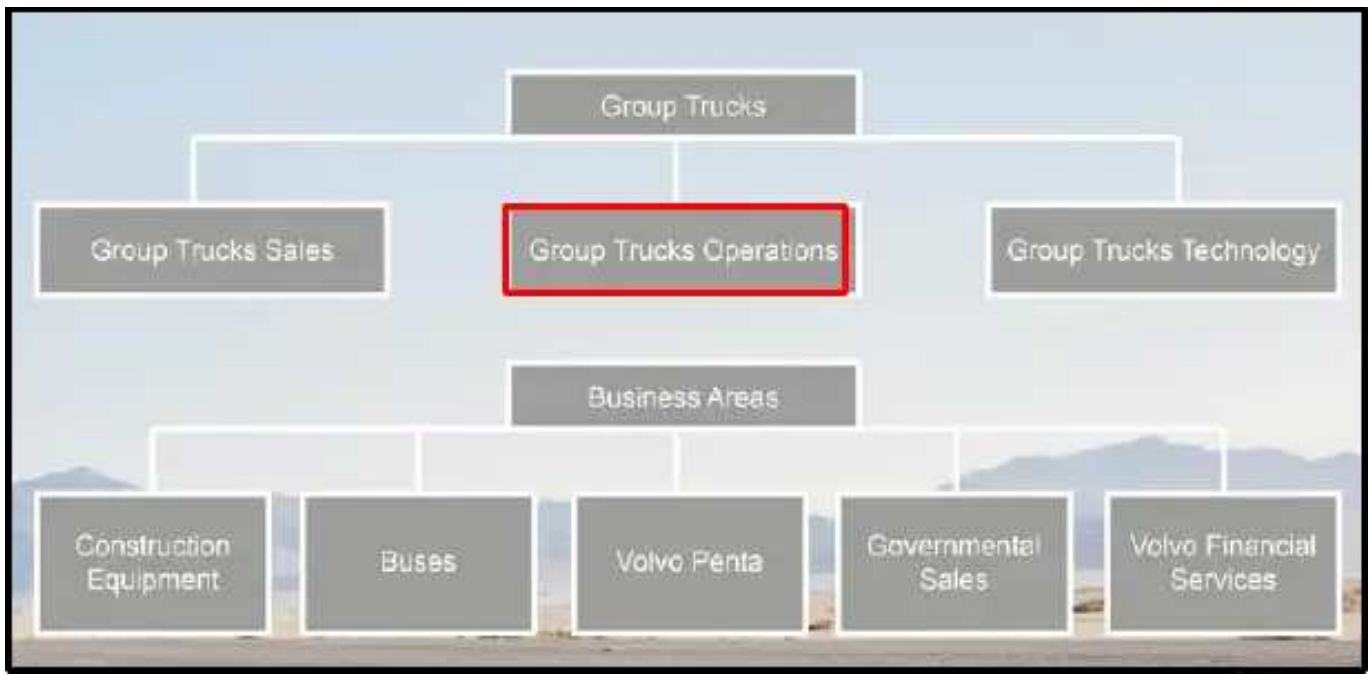


Figure 13 : Volvo Group Divisions [1]

My graduation project took place within the Group Trucks Operations in the Quality and Environment team (Service) having Mr. **David FRÄNNE** as Vice-President.

5. GTO: Group Trucks Operations

5.1 GTO Localization

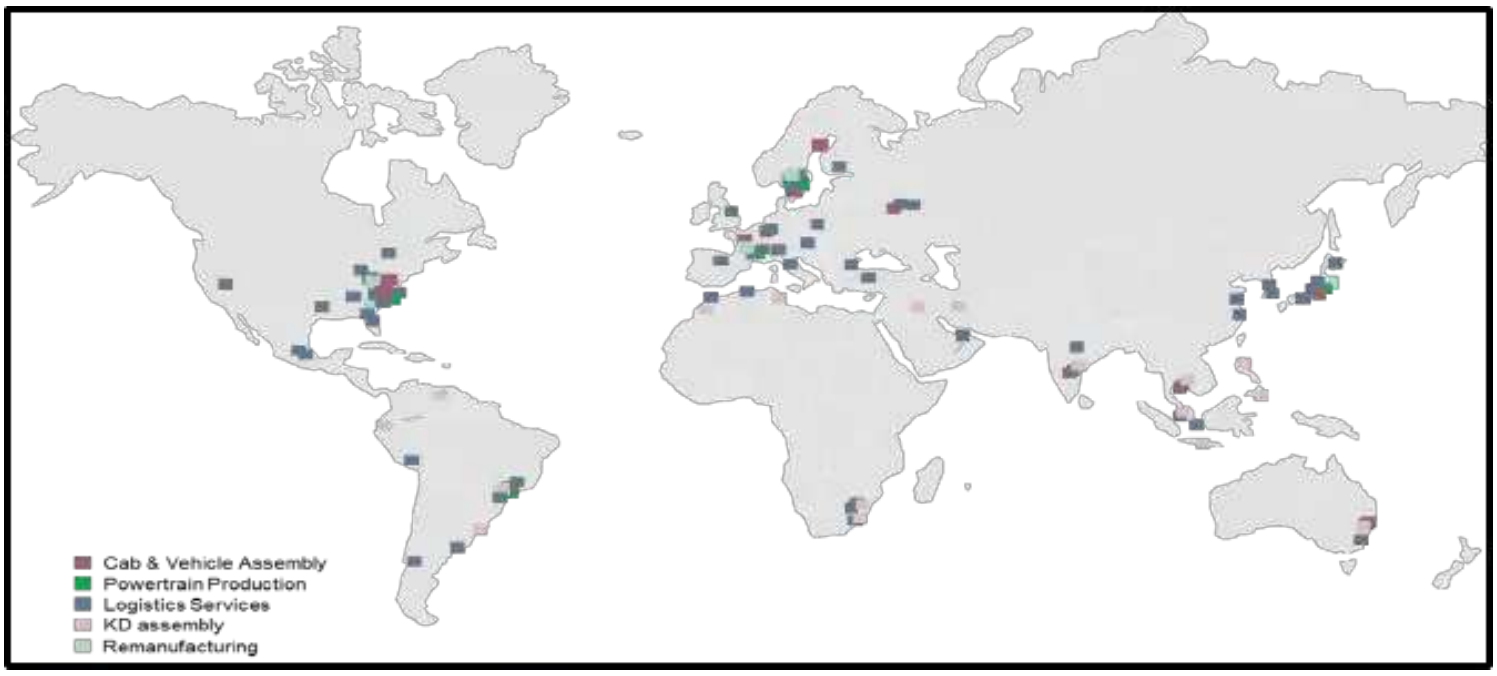


Figure 14 : GTO Plants and Services localizations [1]

There are six regions (operational units) on which we work in the GTO division, three of which are geographical and three operational:

- Europe & Brazil Manufacturing (**E&BM**): assembly plants in Europe and Brazil.
- North America Manufacturing (**NAM**): assembly plants in North America.
- International Manufacturing (**IAM**): assembly plants in APAC (Asia and Pacific Cab & Vehicle Assembly).
- Powertrain Production (**PTP**): engines and transmissions production plants
- Logistic & services: services and logistics
- Purchasing: purchasing services

Working on a global level, means taking into account all regions (these six regions).

5.2 GTO Role in the Volvo Group

The missions of the Volvo Group Trucks Operations division:

- Manufacture of cabs and trucks for Volvo, Renault, Mack and UD brands.
- Manufacture of engines and transmissions for the whole group.

- Supply of spare parts for Group customers and logistics operations
- Health and safety for the whole group.



Figure 15: GTO Role in the Volvo Group

We also highlight that some activities of GTO are not only related to the trucks but can extend to the buses, maritime products etc.

As shown in the graph above, engine production in the GTO division is not restricted to trucks but also to products in the Business Areas division.

5.3 GTO Organization:

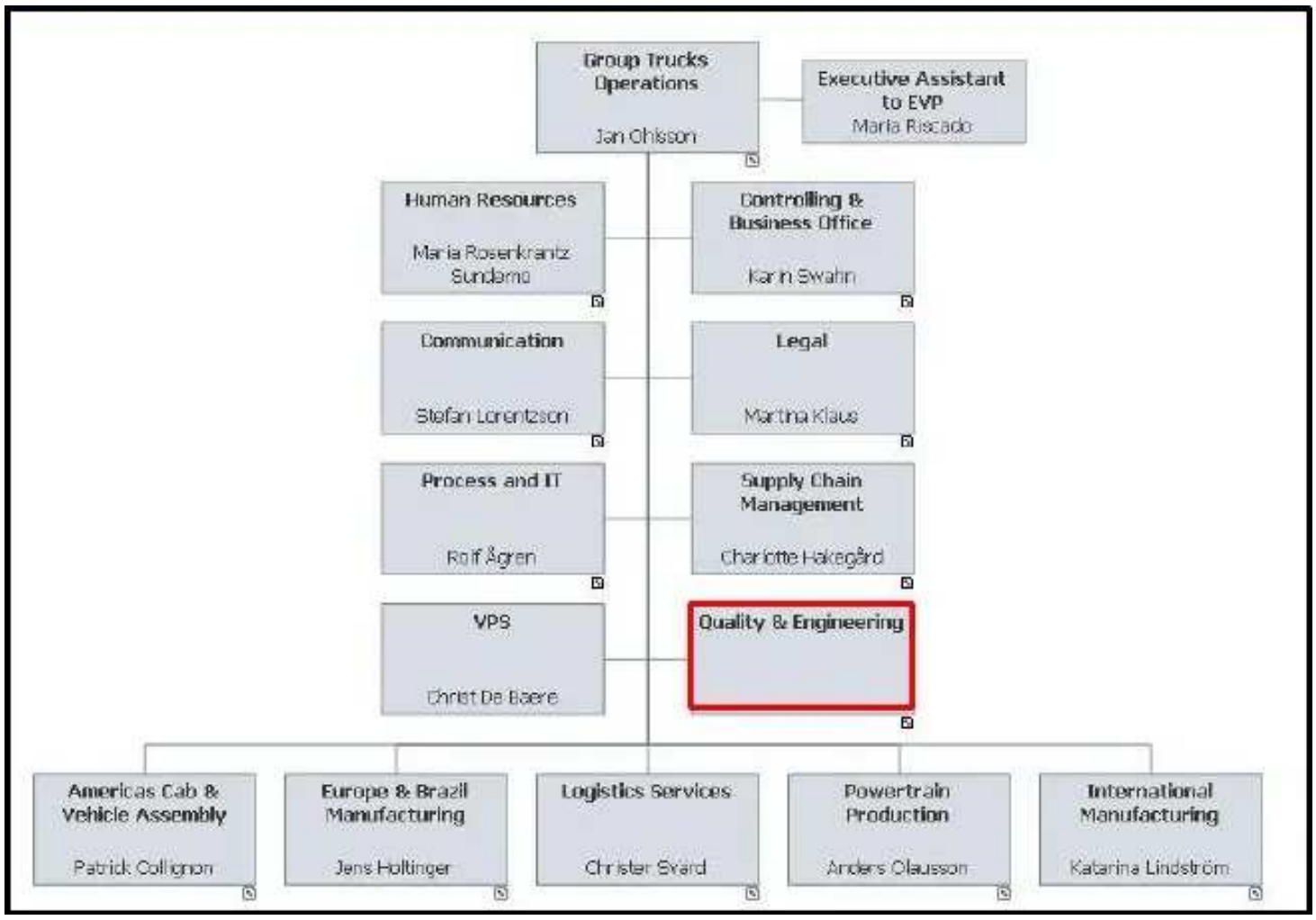


Figure 16: GTO Organigram [3]

Quality & Engineering, CKD is an organization that operates on a global level. It represents a support for current projects in quality and engineering. My internship takes place in the department GTO-Global quality and environment.

AMERICAS Cab & Vehicle Assembly: Manufacture and assembly of cabs, complete vehicles and road tractors for the Volvo and Mack brands in North, South and Central America

Factories: Dublin, Macungie (United States): NRV (New River Valley and LVO).
Employees: 5000

International Manufacturing: Manufacture of a wide range of products for all Volvo Group brands.

Factories: Ageo (Japan), Bangkok (Thailand), Hoskote (India), Wacol / Brisbane (Australia).
Employees: 2000

Europe & Brazil Manufacturing: Manufacture and assembly of cabins, complete vehicles and vehicles kit for Volvo, Renault and UD in Europe, Middle East and Africa.

- 15 factories.
- The biggest plants are at UMEA, Tuve (suede), Kaluga (Russia), Ghent (Belgium), Bourg en Bresse, Blainville (France).
- Employees: 12000

Logistics & services:

Design, handling and optimization of the logistics chain for all Volvo Group brands.

- 55 sites worldwide
- Employees: 5000

Powertrain Production:

Volvo plants engines and transmission lines. One of the biggest manufacturers of high power (diesel) machines for commercial vehicles.

- Factories: Ageo, Hanyu, Konosu (Japan), Charlotte, Hagerstown & Middletown (U.S.), Curitiba (Brazil), Flen, SKOVDE and Koping (Sweden), Limoges & Vénissieux (France).
- Employees: 8000

GTO Purchasing:

A global function covering the purchase of automotive and non-automotive products and services for the Volvo Group.

5.4 GTO Global – Quality & Environment

The quality and environment department is responsible for defining and managing the medium and long term quality and environment work methods in the GTO division. It also supports projects on the global, regional and local level with quality and environmental expertise. It is, among other things, the main interface with other global functions and ensures good cooperation between GTO, GTT and VGRE.

It coordinates and share quality and environment networks with the aim of optimizing, standardizing, accelerating performance improvement and creating a team spirit between the different operational units.

Its mission is to develop a culture of quality with a spirit of zero fault mind set allowing a quick

identification of the problem, correction and prevention of customer claims while reducing the cost of warranty calls.

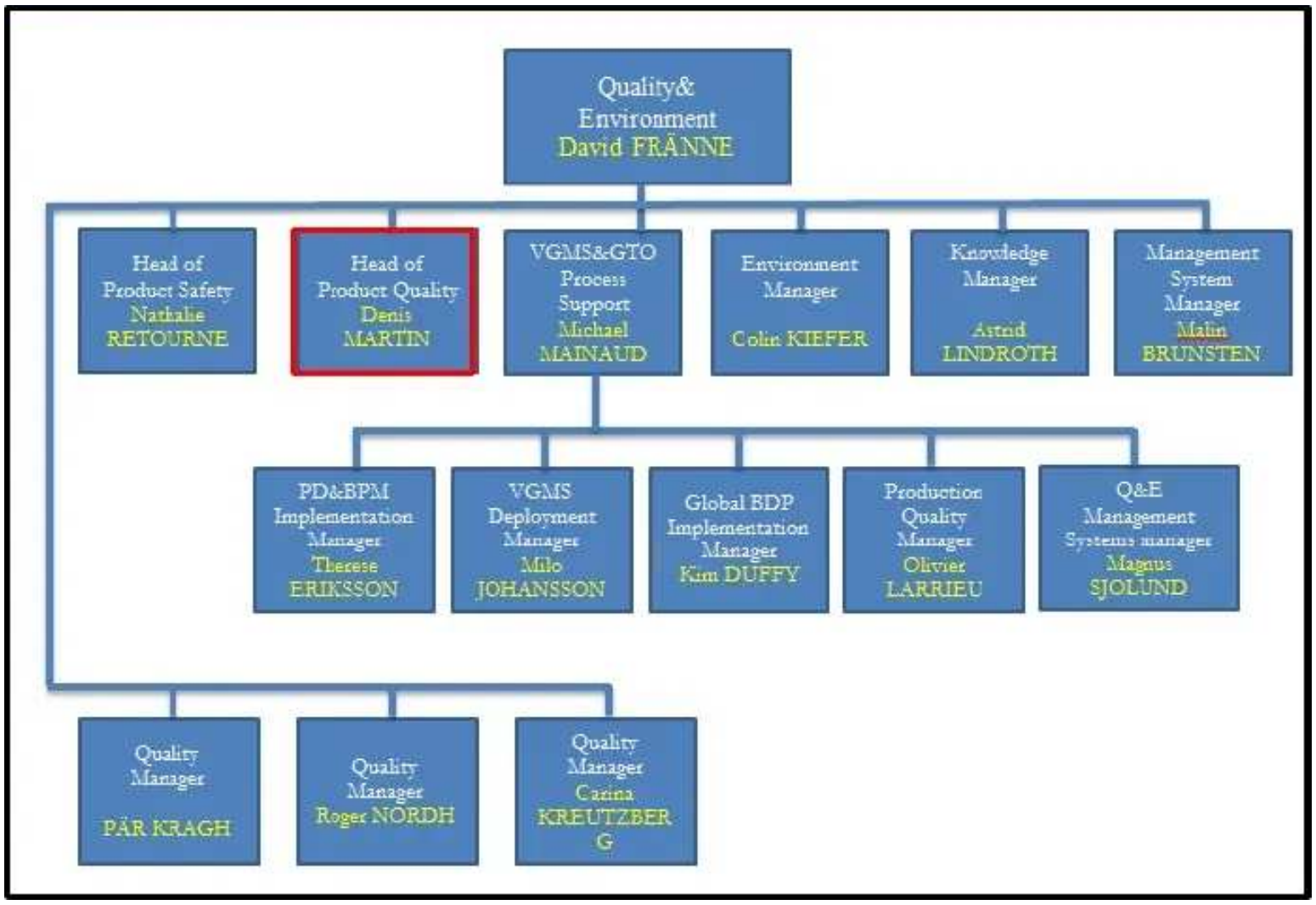


Figure 17: Quality & Environment Department Organization

We worked with Mr. Denis Martin (Head of Product Quality). Our entity handles the global product quality of GTO globally and ensures homogeneity in each region. This is explained by:

- Implement processes to improve product quality by managing the effectiveness of preventive and corrective actions related to the Cost of Non-Quality
- Develop quality strategy, standards and processes.
- Ensure that the quality of the product meets the requirements and the satisfaction of the client and the authorities.
- Produce and deploy robust quality standards
- Reduce the cost of Non-Quality
- Provide quality improvements
- Share lessons learned and best practices.

The above points are achieved through four key activities or belongings:

- Owning the GTO corporate KPI : the FF1 : Fault Frequency 1 month
- Product Audit: as it is the last Volvo check point before that the product reaches the end customer.
- Handling Customer Quality Issues (through QSPs: Quick Solving Process & QJs: Quality Journals).
- Internal campaign process management.

III. Project specifications

Since the first software was developed and implemented in a car, back in the 80's, software have been knowing an exponential growth considering its size by counting the number of lines of codes.
A modern vehicle's calculators, nowadays, may contain up to 400 Million lines of codes.
In addition to that, over 40 different ECU can be installed in today's Volvo Trucks.
These two facts imply a growing risk of non-quality issues related to software, which is also confirmed by the strong contribution of software claims to the overall customer claims.

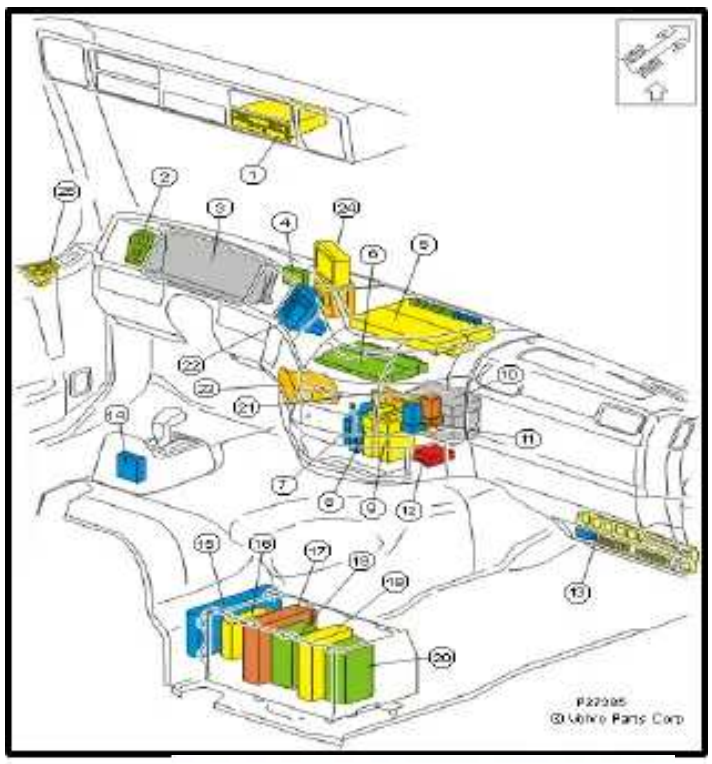


Figure 18: Truck ECUs [4]

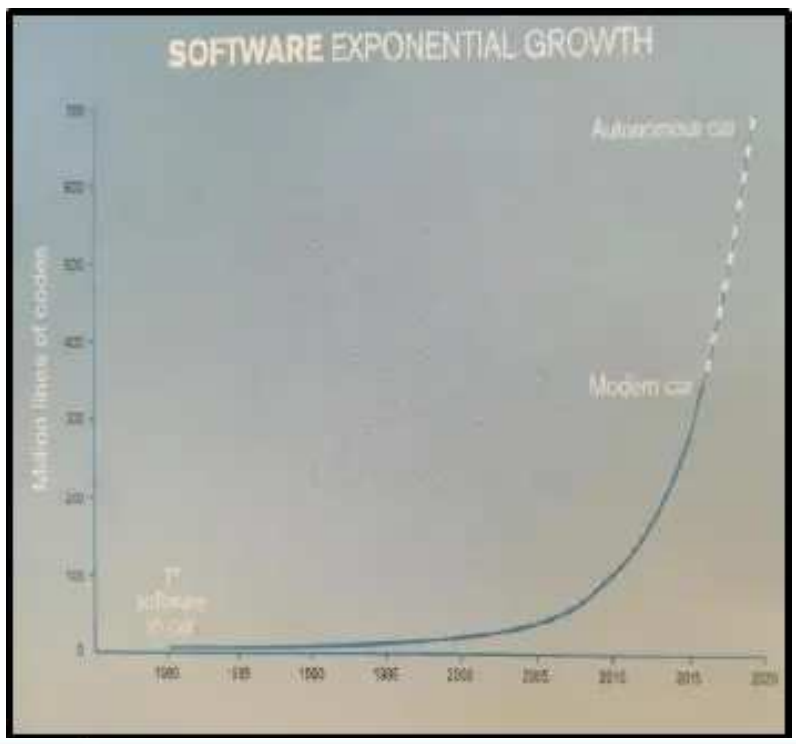


Figure 19: Vehicle Software Exponential Growth [3]

8

By that, software claims are affecting strongly the FF1(GTO's corporate KPI) which is our main responsibility.
Taking that into account , I was asked to initiate and lead two working groups (Europe Brazil manufacturing and North America Manufacturing) in order to ensure the improvement of product quality from the embedded software point of view and subsequently reduce its contribution to customer claims.

This task includes:

- ✓ Running analysis on software claims in order to assess the current situation.
- ✓ Communicating the first degree analysis to the top management in order to have their implication and commitment to the subject.
- ✓ Initiate the two working groups (task force) by gathering around the table, representatives from all the areas involved in the software issues (from Design, Manufacturing to Warranty).
- ✓ Carry out detailed analysis on the software claims and establish an action plan whose aim is to reduce software claims by having more reliable trucks.
- ✓ Monitor and follow up (Follow up meetings) the action plan progress, establishing a continuous improvement strategy.

Having a customer focus mindset, the target style for handling non-conformity and warranty claims is to continuously challenge ourselves to secure customer-related interfaces and establish the most efficient connections with other parts of the Volvo Group.

Speed and agility in solving customer problems and internal problems that may affect also the customer are key. Despite all preventive measures, problems will occur which may affect customers and road users. The benefit of common platforms (product-process-supplier) can be jeopardized when problems occur as they impact a wider range of products and markets. Therefore, reactivity, traceability and cross-site information sharing and actions are crucial to contain issues responsibly.

In this context, my main tasks are:

- ✓ Analyze handling non-conformity processes and their categories.
- ✓ Work on the enhancement of QULIS: Quality Information System, in order to deploy it in all the GTO plants.
- ✓ Set Handle Non-Conformity performance indicators whose main purpose is to evolve operational excellence and monitor Non-Conformity treatment actions (Draft Proposal)
- ✓ Present the Draft to all Operational Units in order to get their approval and start developing the dashboard.
- ✓ Develop and implement the new dashboard in all GTO plants.
- ✓ Set a continuous improvement strategy for this new IT tool.

We should also mention that our graduation project should be carried out in correspondence with the **ISO 9001: TS16949 Quality management systems – Particular requirements for the application of ISO 9001:2008 for automotive production and relevant service part**

organizations and the **ISO 10002: Quality management – Customer satisfaction –Guidelines for complaints handling in organizations.**

On the technological side, we used multiple IT environments for this project such as MS EXCEL, QlikView, PROSIT+, QRAFT and ARGUS.

This specification can be resumed in the following figure as a schematic representation in order to simplify its understanding and analysis.

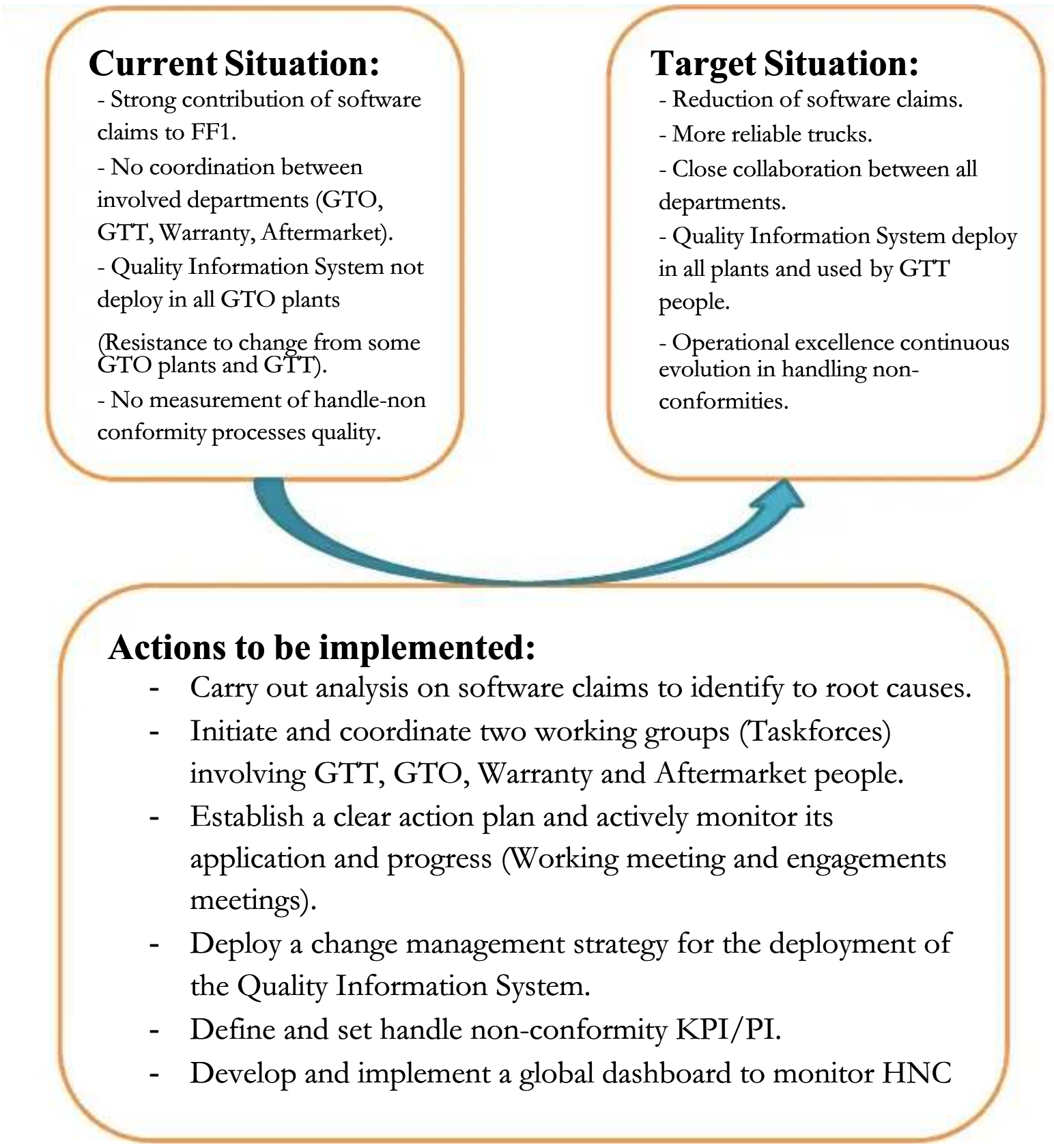


Figure 20: Specifications mapping

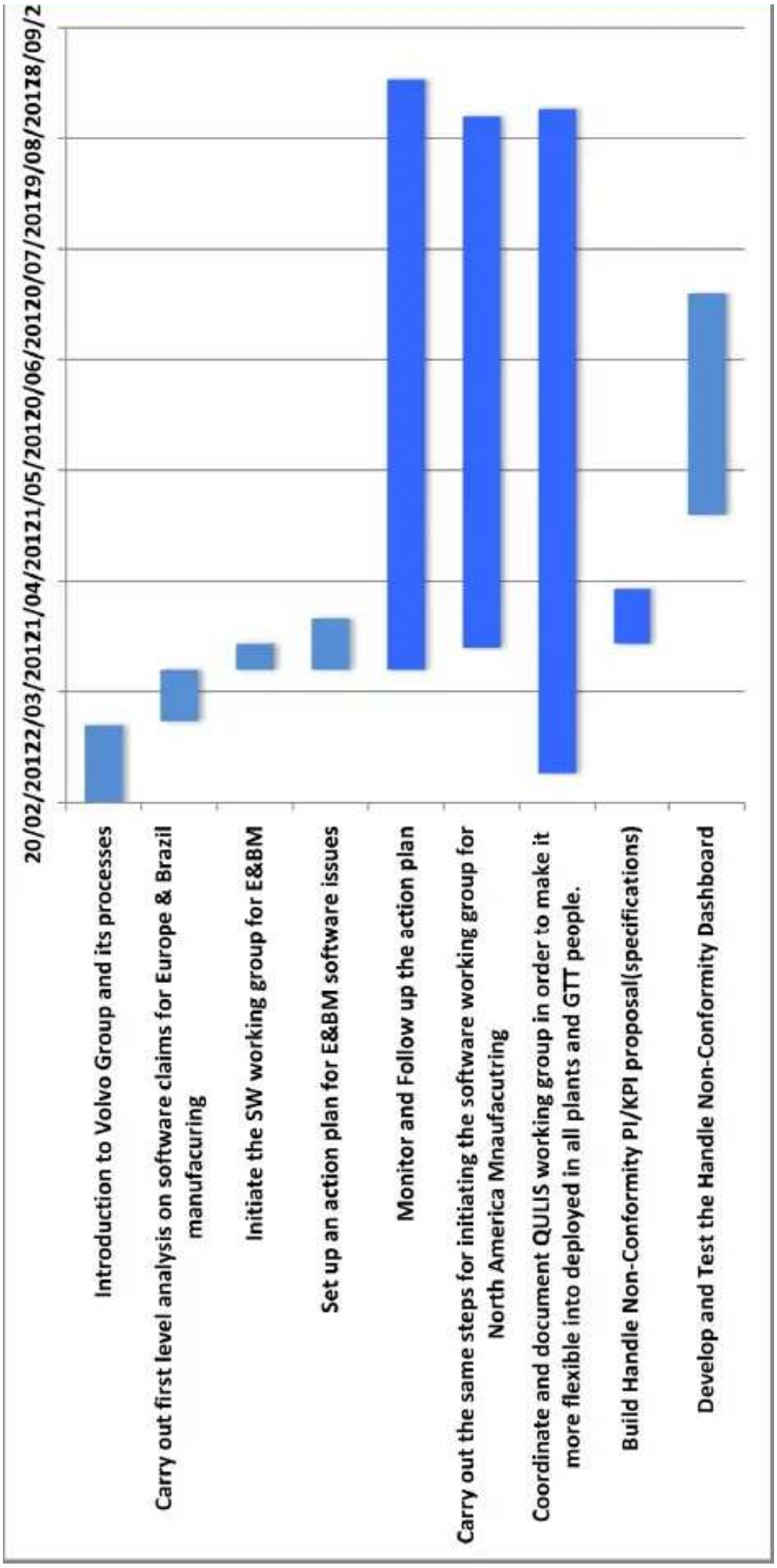


Figure 21 Graduation Project Gantt chart

The actions that took place during our graduation project are defined in the above Gantt chart (Figure **21**).

IV. Conclusion

The environment and the context of the graduation project established in this chapter demonstrate that the software is complex to control activity of the Volvo Trucks Production System and that there is no global dashboard for Handling Non-Conformity Processes. These points show a necessary perspective to master properly these software issues and objectively measure the quality of the provided services.

Having an industrial computing and embedded systems background with some knowledge on quality systems where all processes are optimized and measured to assess performance and allow continuous improvement, I was selected to carry out this mission by bringing a critical eye on the activities around software problems and establishing solutions in terms of productivity and quality management.

The next chapter will be devoted to the study and the definition of the way of working for the main two defined problems in order to set a proposal for solutions.

Chapter II

Current Situation Study

&

Proposed Solution

I. Introduction

In this chapter, we will be presenting the existing situation of Volvo Trucks in direct relation with software issue through the analysis we carried out in order to start the working groups for this kind of customer claims .Therefore, we will first of all present the warranty system till we get to the calculation of our KPI the FF1 which is the main driver of all implemented solutions and working groups in GTO.

The second part will be dedicated to the study of the Handle Non-Conformity Process, its categories and its support IT solution QULIS (Quality Information System) the possibility to establish global performance indicators and a dashboard to measure and monitor Handle Non-Conformities cases and by that build an operational excellence continuous evolution system.

II. Software claims

To describe properly the state of the art at the beginning of the project, we will start by explaining the process of handling customer claims, how we calculate our main KPI the FF1, to focus on the software problems and how we shall proceed with this issue.

1. Process Approach

In correspondence with what the ISO 9001: TS16949 is promoting, the Volvo Group is adopting a process approach when developing, implementing and improving the effectiveness of its quality management system, to enhance customer satisfaction by meeting Customer requirements.

An advantage of this process approach is the ongoing control that it provides over the linkage between the individual processes within the system of processes, as well as over their combination and interaction.

When used within a quality management system, such an approach emphasizes the importance of

a) Understanding and meeting requirements,

b) The need to consider processes in terms of added value,

c) Obtaining results of process performance and effectiveness, and

d) Continual improvement of processes based on objective measurement. [4]

As shown in the next figure, these are the Volvo Group Trucks Core Processes.

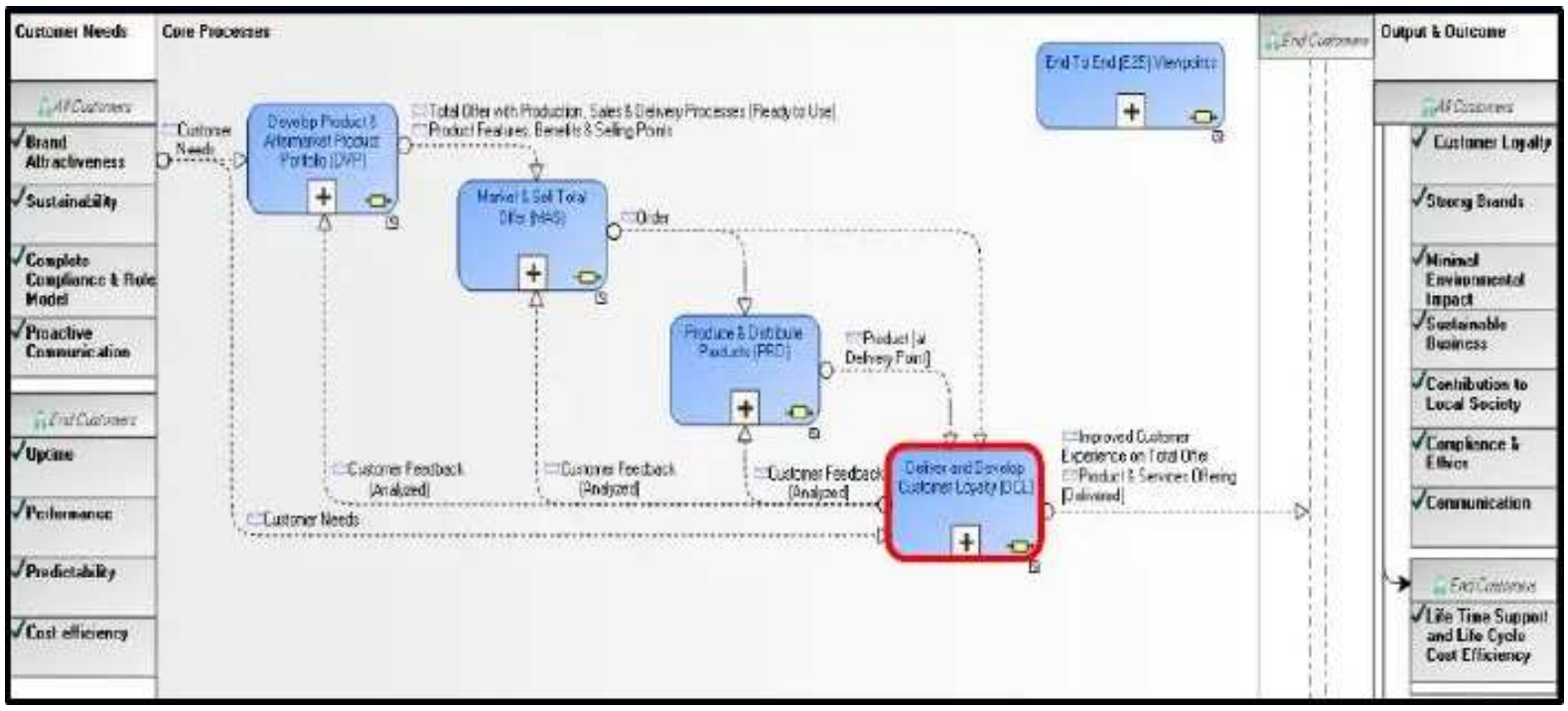


Figure 23 : Volvo Group Trucks Core Processes [6]

Our work took part of the "Deliver and Develop Customer Loyalty" processes and more specifically in the "Collect and Analyze Customer Feedback and Improve Quality" process (Level 2).

2. Customer Claims Information Flows & the way it is proceeded

To commonly understand the customer claims incoming flows and its processing we have decided to resume it in the **Figure 23:**

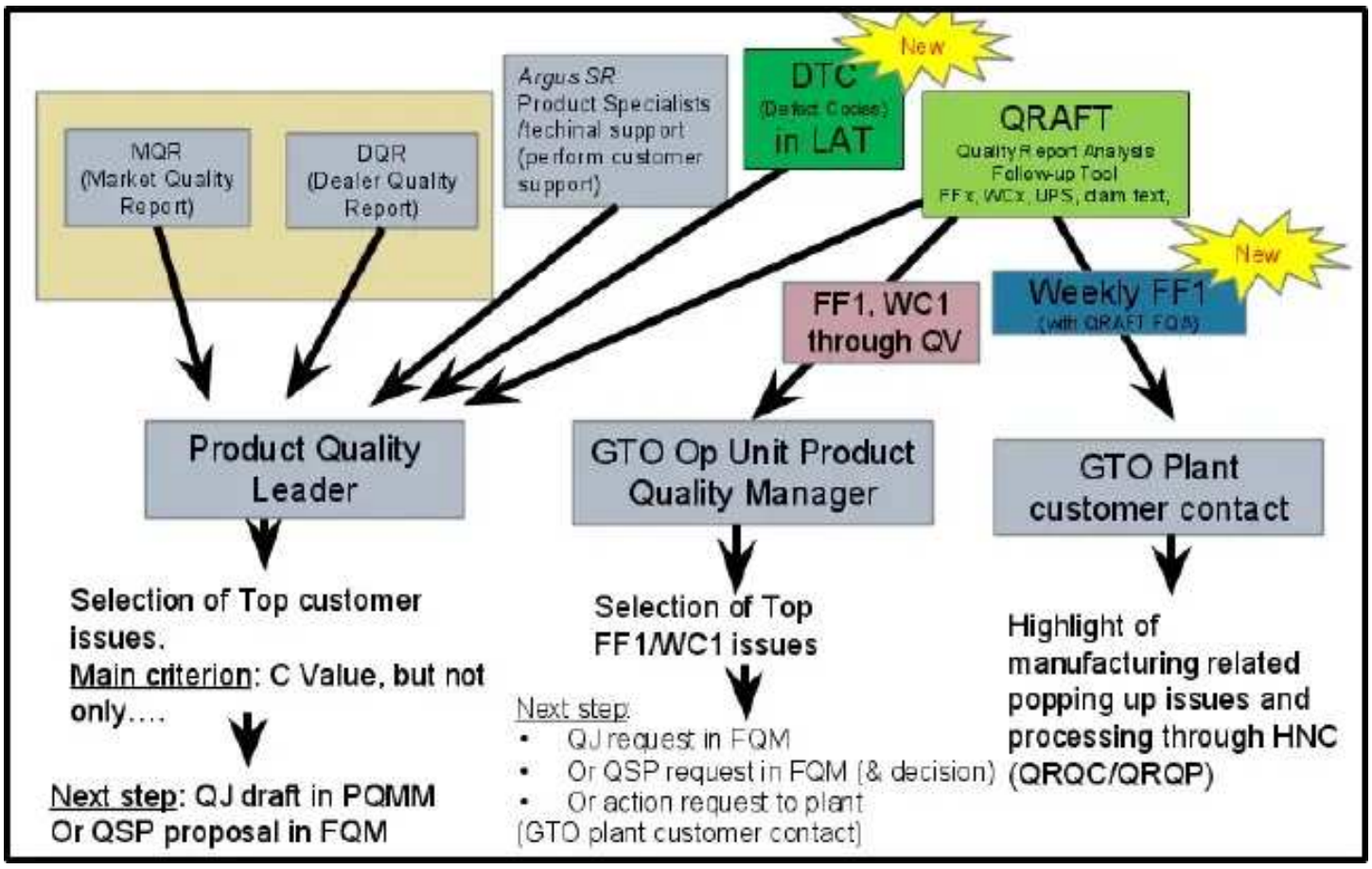


Figure 24 : Customer claims flows and processing [7]

We have three possible roads for handling customer claims:

- The Product Quality Leader (PQL who is a member of the GTT Q&CS Team) has all possible inputs to analyze and decide to deal with an issue through a QJ Draft or a QSP Proposal during the FQM (Field Quality Meeting).
  The **C Value** is the PQL's main criteria to decide whether to deal with the problem or not.

  The Next figure is a definition of the C Value.

Figure 25: C value Calculation and Criticality [7]

- Through GTO Operational Unit Product Quality Manager based on the FF1 and WC1 data.
- Through GTO plant customer contact using weekly FF1 data.

The Data Bases mapping for customer claims is described in the next figure.

Figure 26: Warranty Mapping [7]

UCHP: Unified Claim Handling Process is the web based application developed by Volvo Truck in correspondence with the **ISO 10002: Quality management – Customer satisfaction – Guidelines for complaints handling in organizations,** to support customers and Volvo Trucks processes.

Therefore, UCHP is used by dealers and end customer to write a claim about a certain quality problem that occurred.

PS: We should note that software related problems are usually automatically approved by UCHP since they are claims of low cost compared to hardware issues and that the re-upload of the software resolves a decent amount of the claims but nevertheless the problem may show again.

3. FF1: Fault Frequency 1 Month

3.1 FFx

The Fault Frequency (FF) are indicators used to monitor the quality. They are calculated using the number of claims received for a sample, divided by the number of trucks in service within this sample.

If the FF is low, the quality is good. Otherwise, investigations are set in motion to understand why the quality is not at expected level.

FFx are commonly used in Volvo's quality services, but they don't control the same period. FF12 takes into account the 12 first in-service-months of the trucks; FF24 takes into account the 24 first months, etc.

These above FF are used by the Sales service to know which parts of the trucks cost the most to the company during the warranty period of the vehicle.

As these indicators are essential to the company, a big database (as shown in the **Figure 24**) exists in which is gathered every information about every product and every claim.

In these data, we have everything we know about the vehicles: identification number, brand, model, assembly plant, in-service date, engine model etc. and about the claims: reference number of the claim, repair date, dealer information, price, causal part, **Defect code**, dealer's comments etc.

3.2 Fault Frequency 1 month

FF1 represents the average number of claims that affects a vehicle (or product) from a specific production month, from the time it reaches the dealer until 1 month in service at end customer.

=> Unit: claim per unit

Calculation:

All claims related to the trucks of a production month that occurred from plant delivery till 1 MIS (1 Month In Service) / Total number of trucks in service produced this month (unadjusted claim rate).

! *Maturity 0 is considered for vehicle: means that a claim is not considered for FF1 calculation as long as related truck is not in service (but it is as soon as the truck is in service).*

Considered claims for FF1 calculation:

- Included claims:
  - All variants included (i.e. customer adaptation)
  - Pre-delivery claims included
  - miss-build and discrepancy (out of specifications)
  - claims related to body builder activities
- Not included claims:
  - Service campaign and safety recalls
  - Goodwill
  - logistic claims (no defined mean to catch them until now)

Conditions to display FF1 (Scope):

Two conditions are needed to display FF1 figure that has a meaning (considered as a « mature »):

1) Minimum sampling → at least 40%* of a production month must be in service
2) Minimum maturity level → 65% of delivered trucks must have reached 1 month in service

Even when FF1 figure is mature for a specified month, it is still not frozen.

Important: as an exception and to avoid confusions, reported global FF1 is always related to « reporting month – 3 months ».

Example: In September, we display August report with Global FF1 related to May.

Note also that:

FF1 per model in monthly report is related to vehicles delivered in the geographical region in which they have been manufactured.

FF1 per plant in monthly report is related to vehicles delivered everywhere.

The **Figure 26** shows the FF1 figures of the Volvo FH as displayed in June 2015 :

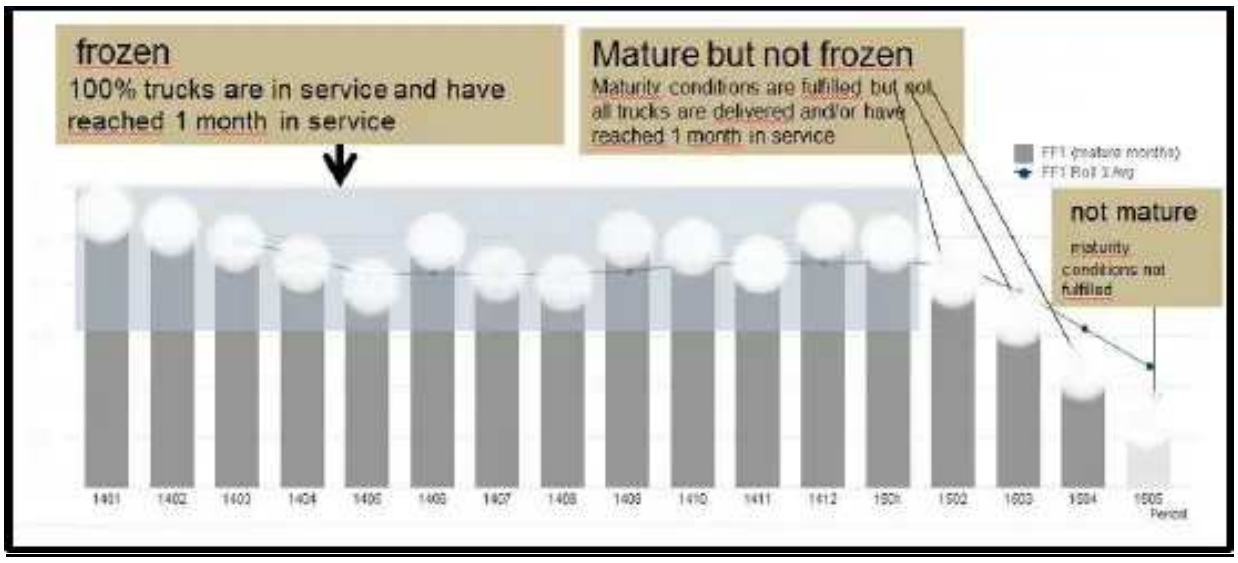


Figure 27:FF1 figures of Volvo FH (June 2015) [7]

3.3 The FF1 dashboard

The FF1 dashboard was built using a software called "**QlikView**" (a presentation of the software will be exposed in the next chapter). The dashboard is using the claims data to automatically calculate our key performance indicator.

It is set using filters in order to be able to select a scope to monitor and focus on:

Example: we can select as Operational Unit: E&BM, Plant: Tuve by that we will be monitoring the FF1 only for this scope.

It generates also paretos of: Top causal parts, function groups etc.

More details of the Qlikview FF1 Dashboard are presented in the Annex.

4. First High Level Analysis

In order to run our analysis properly, we had to define some points:

- **Objective:** Deliver a high level analysis to highlight the big contribution of the software related issues to FF1 and customer claims in general. And then, investigate deeply the root causes of the quality problem in order to establish an action plan to reduce the software claims and continuously improve trucks reliability involving all departments.
- **Period**: Claims for vehicles built from the 1st September 2016 till February 2017.
- **Data**: Extracted data from the FF1 Database.
- **Team**: **Capucine BOUSQUET (GTO E&BM SW Process Manager), Yamina El HAJEM (Director SW&EL France), Denis Martin (Head of Product Quality) and Baha-Eddine Alayet.**
- **Standard**: Volvo Group Standard **STD 920-0004: Defect codes.**
- **Software**: Excel, PROSIT+.

4.1 STD 920-0004: Defect codes

This standard covers definitions of defect codes. It also defines a structure which shall be used by entities in Volvo Group systems & organizations that work with defect codes, including for example claim handling, calculations, warranty, technical service, service contracts, commercial goodwill and extended coverage.

The standard's main purpose is to support a common way of working by referring to common defect codes, so that systems can utilize defect code data from different parts of the organization. Additionally, the standard defines a routine for change management of its contents.

PS: During our project, we will be focusing only the defect code "Software". There are 120 possible defect codes (Short circuit, leak, software etc.).

4.2 PROSIT+:

PROSIT+ stands for **PRO**duction **S**ystem **I**ntegrated **T**ools; it is composed of a set of tools (software and hardware) that allow multiple tasks:

- Read/Erase fault codes (fault codes or DTCs are code defined by GTT that refers to a problem in the vehicle: accurately define the problem.).
- Program the ECUs.
- Test.
- Calibration.
- Check.
- Alter Parameters. [4]

We used PROSIT+ central database in order to check for each claim (vehicle) whether it had a software problem during the production process (having as an aim to detect if there is a gap or more like opportunity to enhance the verification on the manufacturing side).

In correspondence with the International Standards all executed tasks and all related results using PROSIT+ are stored for traceability reasons.

PROSIT+ Environment is presented in the Annex A.

4.3 Analysis Results

Using the defined extracted data on Excel and the pivoted tables' functionality and "Software" as a defect code filter, we were able asses the contribution of software issues from a global view as shown in the **Figure 27**.

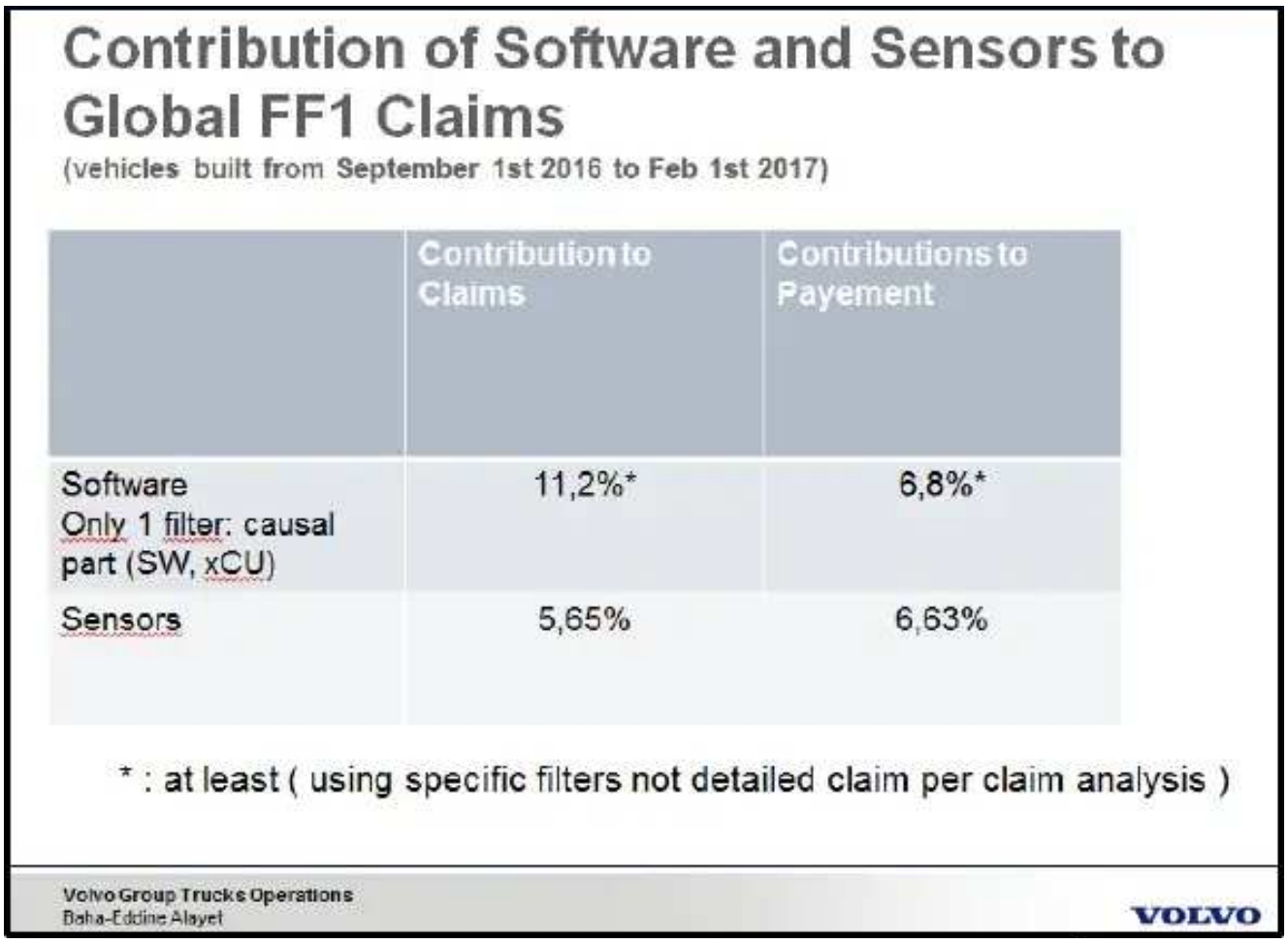


Figure 28: Software claims contribution to FF1 and claims payments

We can see by that the huge contribution of software to customer claims and by that to quality issues as only software issues represents 11,2% of total FF1 claims having in mind that there is 120 possible defect codes and only one of the 120 has such a contribution.

The **Figure 28** shows the bad trend of the contribution of software claims since September 2016.

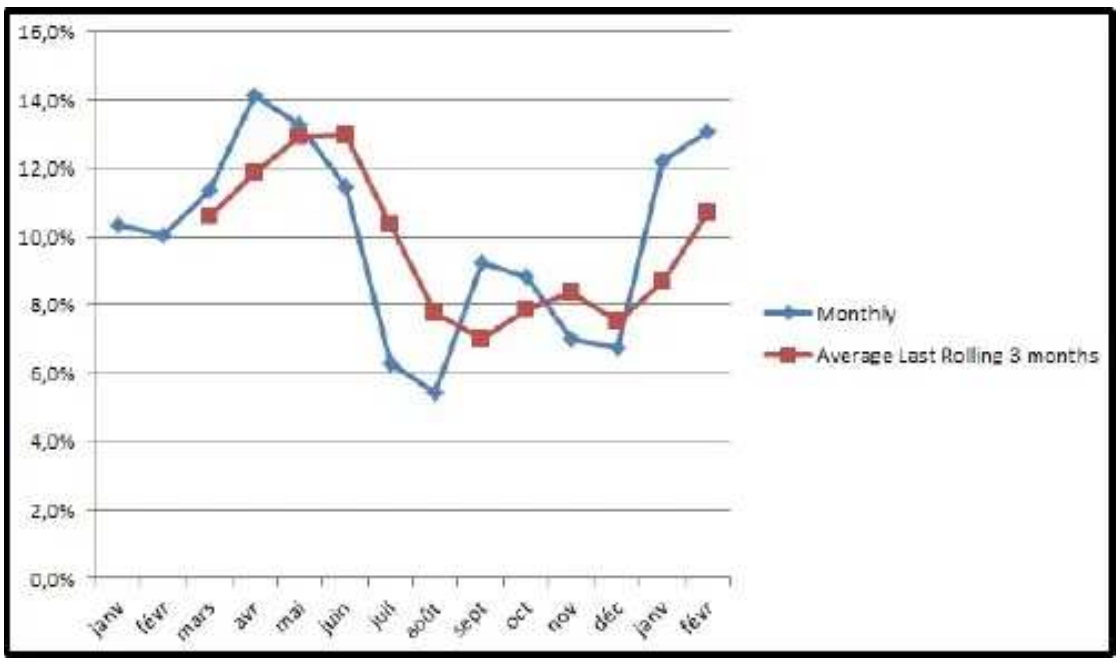


Figure 29: SW claims trend before starting the WG

Going further in our analysis, we decided to use the pareto analysis in order to set the main contributors in software issues for each operational unit (E&BM and NAM).

Having that in mind we started by drawing the pareto of the contribution of each production site to software issues.

This pareto is presented in the coming figure.

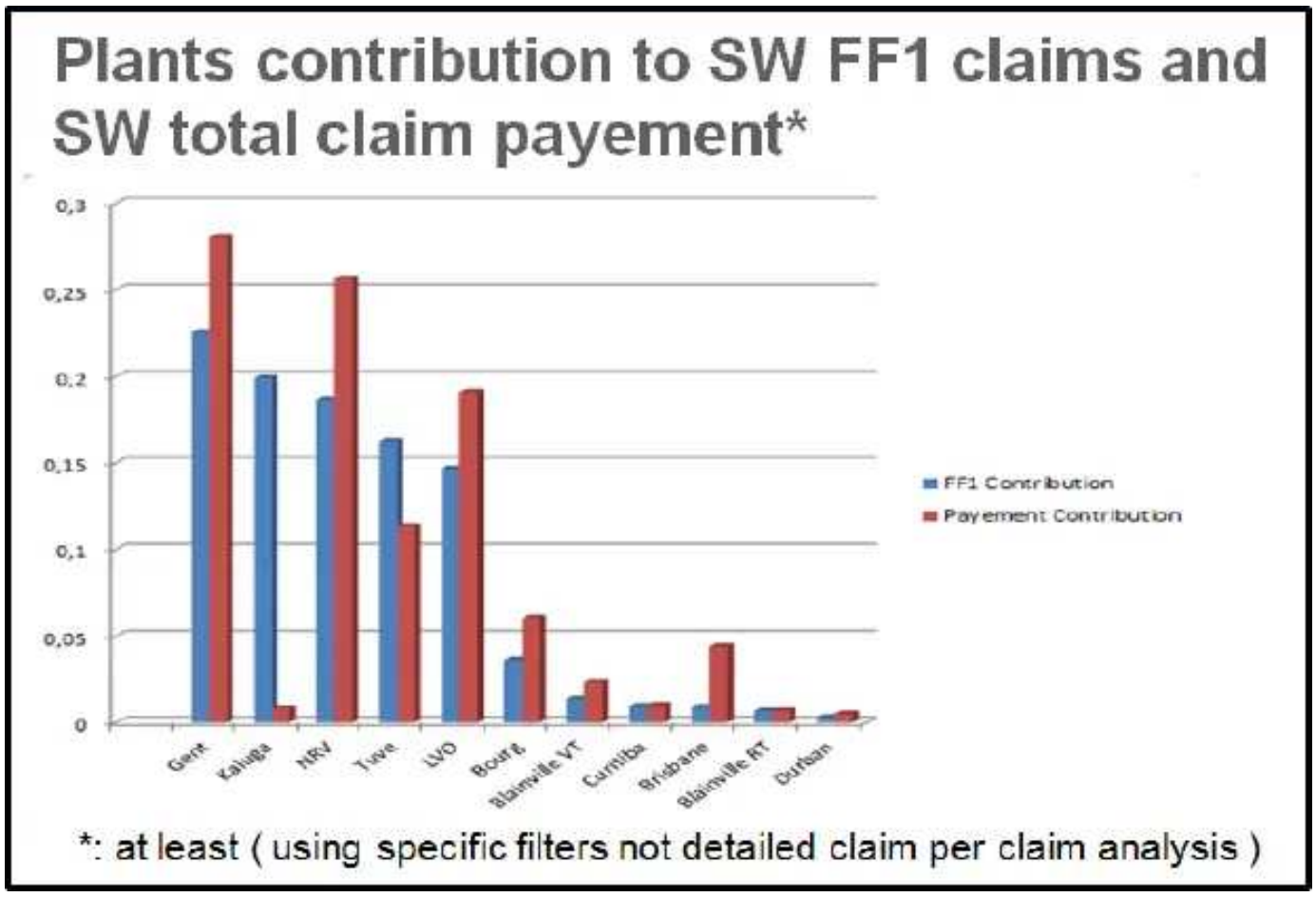


Figure 30: Plants SW contribution to FF1 claims

As shown in the **Figure 28,** for Europe & Brazil Manufacturing, the leading contributors are Gent (Belgium), Kaluga (Russia) and Tuve (Sweden).

And for North Americas Manufacturing NRV (New River Valley-USA) and LVO (USA).

- Based on this pareto and knowing the impossibility to manage detailed analysis and working groups related to multiple plant, we decided to carry on our analysis for Europe & Brazil Operational Unit considering **Gent plant** as a **pilot** (since it is the biggest contributor to software claims. We should also mention that Gent is the biggest Volvo plant in the world with a production capacity that goes up to more than 4600 truck per month).

  All actions will be applied to the rest of the plants.

  The SW claims are causing financial losses of **8K euro** per month (Gent).

- For North Americas Manufacturing our analysis will take into account both plants: NRV & LVO.

  To have a financial overview these SW claims costs **1077 K SEK (121K$)** to the warranty budget for the last 6 months => average of **20K$** per month (for NAM).

The Volvo Trucks work is divided into multiple competence areas called "Vehicle Module Team" during our first analysis we detected that only 3 VMTs which are:

- 20: Electrical components
- 21: Electronics
- 50: Engine

detain 80% of the software claims , so we reduced our scope to take only this three function groups into our loop.

To reduce our scope for E&BM, we selected these filters:

- Defect code : Software
- Plant: Gent
- Function Groups: 20, 21 and 50.

Considering that we were able to reduce our scope to select **134** customer claims referring to the top 3 causal parts:

- 3651-Electronic Control Unit.
- 3111-Battery Complete Energy Absorber.
- 2849-Miscellaneous (Control system, Fuel supply).

These 3 causal parts contribute to 70% of Gent software claims.

The next step was to run more focused and detailed analysis on these 134 issues in order to assess better the situation and define a working strategy.

This task was done using PROSIT+, the customer/dealers comment fields and our expertise in the embedded software sector.

The main core of the team is composed of: **Denis MARTIN**, **Capucine BOUSQUET and Baha-Eddine ALAYET.**

The final results are summaries in the next figures:

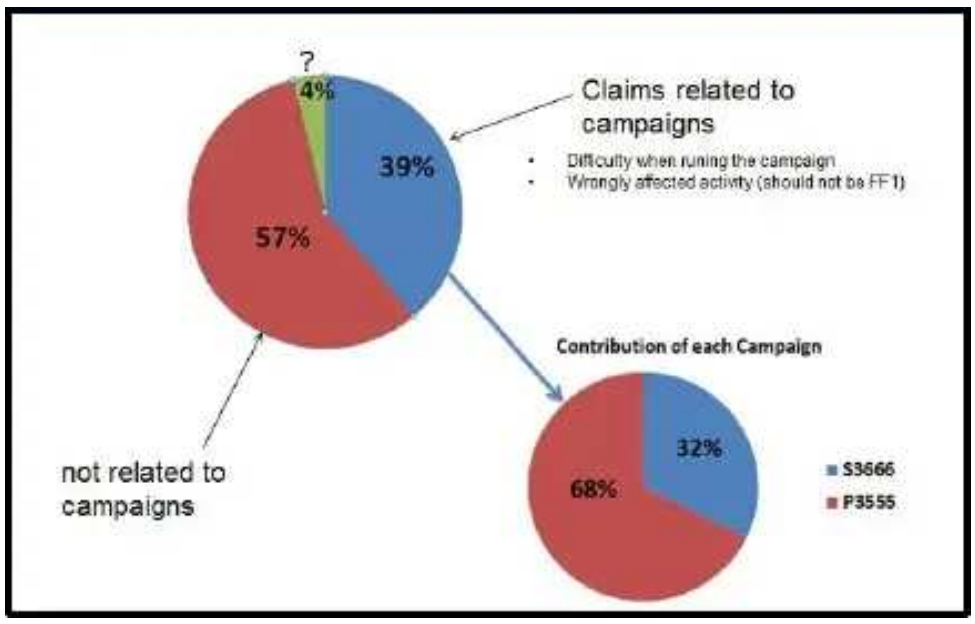


Figure 31 : Impact of campaigns on FF1 software claims

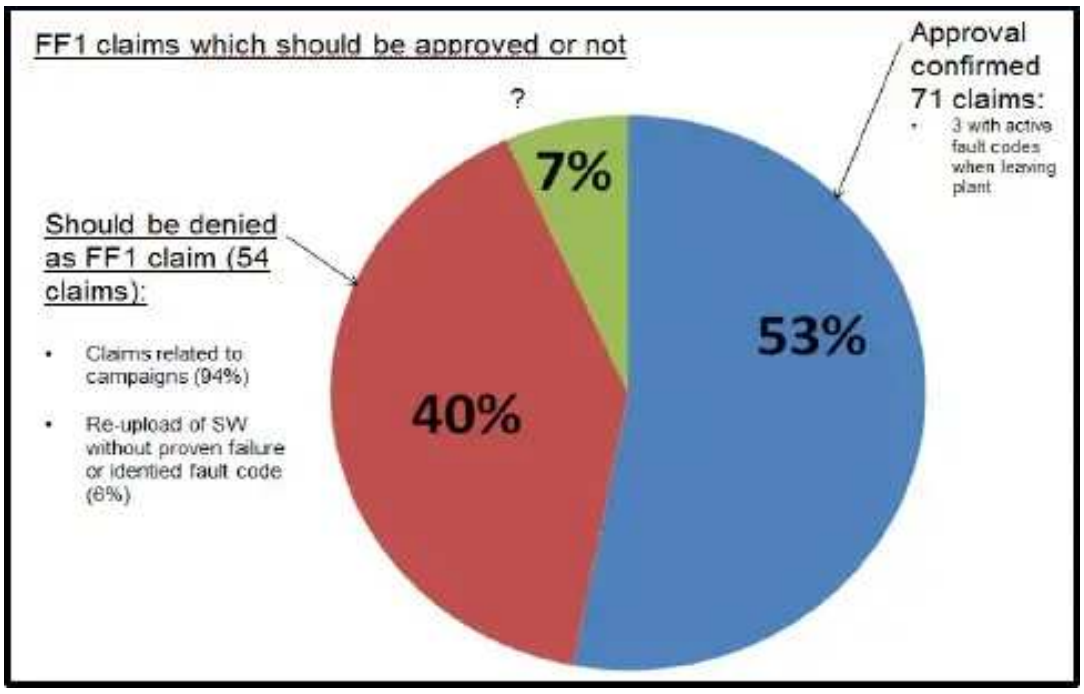


Figure 32: FF1 Claims that could be approved or not

As shown in the **figure 30** and **figure 31,** we were able to elaborate some statistics highlighting the main takeaways from our analysis.

To have a better view we choose to clearly define these points as below:

1) High contribution from External Campaign claims (39%)

a) Wrongly affected debit code (37, 6 %):

External Campaign claims as defined should not be part of the FF1 database as they are whether claims of already solved problems or claims with an existing solution. Having this amount of claims in the FF1 Database in not only causing a miscalculation of our KPI presenting a misleading image of the actual status but also making it harder to the solving team to process all this data(waste of time and fatigue).

b) Difficulties to run requested campaign

2) Unjustified FF1 claims

a) See 1)a

b) Few claims with SW re-upload without any fault code (1b + 2b = 2, 4%)

Some customers and dealers are claiming (and getting also paid) for re-uploading the ECUs Software whether to upgrade it (even though there is no need and even doing that should not be considerate as a claim) or to solve an issue with the truck without properly identifying it.

3) Trucks which have left the plant with still activated fault code (2, 24 %):

PROSIT+ showed us that 3 trucks have left the plant with a status NOK for have been delivered.

4) Trucks which have had fault code at plant (upload replayed &/or fault code deactivated) but fault code popping up again (8 cases from 134)

Using PROSIT+ also we were able to detect that 8 trucks showed a NOK status for some software operations in the plant, operators were able to solve the issues back then to have an OK status but

the same trucks were the cause of 8 claims from the end customer in relationship with the same operations.

5) 53% of claims « justified » but further investigations needed

More than half of the analyzed cases were for us justified and needed further investigation with the help of software specialists.

PS: The same analysis and results were done for North Americas Manufacturing considering the same filters (Plants: NRV&LVO, defect code: Software, Function Groups: 20, 21 & 50).

As a result we got 4 causal parts contributing to more than 80% of NAM SW Claims:

- 2841-Electronic control Unit
- 3982-Wireless Electronic Control
- 3649-Miscellaneous(Electrical Item)
- 3811-Combined Instrument

5. Next step

Using these analysis takeaways, we resumed it in a PowerPoint presentation.

This presentation will be used as a communication material during a first meeting with representatives from the different departments who will be if convinced a part of our "software working groups", helping in having more information about the defined points and setting actions (an action plan) to continuously improve the quality of our products and solve customer issues.

The detailed information about our SW taskforces and the action plan defined action plan will be presented in the 3rd Chapter.

We should also mention that this analysis was done manually using excel sheet but we have started working with Linda ÖWRIN (Process Project Manager SW&EE Technology) to build a QlikView dashboard using FF1 and PROSIT+ databases in order to automatically generate these analysis on a regular basis (monthly) and be able by that to report back easily to design and manufacturing departments the issues and gaps for improvements.(Using the chassis number as a link between the two data bases).

III. Handle Non-Conformity Process & QULIS

1. Definition

The "Handle Non-Conformity" process is a Level 3 process which is part of a bigger process (Level 2) called "Produce Product"

1.1 Produce Product Process

Produce Products add value, step by step, to the physical product until it is in accordance with the product specification and the production order, thereby fulfilling our customers´ and stakeholders´ needs.

The main purpose of this process is to produce a product according to specification, on time and in an efficient way. [6]

The process diagram is presented in the **Figure 31**.

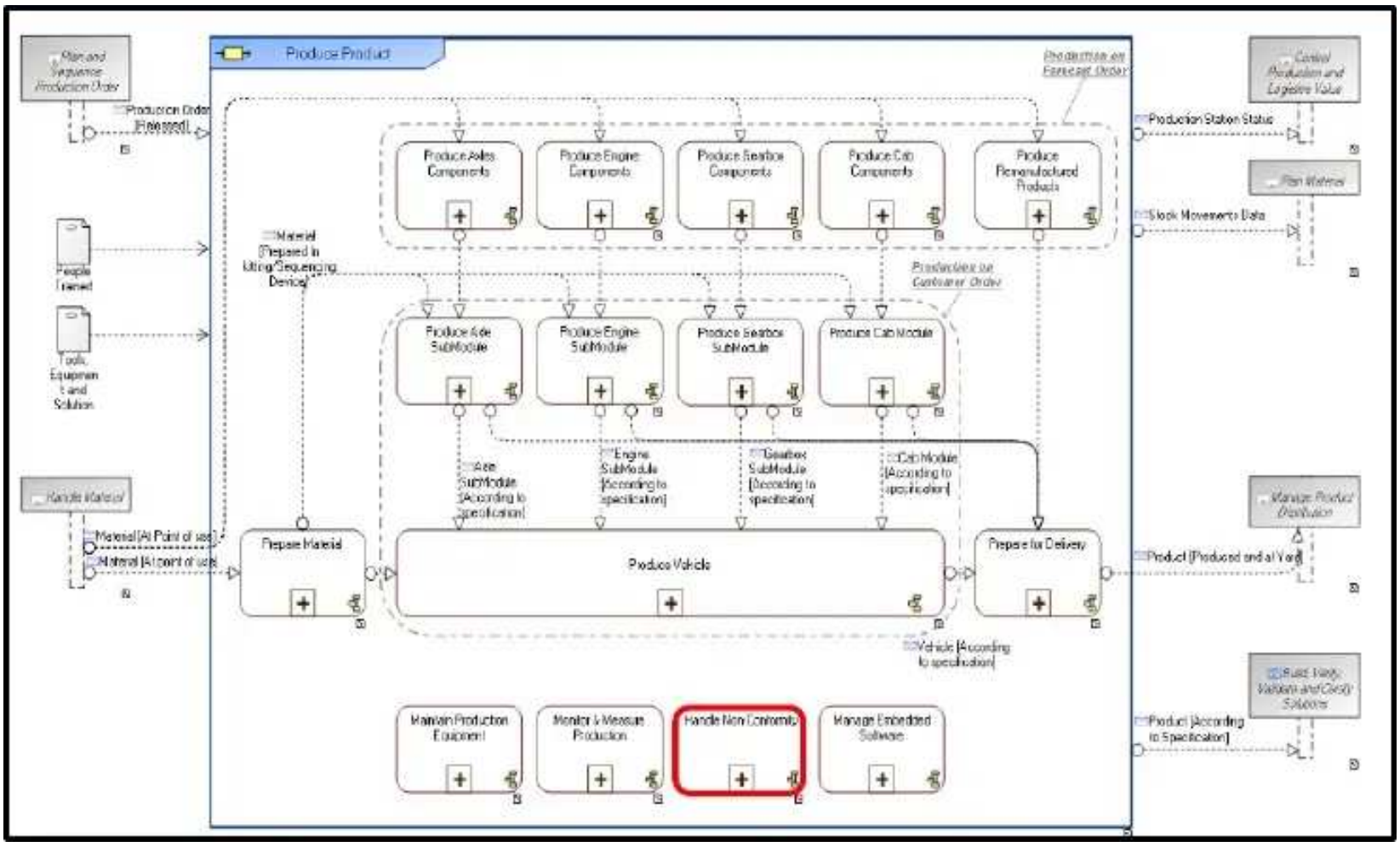


Figure 33: Produce Product Process Diagram [6]

1.2 Handle Non-Conformity Process

Starting from issues on product or on manufacturing process product related, the organization will grasp non-conformities by containing the unexpected effects to protect the customer, implement effective action after proving the root cause and finally expand best practice where possible to improve overall efficiency.

As for that the main purpose of this process is to develop Customer satisfaction by ensuring all non-conformities on the product are managed prior to delivery and to increase manufacturing efficiency by expansion of proven effective corrective actions to avoid recurrence of non-conformities. [6]

The **Figure 32** shows the Handle Non-Conformity Process diagram.

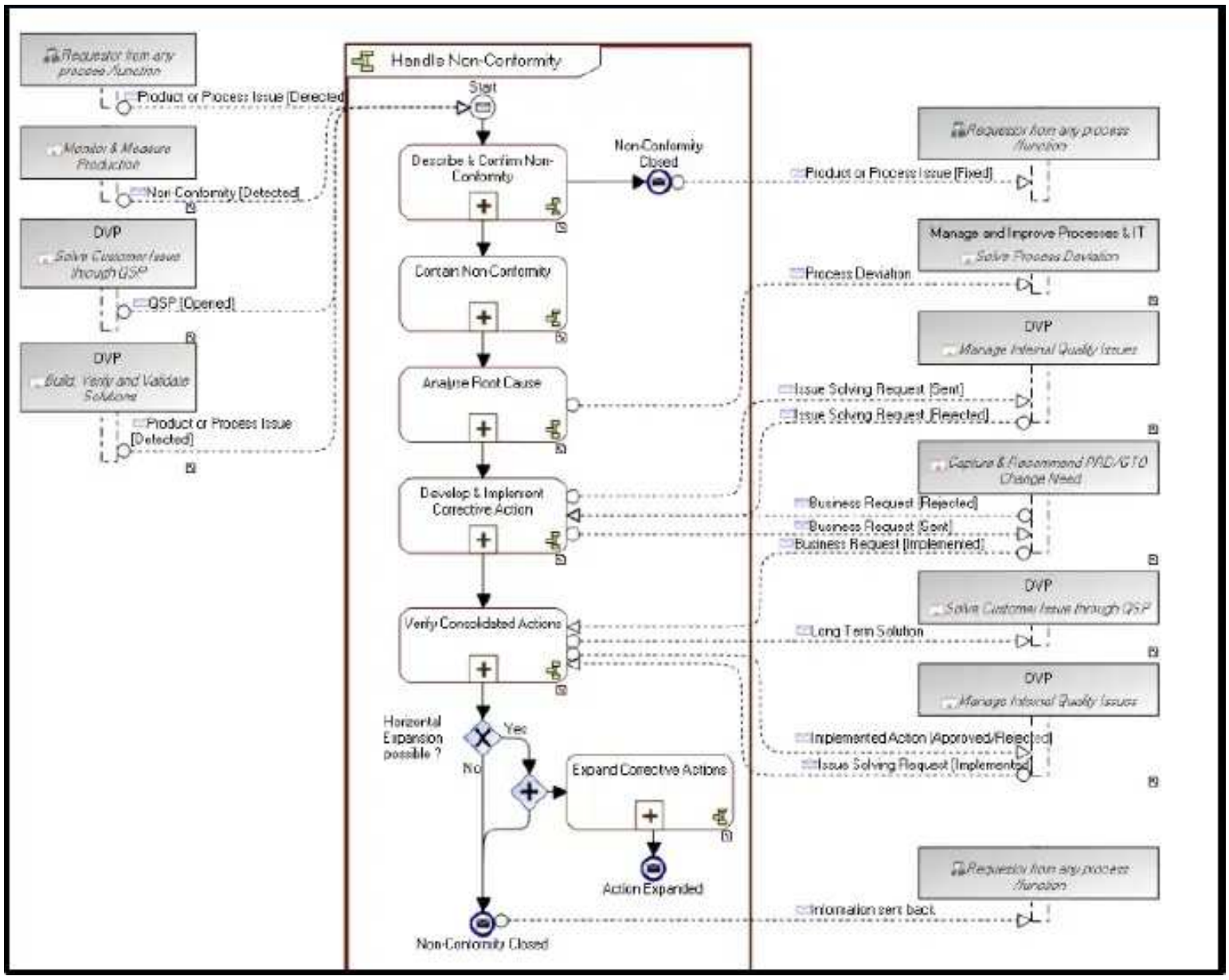


Figure 34: Handle Non-Conformity Process Diagram [6]

The regulation framework for the development of this process is:

- ISO 14001: 2015 :
  - 10.2 Non Conformity and Corrective Action
  - 10.3 Continual improvement
- ISO 9001:2015 (8.7.1 Control of non-conforming outputs)

2. Handle Non Conformity Processes

Handle Non-Conformity is the core process for handling quality issues and depending on the issues (gravity, category etc.) Volvo has developed some specific processes:

- **QJ: Quality Journal**

QJ Process is to be applied to solve Market Quality Issue. It deals with high important items and is applied where there is an urgency/severity when DVP (Develop Product & Aftermarket Product Portfolio) process is not sufficient. It should be used as a response to meet quality targets and when the market is "bleeding".

- High priority resolution to move quickly in order to limit associated costs and disruptions caused by the issue
- Aftermarket publication communication and new spare parts

The QJ Process shall be applied based on facts or prediction when we have:

- Safety related issues
- Non-conformity to legal requirements
- Sudden increase in failure rate, Warranty Costs and Break down
- A need to meet Fault Frequency (FF) and/or Breakdown /Unplanned Stops frequency (BD/UPS) targets (short term targets)
- A need to meet Warranty Cost targets.

A QJ is usually opened to deal with a design issue. [7]

- **QSP: Quick Solving Process**

It delivers a solution to the issue by following the 8D methodology. It is a quick way to solve quality issues when

- Potential manufacturing, logistic or supplier improvement action is foreseen to solve the issue
- No design change is foreseen
- High confidence root cause shall be identified

QSP is divided into 4 specific categories:

QSP PL: quality issues affecting end customer, related to Production Logistics and being solved through QSP process.

QSP MSL: quality issues affecting end customer, related to Market Service Logistics and being solved through QSP process.

QSP Manufacturing: quality issues affecting end customer, related to manufacturing process and being solved through QSP process.

QSP Purchasing: quality issues affecting end customer, related to supplier manufacturing process and being solved through QSP process. [8]

- **Other Handle Non-Conformity Processes**

Internal Provider Cab: used for claiming between GTO plants for issues related to Cab.

Internal Provider Frame: used for claiming between GTO plants for issues related to Frame.

Internal Provider Powertrain: used for claiming to GTO provider plants Powertrain

Internal Provider other: used for claiming between GTO plants for issues related to other areas (excluding Logistics, Powertrain, Cab and Frame).

External supplier Q: material defect from external supplier.

General (Site Internal): fault highlighted and generated by plant manufacturing process (inside own factory).

End Customer claim: issue affecting end customer that is not taken through QSP, for any reason.

Product audit remark: if a fault is found during audit and registered on own factory. [9]

For these processes the solutions are usually simple to implement, nevertheless, all Handle Non-Conformity processes are solved following the 8D methodology.

3. HNC Support IT Solution : QULIS

Being a global company with production sites in 18 countries plus the fact of having multiple brands implies the existence of multiple ways of working and multiple ways for documenting cases and dealing with Non-Conformities.

Having a customer mindset and knowing that reactivity, traceability and cross-site information sharing and actions are crucial to resolve issues responsibly, GTO decided to develop an IT solution to support HNC Process. This IT solution is called QULIS.

In other words, this is the IT solution to support communication and follow up around Quality issues which are related to the manufacturing process.

QULIS stands for Quality Information System. It is a web-based application developed internally using VB.NET framework and an SQL Server database.

Inspired from the 8D methodology with some modifications a QULIS case contains the following steps:

- Add/Update New Case Handling

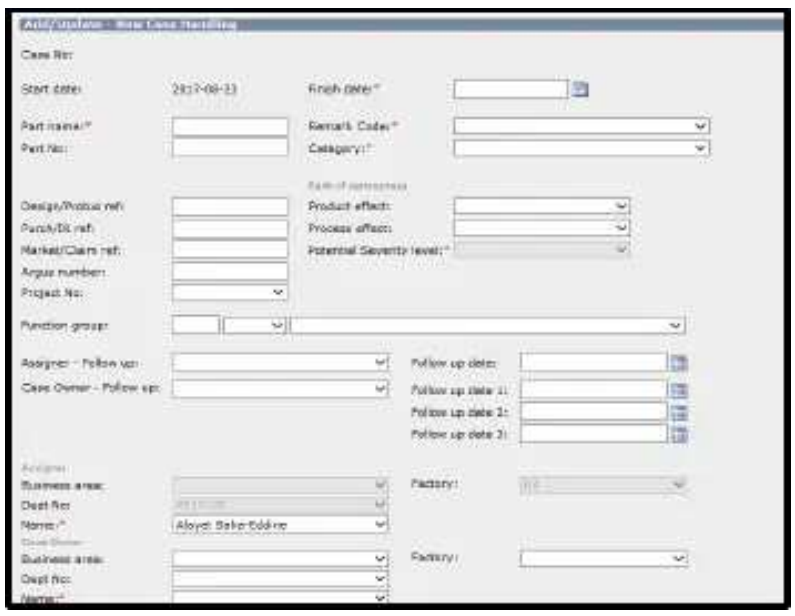


Figure 35: Add/Update-New case handling Step

This step allows us to initiate the case by filling the case information (Part name, category, severity level) and assigning the case to a working group composed of one case owner and at least two assistants.

- Investigation : consists of a first problem description and analysis using the 5W2H methodology

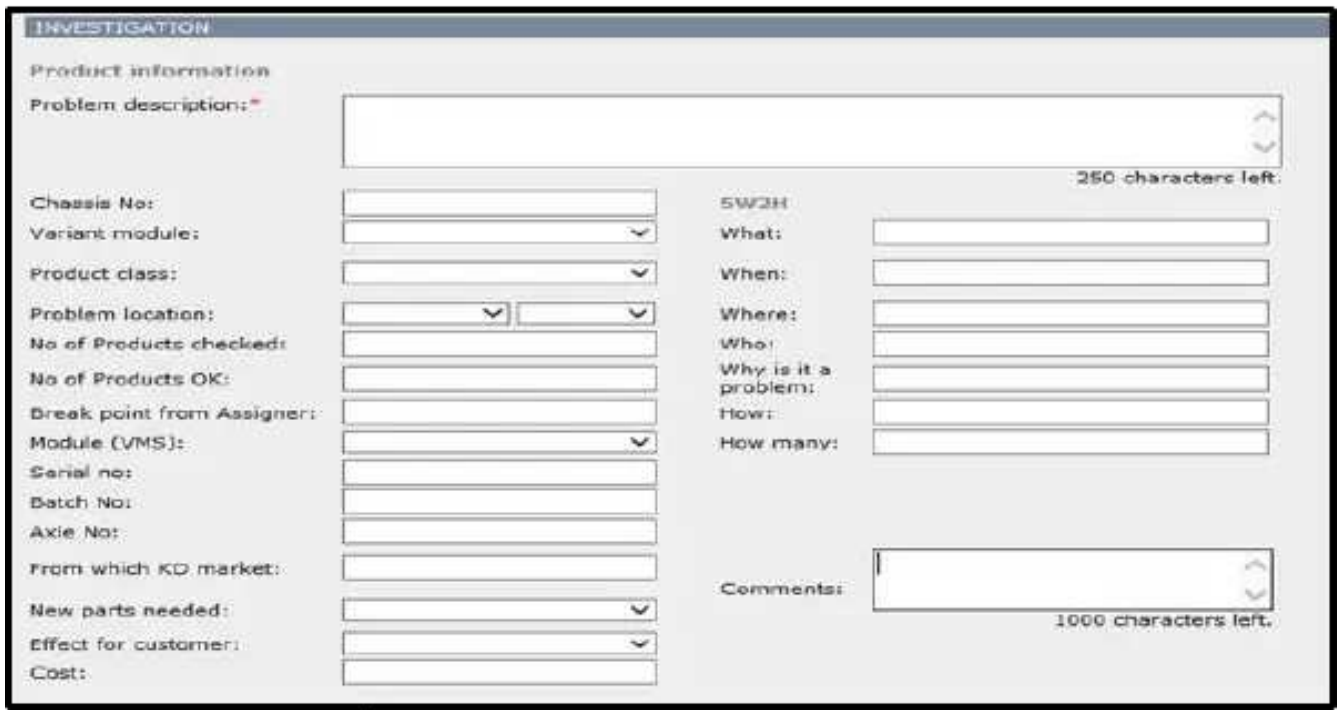


Figure 36: Investigation step

- Containment within 24h:

It is a description of the containment action.

A CA is usually a fast and short term solution to stop the "bleeding" and protect the customer.

At this step, there is no need to clearly define the root cause, the priority is to have a temporary solution till we run a detailed root cause analysis and implement an efficient and long term solution.

The 24h target is mentioned in order to push the staff to implement the solution as soon as possible

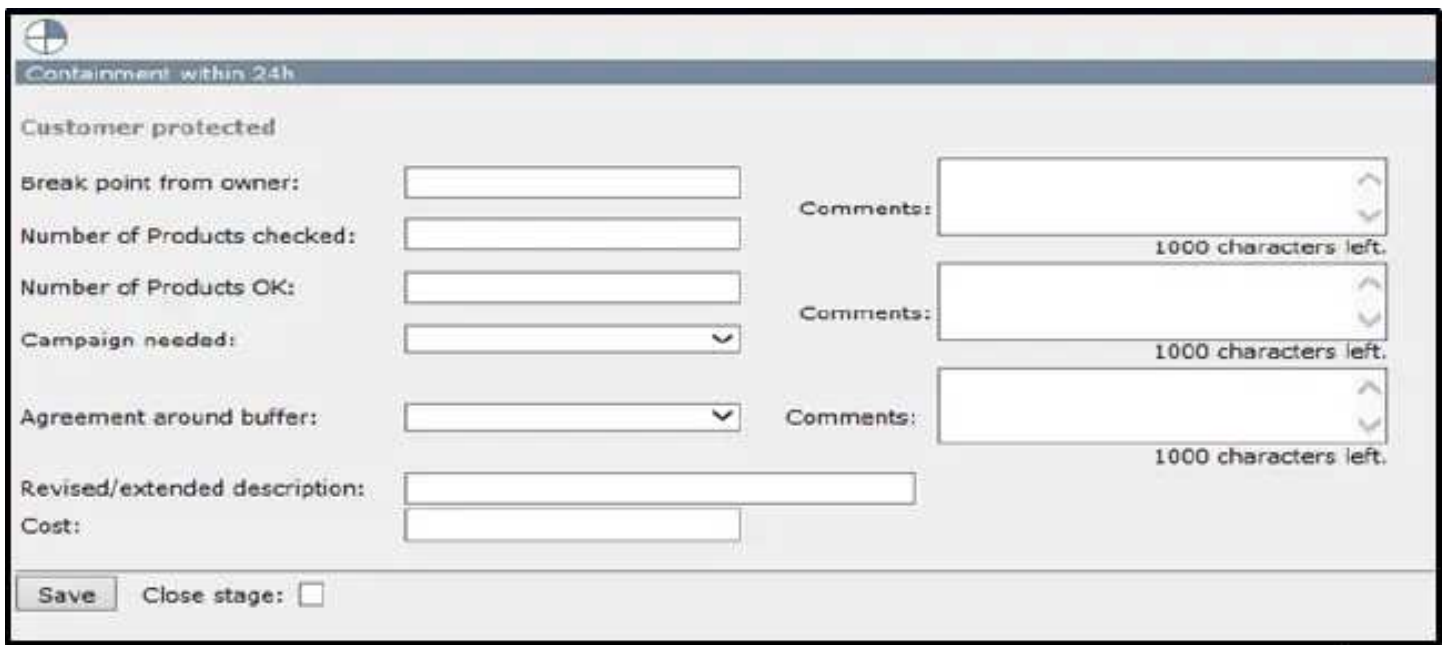


Figure 37: Containment action step

➢ Root Cause Analysis within 7 days:

Identify all applicable causes that could explain why the problem has occurred. Also identify why the problem was not noticed at the time it occurred. All causes shall be verified or proved. We can use Ishikawa diagrams (4M) to map causes against the effect or problem identified.

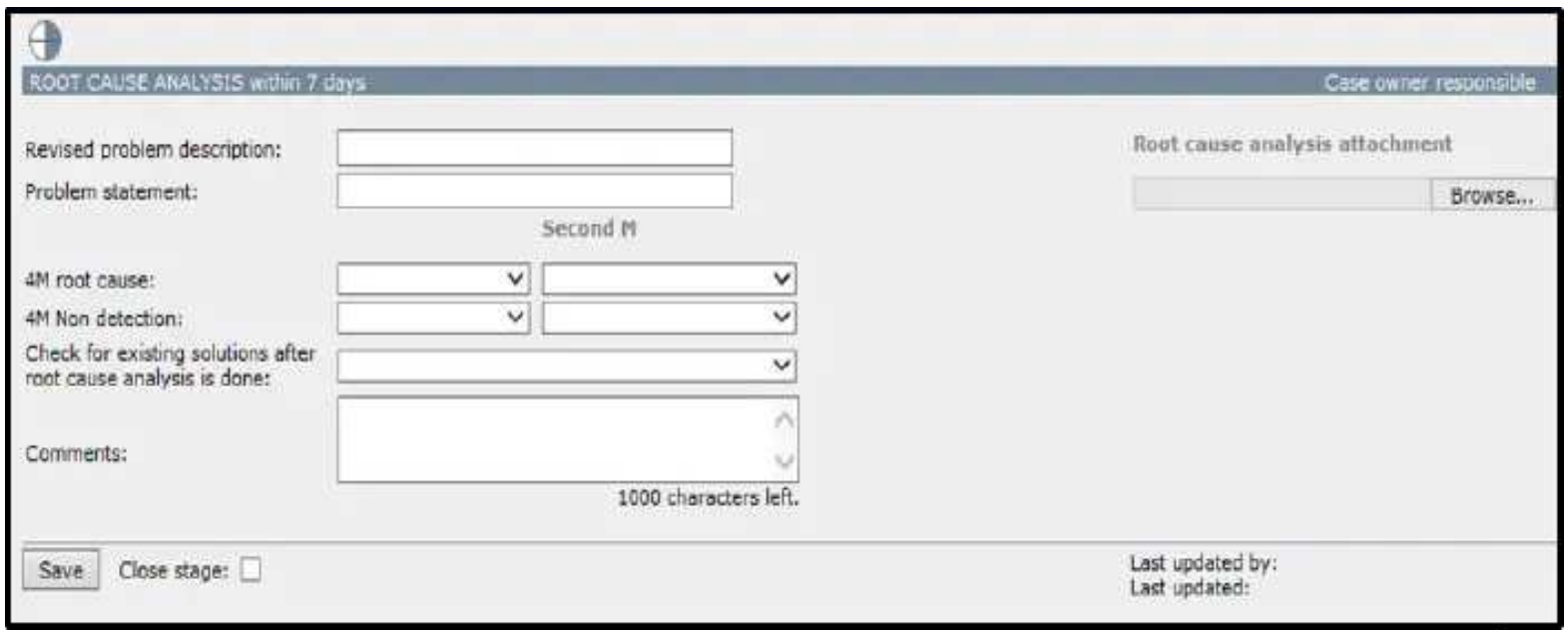


Figure 38: Root Cause Analysis Step

➢ Solution Implementation:

At the stage the case owner should:

- Explain what was done to solve the problem so other colleagues in other factories can get lessons learned.
- Explain how is further detection secured

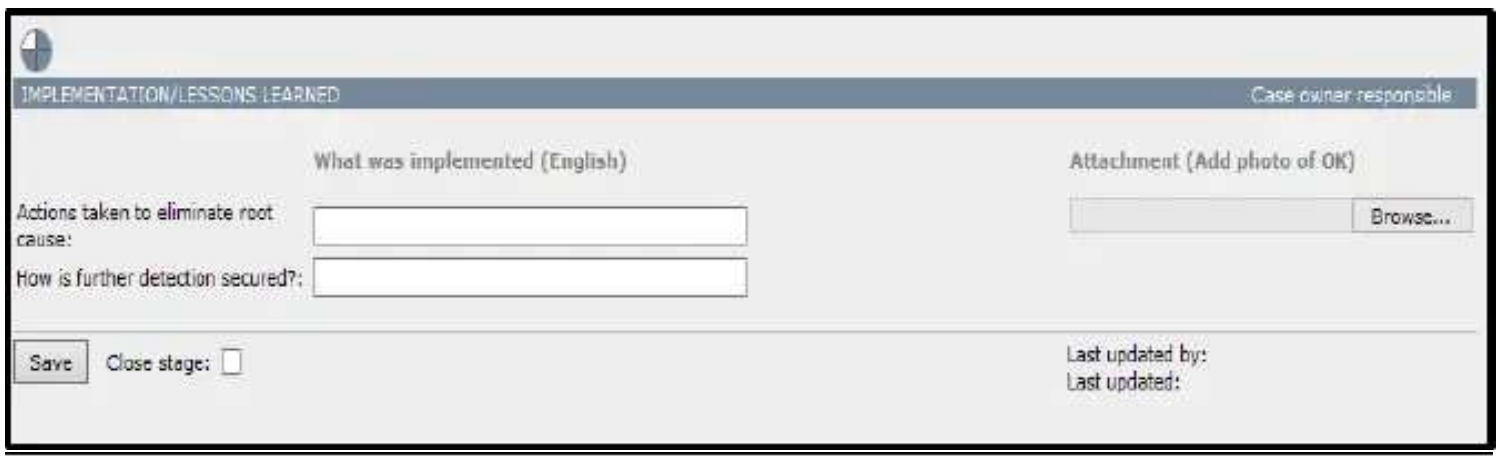


Figure 39: Implementation/Lessons learned step

➢ Verification step:

Case owner should:

- Specify from when have we implemented the final solution
- Reflect if the solution can be used in other areas in the plant
- Attach the result from what you did to verify that the product is OK from now.

- Attach the result from what you did to verify that the process is secured from now

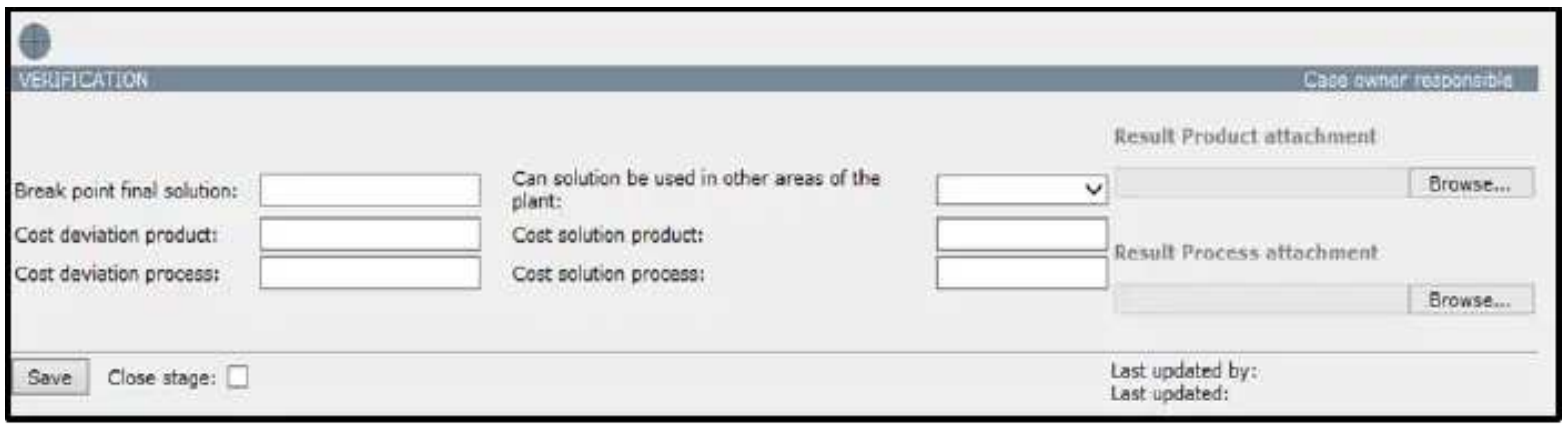


Figure 40: Verification step

➢ Completion within 20 days:

Marking the date of the case finish, case owner should specify:

- the responsible organization for the long term solution
- the responsible support function in the plant to come up with a short term solution and contact organizations outside plants for support

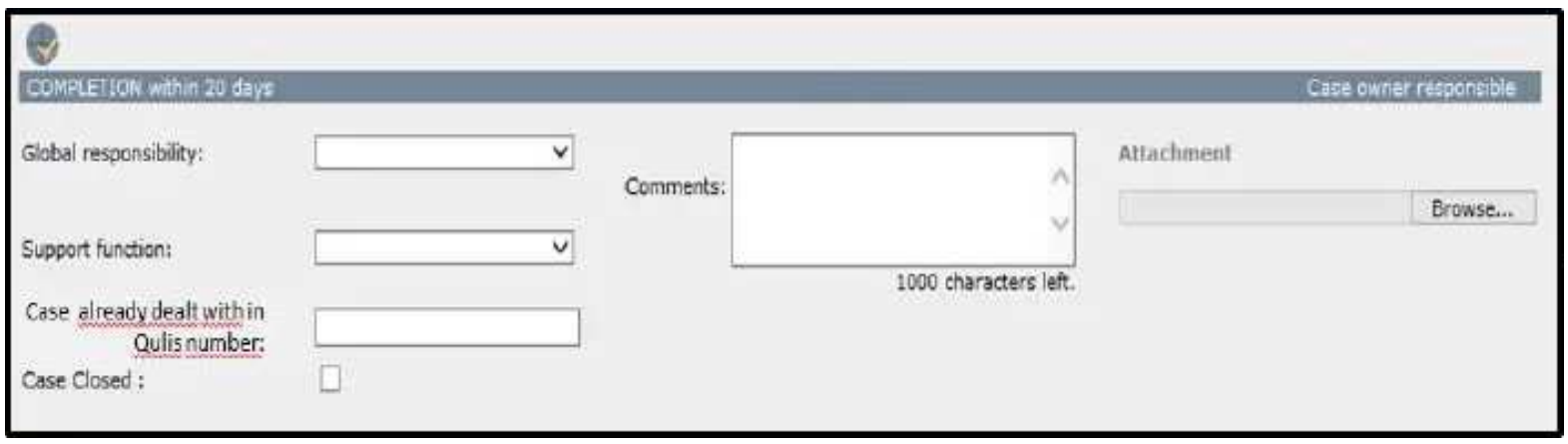


Figure 41: Completion within 20 days step

4. Main takeaways

After analyzing the Handle Non-Conformity Process and its support IT solution we came to two conclusions:

- QULIS is not deployed in all GTO plants and a resistance from GTT organization to use it for QSP handling.
- There is no global monitoring for the handling non-conformity cases 'performance and by that we are not securing the continuous improvement and the evolution of the operational excellence.

For the first conclusion, we used the **5 Whys** technique in order to clearly identify its root cause:

"QULIS is not deployed in all GTO plants & Resistance from GTT"

1- **Why?** : Some plants didn't accept to use it.

2- **Why?** : Change Resistance.
3- **Why?** : QULIS a new tool to them.

4- **Why?** : Some plants have already developed their own IT tools for supporting HNC cases.

5- **Why?** : Each plant has its own way of working for HNC and communicating around it & QULIS is not flexible enough to support all HNC categories.

⇨ A need to make more flexible and user-friendly so that GTO plants accept to deployed.

For the second, we saw an urgent need to set KPI/PIs to monitor the HNC cases around QULIS in order to actively control the cases and have both global and specific view on our situation so that we can react according to those indicators.

5. Proposed Solutions

- Form a "QULIS Working Group" whose objective is to continuously improve this IT solution by:
  - Assigning pilots for the deployment of QULIS in each plant.
  - Weekly follow-up meetings to get plants feedback and change requests.
  - Monthly Working Group Meetings to assess the current situation and prioritize changes on the system according to the urgent needs and the allocated budget.
- Defining a new HNC KPI/PIs and developing a global dashboard to be used by people ranging from global organizations to quality engineers.

  To do that, we should:
  - Set a list of all the data that we can extract from QULIS Database.
  - Build our performance indicator.
  - Write the specification of the connection between QULIS and the chosen software for the dashboard.
  - Develop the dashboard & test it.

IV. Conclusion

In this chapter we have presented a first diagnosis of the current situation in relation with software claims and HNC process and its support IT solution.

For the software issues we were able to assess to situation by running analysis using feedback for end-customers and dealers (Claims) in order to start our working groups to reduce these claims by setting and following a clear action plan.

As for Handle Non-Conformity process and QULIS we did identify the few difficulties it is facing and set down a strategy to overcome it. The more important point is the decision to set new HNC performance indictor and a dashboard to display it and actively monitor QULIS cases.

This way we can ensure a more reactive organization but more than that a continuous evolution of the operational excellence dealing with non-conformities.

During the next chapter, we will present the development of the proposed solutions.

Chapter III

Solutions & Implementation

I. Introduction

After the diagnosis that we have presented in the previous chapter, the actual chapter details the implemented steps, actions and development we did to achieve the target situation as defined the our project specifications, taking into account the both subjects : Software product quality issues and handle non-conformity process quality improvement.

II. Software Working Groups

Considering the first analysis we ran, showing the big contribution of software quality issues to customer claims. It is time to form the appropriate taskforce to handle these issues.
We decided according to that, to gather two working groups; one for E&BM and another one for NAM.
PS: We will only present the work for E&BM with a view on NAM.

1. Working Group Team Members

To have an efficient results, we decided to form a cross functional group in order to ensure the share of knowledge but more than that to secure the communication between all the organizations and be by that a facilitator to the success of this mission.

- Denis MARTIN : Head of Product Quality
- Baha-Eddine ALAYET: Product Quality Engineer Trainee
- Yamina El HAJEM (Director SW&EL France)
- Capucine BOUSQUET (GTO E&BM SW Process Manager**)**
- Jennie LARSSON(GTO E&BM Product Quality Manager)
- Loan BESSE(Product Quality Engineer)
- Anna-Karin FLOOD(Director Warranty Management)
- Jenny GUSTAFSSON(Director Customer Quality & Campaign Management)
- David HAUTEVILLE(ESW Product Specialist)
- Hans LEMNER(Lead Engineer Electronics & Software)
- Caroline NASLUND(Software Product Quality Leader)

2. Way of Working

In order to get the working group members full engagement into this issue we had first of all to present the first analysis we have done (2nd Chapter) in order to convince them of the importance of software into reaching customer success and by the Volvo Group target and to clearly define the purpose of the working group.

After that, we started discussing each point defined as outputs from the 2nd chapter in order to set clear an effective action resumed in a global action plan.

We organized the work by setting global follow-up meetings (once in two weeks) and specific follow-up meetings for each action (depending on the action itself)

3. Action Plan

Following the outputs of our analysis we were able after some decisions and more discussion to set the following action plan for each point by defining the main activities to do, expected achievements and we have appointed a team leader and people to involve.

3.1 High contribution from External Campaign claims (39%)

- Wrongly affected debit code (37,6 %): DC10 (factory warranty)

Team Leader: Anna-Karin FLOOD

People to involve: Loan BESSE, Jenny GUSTAFSSON, Sylvie DENIS (Quality & Warranty Director for Renault Trucks), David HAUTEVILLE.

Activities: - Carry investigations to know whether related vehicles were inside the P-Code period or not and act consequently with dealers. Only claims related to trucks with this P-code are allowed to be part of claim.

- How to handle with Op Unit Quality network
  - Expected achievement: Better Debit Code Management.
    Follow up (correct DC used by dealers)
  - Decided actions: Implementation of a monthly SW meeting for E&BM to be led by Loan to monitor the dealers' behavior and act according to the situation.
  - Status: Done (Monthly Meetings Implemented but needs a continuous follow up).
  - Expected completion date: week 34 (2017)

- Difficulties to run requested campaign

Team Leader: Denis MARTIN

People to involve: David HAUTEVILLE.

Activities: - Question AM (After Market): what's the major difficulties when starting a SW campaign?

- Expected achievement: support the dealer in SW campaign execution with better definition of what is to be done.

- Decided actions: Ensure that all requirements are available and well defined when running a SW campaign.
- Status: Conscious decision. Single case. Decision to park it but actively follow up.
- Comments: Problem with specific campaign (timed out at 99%).
- Expected completion date: Week 20 (2017).

3.2 Unjustified FF1 claims

➢ Few claims with SW re-upload without any fault code(2,4%) : we should mention that an investigation we did with the help of Jennie LARSSON and Loan BESSE showed that the contribution of this kind of claims for trucks built in Tuve(Sweden) and Blainville(France) and delivered to the Scandinavian market are way higher (up to 15%)

Team Leader: Yamina El HAJEM

People to involve: Jean Yves DROUHUIN

Activities: - Facilitate an efficient Root Cause Analysis

- ~~Expected achievement~~: Avoid SW re-loading without any identified cause.
- Decided actions: analysis chassis by chassis to know if something specific was decided.
- Status: Ongoing (found the people contact but workshops didn't start yet).
- Expected completion date: Week 50 (2017)

3.3 Justified FF1 claims

➢ Trucks which have left the plant with still activated fault code (2, 24 %)

Team Leader: Capucine BOUSQUET

People to involve: GTT organization and Daniel LENHAGEN (Project Quality Manager Operations.

Activities: - 1st step: Investigations on green OK process (why trucks have left the plant with active fault code??)

- Expected achievement: Avoid delivery of trucks with fault codes.
- Comments: a deeper investigation showed that these kinds of fault codes are allowed since it should be turned off after that the bodybuilders finish their activities on the truck. However, there is no check point when the truck leaves the bodybuilder yard.
- ~~Proposed actions~~: Establish a Volvo check point at bodybuilders or a system to share the validation process for traceability reasons.

- Status: Ongoing (found the people contact but workshops didn't start yet).
- Expected completion date: Week 50 (2017).

- 2nd step: Green OK operation sequence optimization
  - Expected achievement: Align Green OK checks on best practice.
    Investigation on Green OK with GTT (Quality + Engineering)
    Working plan to be established
  - Comments: a deeper investigation we have carried out revealed multiple gaps in the Green OK process between GTO Plants.
  - Decided actions: Meetings with GTT SW on what should be checked to get an efficient Green OK.
  - Containment actions: Assess which Green OK process is the most efficient between the plants and standardized for the rest of the plants.
  - Status: Ongoing
  - Expected completion date: end of 2018.

➢ Justified claims but further investigation is needed(53%)71 claims

Team Leader: Baha-Eddine ALAYET

People to involve: Hans LEMNER, Caroline NASLUND, Jairid MOWEN (Director Product Quality Management Mack Trucks USA).

Activities: - Carry out detailed analysis on these claims in order the find out the root causes.
Follow up QJ progress with PQLs and through ARGUS.

- Expected achievements: Better product audit to prevent issues in AM.
  Push GTT to open more QJs in order to quickly solve issues.
- Comments: C value formula may sometimes neglect some issues and usually GTT people use FF12 indicator to decide about quality issues on the other hand we are working on fresher claims.

⇨ Slow reaction to quality issues.

- Activity outputs: The working sessions outputs maybe resumed as below :
  - ❖ 20%: 14/71: Campaign S3666 related.
  - ❖ 15%: 11/71: EATS: After Treatment System issues.
  - ❖ 14% : 10/71 :Hardware issue : related to wiring or BBM(Bodybuilders)
  - ❖ 10% : 07/71 :Calibration issue(Sensor, remote control)
  - ❖ 4% : 03/71 :ESCL issue (Electronic Suspension Controller)
  - ❖ 4% : 03/71 :Issue when performing TSB/Campaign
  - ❖ 3% : 02/71 :Bluetooth (This functionality is still not robust enough)
  - ❖ 1,5%: 01/71 :Handled in **QJ 1-5664571611**
  - ❖ 1,5%: 01/71 :Solved through a **QSP**.

- ❖ 1,5%: 01/71 :Will be handled in **QJ 1-7455049241**
- ❖ 1,5%: 01/71 : No clear link between reprogramming and explained failure

More discussions with Jairid MOWEN revealed that software quality specialist are facing major issues analyzing software claims and finding root causes since that based on the data they have , customers and dealers didn't write the fault codes the tool showed in their claims for **74%** of the cases for the past ~~**12 Months**~~.

- Containment actions: To have a temporary solution for the missing fault code issues we were able with the contribution of Vince LINDELY (Director Quality, Reliability & Warranty USA) to implement an auto exception rule in UCHP. This rule will ensure that software claims won't be treated unless the claimer will enter the raised fault codes and will have by that two main effects:
  - Decrease software claims and software re-upload without proven failure.
  - Increase the percentage of software claims with fault codes.

The **Figure 41** shows the description of the implemented auto exception rule in UCHP for software claims warranty handling.

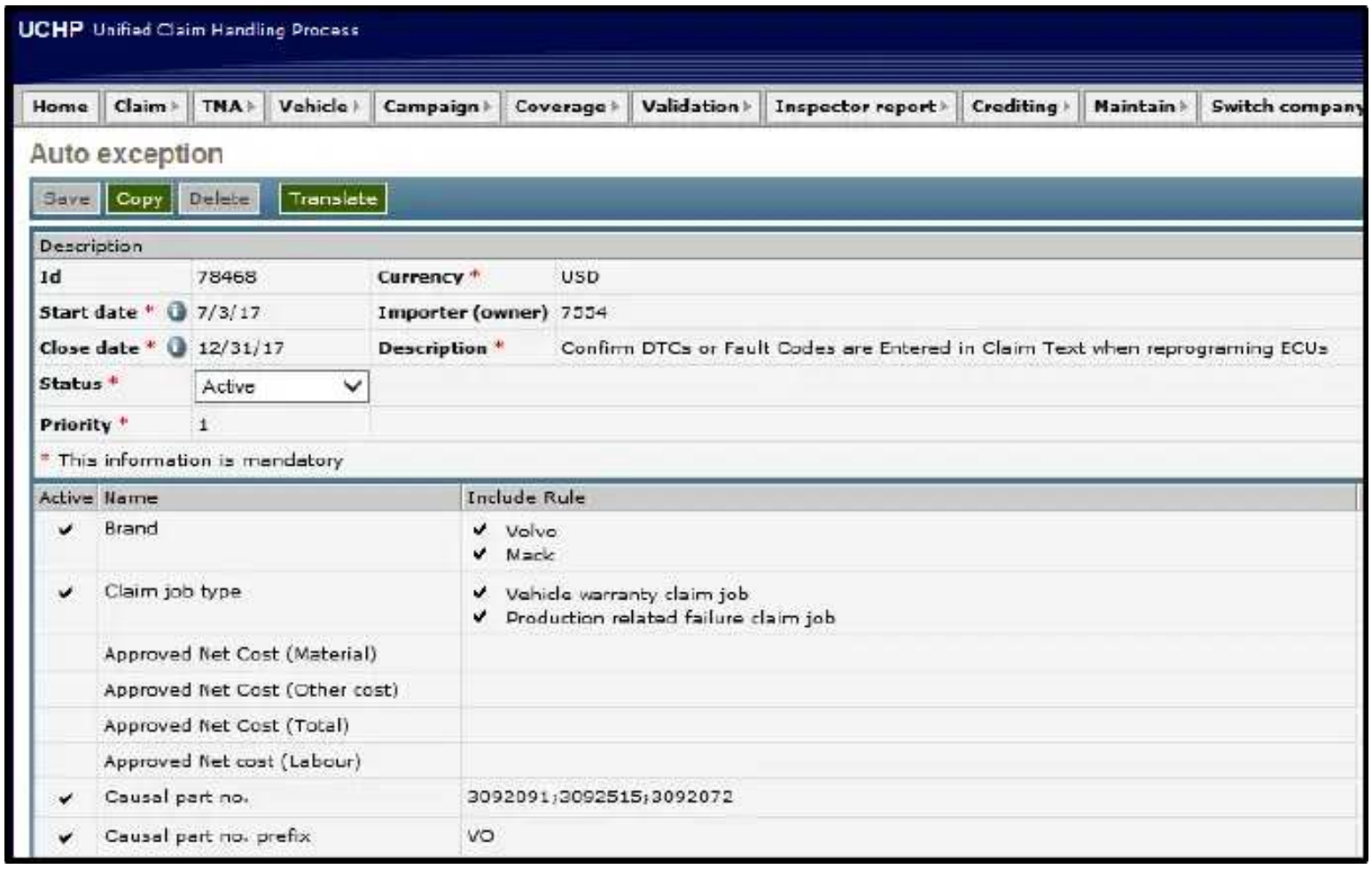


Figure 42: Software UCHP new auto-exception rule

PS: The rule was only implemented for the top 3 dealers causing 70% of the claims with missing fault codes and for the top 3 causal parts contributing in 70% of software claims both in NAM and E&BM.

- Long Term (Sustainable) Solution: After starting this working group and in order to avoid missing fault codes in reported claims and to ensure a continuous improvement of our reactivity toward costumer problem and even to anticipate the occurrence of an issue, GTT organization started the development of a revolutionary solution that will be tested by the end of September 2017.
- Other decided actions: Carry on follow up meetings with GTT to assess the situation and actively push them to considerate and prioritize new issues. And to do so we are extracting weekly fresh data to analyze during our working sessions.
- Status: Ongoing.

The action plan is summarized in a shared excel file to have a follow-up on each action and to communicate with other parties.

4. Working Groups & Action Plan Results

This paragraph will present a summary of the decided actions, the ones who already are in place and their effects/results:

- **Action 1:** Implementation of a monthly SW meeting for E&BM led by Loan to monitor the dealers' behavior and act according to the situation.

  Effects: - Better Debit Code Management

  - Reduction of the existence of claims related to campaigns in our data **(From 39% to 0%).**

  ➔A filtered key performance indictor with no misleading data.

  ➔FF1 is reflecting the real situation, displaying real quality issues detected during the first vehicle's month in service.

  ➔Less overwhelming data to process.

- **Action 2**: Audit Software campaign execution.

Effects: - Ensure that all requirements are available and well defined when running a SW campaign

- **Action 3:** Continuously audit customer claims handling process and its warranty system.

  (According to the paragraph **8.5 Auditing of the complaints-handling process of the ISO 10002:2004: Quality management-Customer satisfaction-Guidelines for complaints handling in organizations.)**

  Effects: - Improvement of UCHP rules by setting the new auto exception one, an analysis we did showed a quick increase of the percentage of software claims with assigned fault codes from **(26% to 79%): reported on august 31st .**

  ➔ Provide PQLs and Software specialists with the needed information to clearly understand to claimed quality issues and to act accurately according to that.

  ➔ Avoid SW re-loading without any identified cause.

  ➔ Continuously improve the claim handling process (ISO 10002:2004).

  ➔ Ensure communication and cross functional work between warranty and design organization (GTT), helping to address issues to the right people.

  PS: GTT organization was aware of the missing fault codes, but never found the way the properly address this issue to someone who can resolve it.

- **Action 4:** Analyze and do a benchmark about the differences of Green OK processes from a plant to another.

  Effects: - Identify gaps at checks done at end of line.

  - Select the most efficient and optimized Green OK Process to run it as a pilot for all plants till GTT establishes a clear definition of a global Green OK Process.(Align Green OK Process Checks).

  Results presented in the **Annex B**.

- **Action 5:** Carry out working sessions with PQLs and Software Specialists for detailed analysis on SW claims.

  Effects: - Clearly identify the root causes of SW quality issues.

  - Monitor the claim handling process from the PQLs side.

  - Engage GTT to open more GJs to solve the identified issues.

- **Action 6:** Build a Qlikview dashboard to connect FF1 and PROSIT+ databases, using taking advantages of manual work we have performed.

  Effects: -Automatically generate on a monthly basis similar analysis to the ones we have performed.

  -Report and track down to design and manufacturing gaps and areas of possible improvement.

5. Software claims status

Extracting the FF1 warranty claims by the beginning of September we were able to draw this chart showing to evolution of the number of software claims since January 2016 till our last mature month (May).

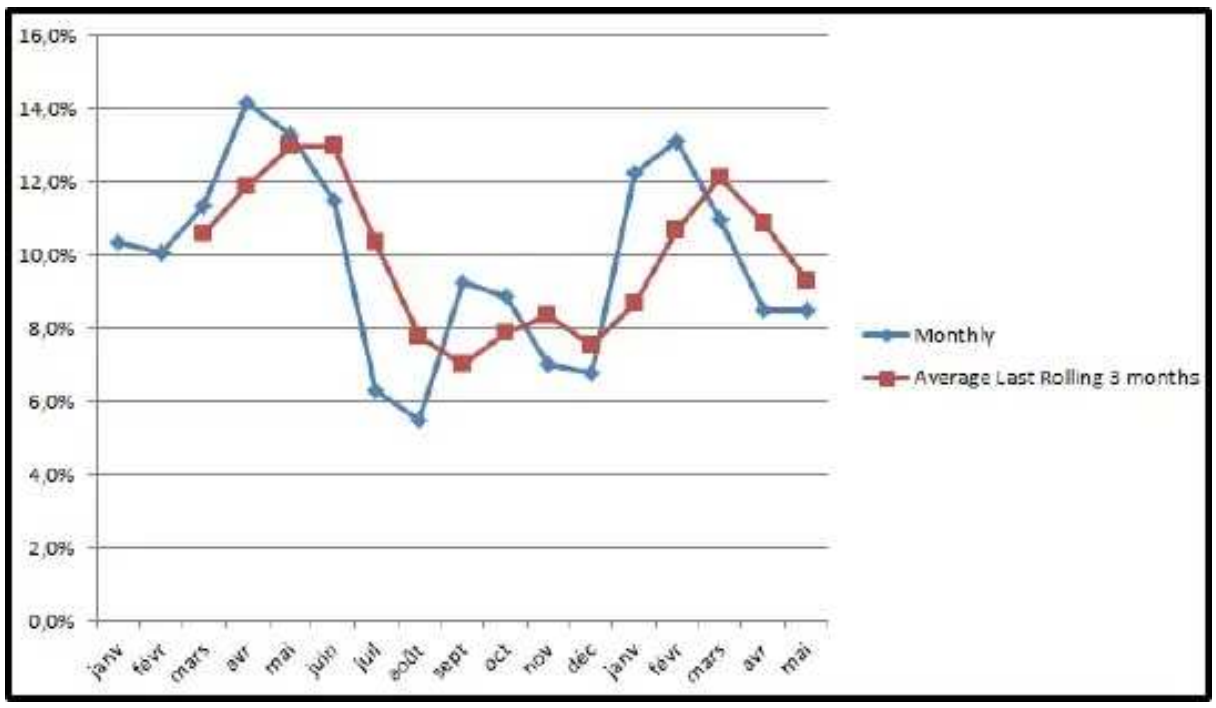


Figure 43: SW claims trend after starting the working group

We can easily see a noticeable decrease in SW claims contribution after March's 12%.

Focusing on the last two mature months (April and May), we can clearly identify an improvement of 40% compared to the same months of the previous year.

The **Figure 43** display the evolution of the contribution of SW claims to total claim payments since 2016 and highlighting some improvements, reaching 5,4% by May (February 6,8%) as a decrease of 1,4% which is almost equivalent to 150K SEK (15K euro).

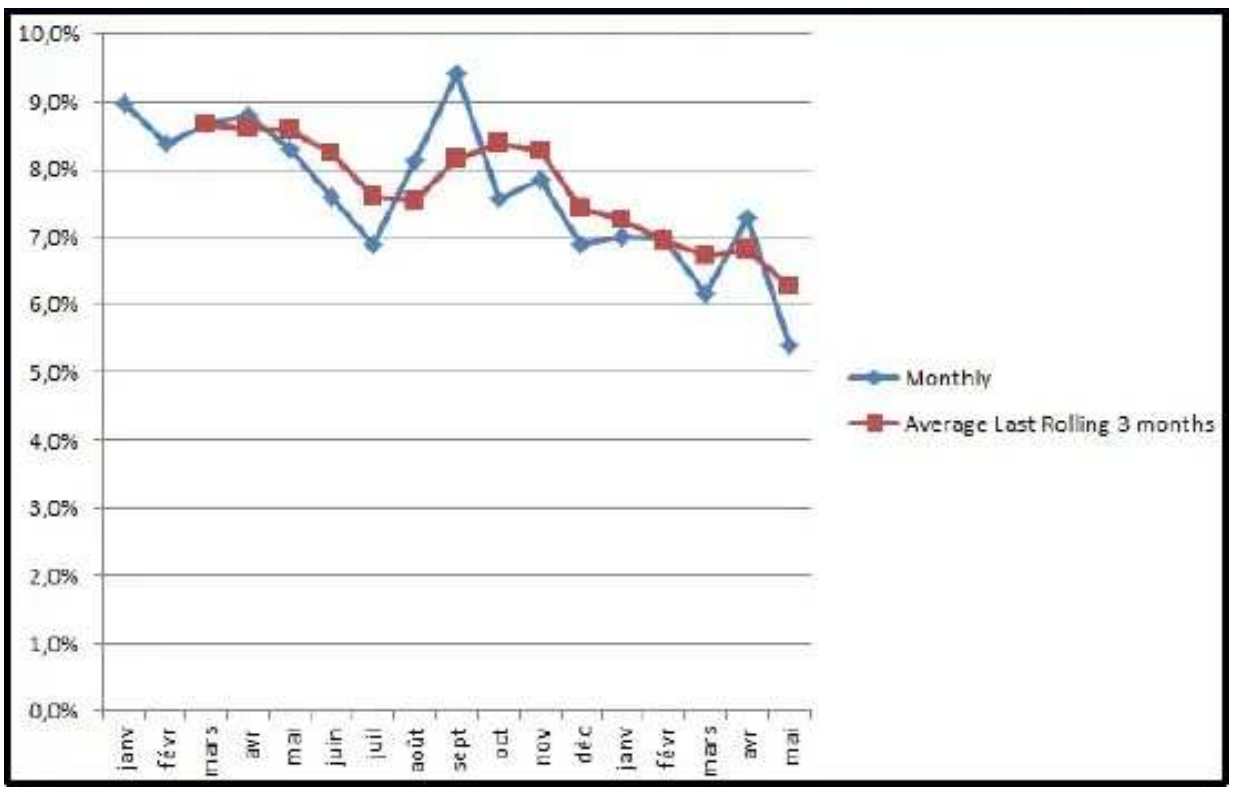


Figure 44: SW claims payment trend after starting the working group

That to be said, the software issues are still unstable and we are certain that the implementation and follow-up of the actions we have set will contribute enormously in its reduction, ensure our trucks reliability and avoid customer claims.

III. Handle Non-Conformity Processes Improvement

This part will be dedicated to present the work we have carried out in order to establish a continuous improvement system for the handle non-conformity cases by ensuring a sustainable evolution of its operational excellence.

But before detailing the chosen PI/KPIs and the developed dashboard to monitor quality issues handling processes and help in the decision making, we should, first of all, show the strategy we have implemented so that QULIS would be deployed by all GTO plants and used by GTT Quick Solving Leader for QSP Processes.

1. QULIS Deployment Strategy

After assessing QULIS situation and the resistance it is facing from some GTO plants and especially GTT Quick Solving Leaders we have decided to act according to the following points:

- Run pilots in GTO plants and get their feedback/Change Requests: Assigned PSKUs (QULIS Key Users) will ensure the usage of the IT support, collect involved parties feed-back and proposed enhancements to easily work with QULIS.
- Run QSP Pilots with weekly feedback.
- Working sessions with GTT Quick Solving Leaders (Led by Didier VIVIAN: Process Manager & Business Process Developer Renault Trucks SAS) to assess the situation from GTT side.
- Gather all feedback and change requests in one common team place.
- Prioritize Changes according to :
  - Whether it is a stopper or not for QULIS deployment.
  - IT budget.
- Coordinate the work with the IT developers (Located in India) to clearly define to wanted solution.
- Run changes 'tests in the Quality Assurance application to give the OK to promote it in the Production version.

Following the above steps, we were able to transform QULIS to a more flexible and user friendly IT solution and ensure a continuous improvement process for it.

For the period of the internship, we able to promote into production mode about 15 changes coming mainly from Tuve plant (Sweden) which was the most resistant plant for QULIS deployment and by that we are getting progressively more engaged plants in the usage of the HNC IT solution.

By beginning of September 2017, we had a confirmation from both GTO plants and GTT Quick Solving Leader that they are going to use QULIS for all handle non-conformity cases.

The **Figure 44** is the result of two separate working sessions with Bernard PESEZ (Purchasing SQA Customer Satisfaction Vehicle Manager –Renault Trucks SAS) in order to understand purchasing needed specifications in QULIS.

Date	Issued by	Problem	Motivation for change	Mandatory for QSP handling
02/05/2017	Bernard Pesez	QSP Purchasing : need supplier & parma code in the initiation phase (close to part number)	Prepare connection to Purchasing System (EPS) ; Possibility to sort QSP per supplier	important but not mandatory. Expected 2018
02/05/2017	Bernard Pesez	When receiving an email announcing that a case was opened , it would be great having a link leading directly to the case in the system	Time and effort saving. Already considered for 2017 Q2 release (tbc by Magnus). It excists in test version	Done Q2 2017 release
02/05/2017	Bernard Pesez	email sent to inform of any updates of the Qulis Case to Assigner/Owner/Assistant	important in order to have a good solving pace and avoid emails.	important but not mandatory. Expected 2018
07/07/2017	Team	Add acceptance date	For better leadtime calculation	important but not mandatory. Expected 2018
16/06/2017	Bernard Pesez	In the acceptance box, there is a need to be able to write a few word explaining why the case is not accepted. => "Why not accepted: ....."	For purchasing QSP, it happens quite often that supplier needs defective part to understand the cause of the problem.	important but not mandatory. Expected 2018
07/07/2017	Bernard Pesez	For Pur QSP, In accepatance are should be written " For Purchasing QSP, acceptance date = faulty part at supplier = start of investigation"	In case supplier first analysis shows that QSP criteria are not fulfiled, then QSP will be rejected. Aim is to monitore part expertize time.	important but not mandatory. Expected 2018
07/07/2017	Bernard Pesez	Distinguish: . MR date at supplier . MR date Volvo plant	there can be significant time between MR at supplier and MR at Volvo plant.	important but not mandatory. Expected 2018

Figure 45: QSP Purchasing Changes Requests

In the **Annex C** we will be presenting the QULIS deployment status (Beginning of September 2017).

2. Handle Non-Conformity Performance Indicators

After analyzing the HNC Processes' specifications and their IT solution QULIS supporting the communication and follow up around quality issues which are related to the manufacturing process, we have decided to define and build performance indicators around QULIS.

The reason behind that is not only to monitor the solving process but more than that to actively challenge ourselves into reaching customer success and by that achieve the group's target image.

And by challenging ourselves, we will be by that ensuring the evolution of the operational excellence in solving quality issues.

As a first step, we have extracted all the data in the QULIS database in order to get a view on the fields and information that would be useful to use in order to build our PIs.

After one working session with our direct manager, we were able to define the following KPIs presented in the **Table 3**:

Table 3: QULIS HNC Key Performance Indictors

Area	Name	Purpose	Definition	Calculation details	Unit of measurement
Lead Time	Time to Protection	Measure reactivity of the organization to protect Customer with implementation of effective containment actions	Time from issue detection up to start of execution of approved containment actions	Containment action finish date – Start date	Hours
Lead Time	Time to Root Cause	Measure reactivity of the organization to find/understand/approve the root cause of the Non-Conformity in order to enable development of effective corrective actions	Time from issue detection up to approval of identified root cause	Root cause analysis finish date – Start date	Days
Lead Time	Time To HNC	Measure global time to solve quality issues	Time from issue detection up to Non-Conformity closed	Completion finish date – Start date	Days
Lead Time	QRQC Lead Time	Measure time needed for QRQC problem solving process	Time from QRQC opened to verification of effective actions	QRQC Finish date – QRQC start date	Days
4M	4M analysis	Assess the contribution of each M to quality issue detection and no detection		Number of occurrences(per M)/Total cases	

3. Handle Non-Conformity Dashboard

In order to build not only a conventional dashboard but more than that a powerful tool for monitoring HNC cases, decision support and reactivity improvement on a global scale, we had first of all to present a Proof of Concept (Built using Excel) of the future tool in order to get approval from all the operational units and get also their suggestions.

After some discussions, we got the OK to start the development.

We will be now presenting the developed dashboard and the used environment.

3.1 QlikView

Qlik is a software company based in Radnor, Pennsylvania, United States. Qlik is the provider of QlikView and Qlik Sense, business intelligence & visualization software.

Qlik (previously known as QlikTech) was founded in Lund, Sweden in 1993 as a software company in business intelligence (BI). Its PC-based desktop tool was called QuikView. "Quik" stood for "Quality, Understanding, Interaction, and Knowledge." Initially the software was sold only in Sweden.



Figure 46: Qlik Official Logo

According to the Visual Intelligence Website:

"QlikView is the most flexible Business Intelligence platform for turning data into knowledge. More than 24,000 organisations worldwide have enabled their users to easily consolidate, search, and visually analyse all their data for unprecedented business insight using QlikView's simplicity. Effective decision-making is based on having the right information available and easy accessible. Taking just minutes to learn, the automatic associations of QlikView create endless possibilities for making ad hoc queries without requiring tediously defined structures and hierarchies, as is typical in other data analysis tools. QlikView promotes unrestricted analysis of application data, helping users make time-saving and accurate decisions. It brings a whole new level of analysis, insight, and value to existing data stores with user interfaces that are clean, simple, and straightforward."

The **Figure 46** presents the magic quadrant for business intelligence and analytics platforms proving that Qlik is a leader in the data visualization sector. [9]



Figure 47: Magic Quadrant for Business Intelligence and Analytics Platforms [9]

Being already used to satisfy other GTO business needs, QlikView was the more convenient and efficient platform to build our HNC dashboard taking advantage of developers' return of experience. QlikView scripting needs only a knowledge of the SQL language with some ready to use functions.

3.2 Qlikview Components

To understand Qlikview standard deployment we should present its main components as displayed in the **Figure 47**:

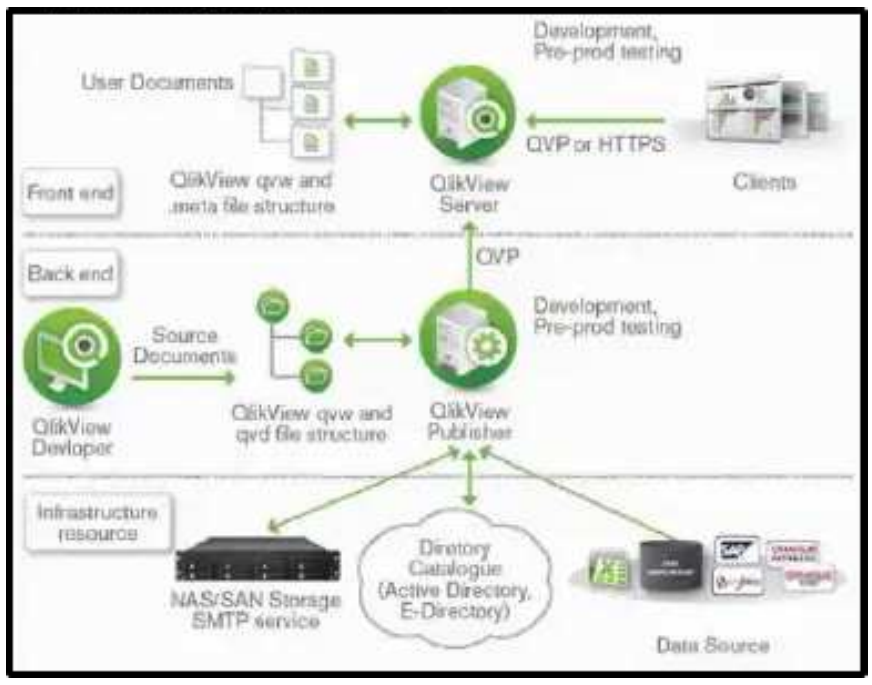


Figure 48: QlikView Layers & Components [10]

Components of QlikView are divided into "Back end" and "Front end":

- Back end contains QlikView Developer & QlikView Publisher :

QlikView Developer creates documents & qvw files by connecting to various source systems and extracting data in order to create a data model described in a qvw file.

This qvw file gets published using QlikView publisher which is responsible for data load and distribution within the Back end.

- Front end is more of a user interface layer in order to interact with applications and documents.

QlikView Server contains the documents that have been published and it uses either HTTPS or QVP (QlikView Own Protocol) to handle the communication between the applications and the clients.

QlikView Server handles the processing that accrues on the documents side once it's published to the web portal though an access point.

The access point is a focal point from which the client can interact with various applications.

4. The developed solution

With the help of *Emeric JAUSSIONS* (External Consultant specialized in QlikView) and the access we have been granted to "QlikView Developer Course" on the UDEMY website, we were able to develop the QULIS HNC dashboard helping to address problems of visibility and follow-up to an overall level of handle non-conformity cases. It brings together all the scattered information, between solving process teams assessing the situation and each milestone. The displayed KPIs and data are automatically updated every 3 days to facilitate management and decision-making during all phases of the solving process.

- **Database:** In order to develop the solution we were given a view (access) on the QULIS SQL database from the team in India.

  The **Figure 48** is a simple view on the used data model for the QlikView Qulis application.

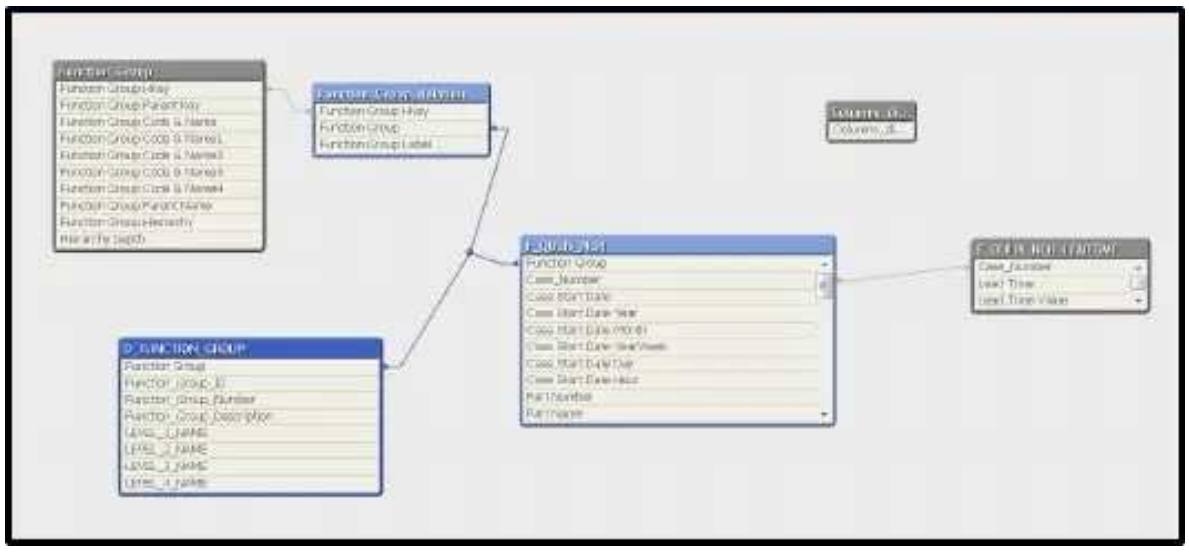


Figure 49: QlikView QULIS Database Schema

- **Overall View:**

The presented dashboard was developed according to well-defined functional and non-functional requirements in a way to satisfy the business requirements but at the same time be user friendly application.

As shown in the following figure the developed dashboard is using some extracted data as filters; these filters help users shift from a global view to a reduced scope according to that data we want to work and specially areas and plants.

In the following example we have selected:

- Brand: Volvo Trucks
- Cases which were accepted by case owner.
- Case status: Open

By that the displayed KPIs and data would be related to only the HNC cases having Volvo Trucks as brand, cases that were accepted by case owner and that are still open.

PS: More filters could be used by clicking on "More filters" button.

To improve the user experience users are given the possibility to save specific combinations of filters as a bookmark so that they don't have to reselect them to get a specific scope.

For example, if Adam WHITE (Director-Core values at Hagerstown: Powertrain Production USA) is only working on subjects related to QSP manufacturing for North America Manufacturing Operational Unit, for Volvo Trucks and Mack Brands and only open cases where he is the assigner, he can save this selection in a bookmark so that by a single click he can see the updates on the wanted data without reselecting the filters one by one.

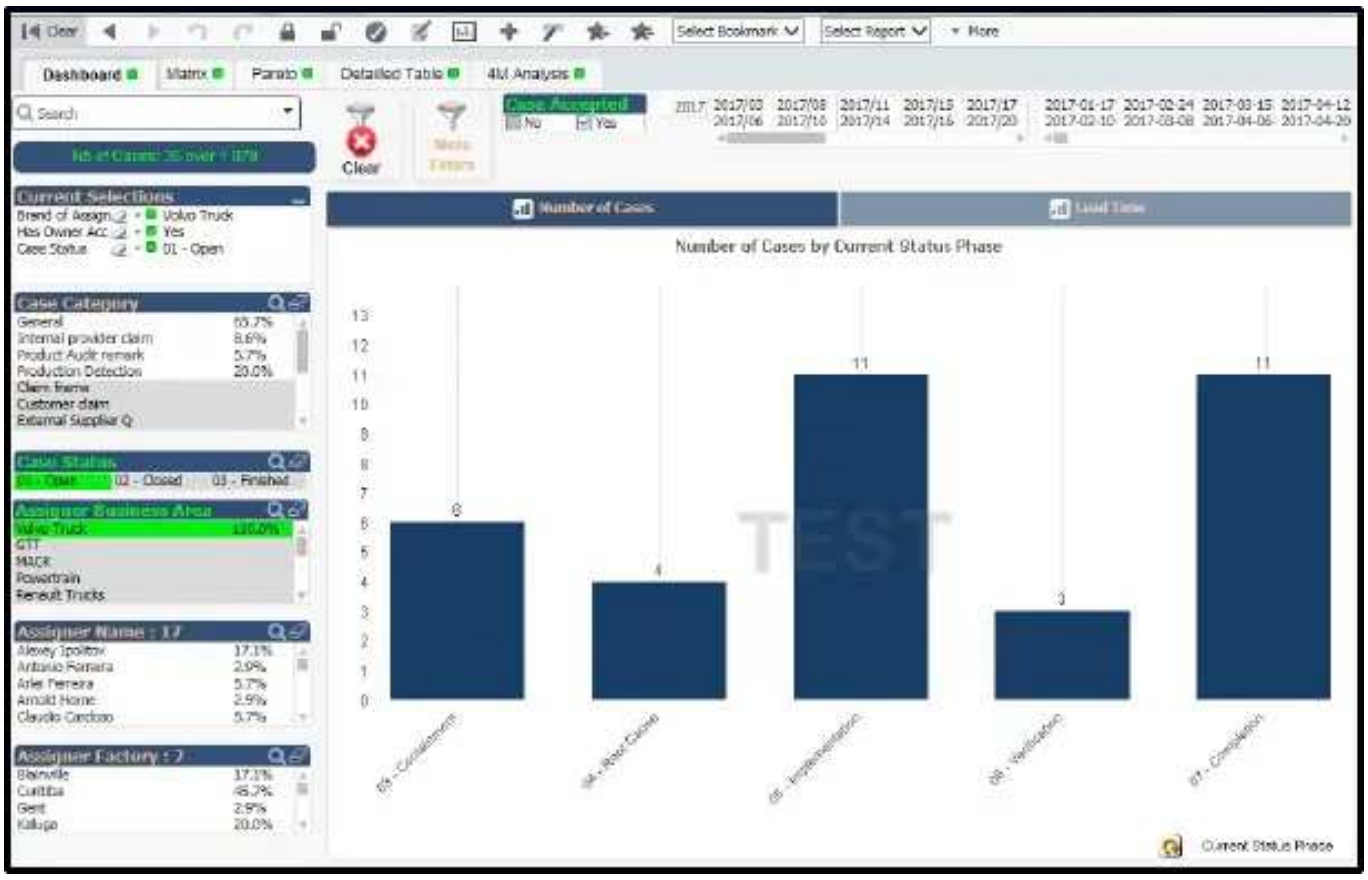


Figure 50: QULIS Dashboard Filters Overview

➢ **The Main Dashboard:**

The main lead-time dashboard displays the average lead times which are the results of multiple combinations as presented in the **Figure 50**:

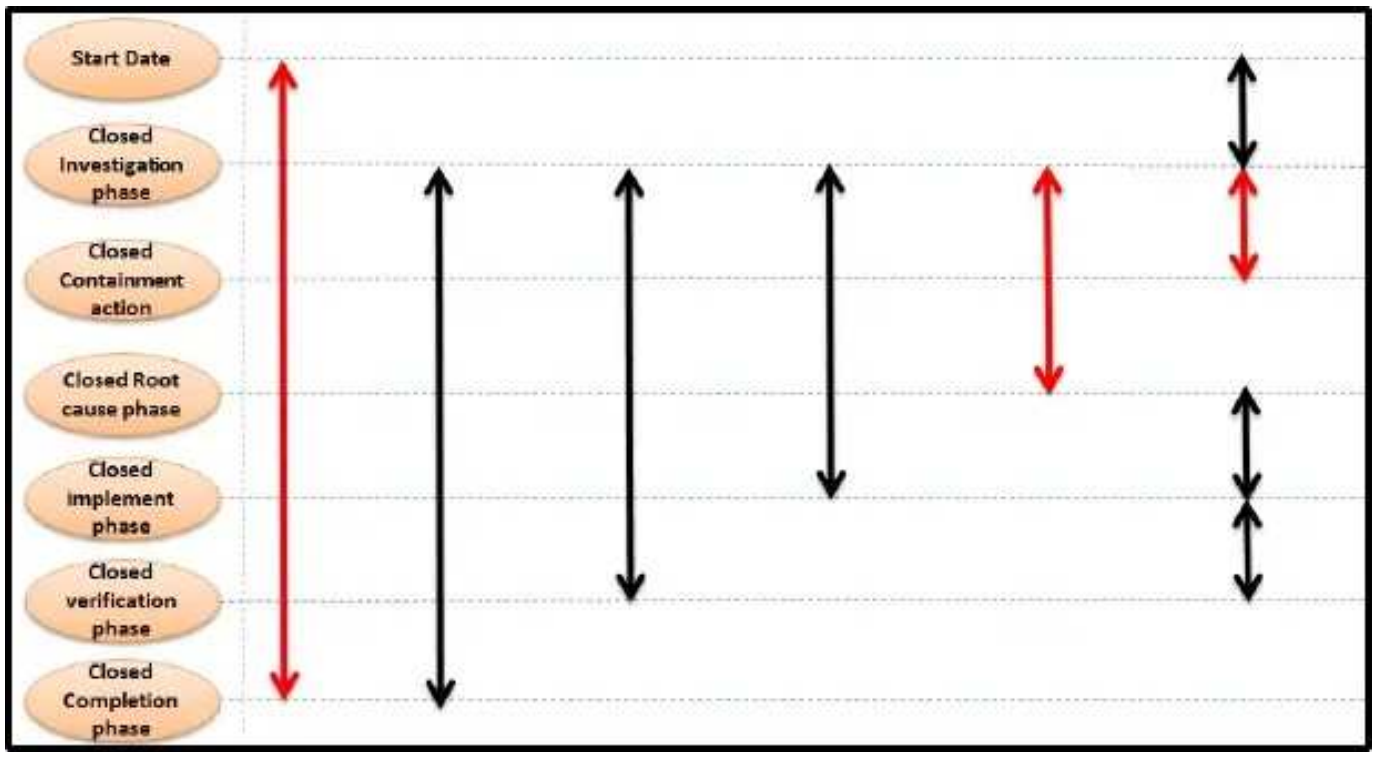


Figure 51: QULIS Lead Time combinations

The three lead-times in red are our 3 lead-time KPIs, the rest are also calculated and displayed on the dashboard but we gave the user the possibility the select the ones he is interested in monitoring. (Lead Time can also be filtered by HNC category: QSP Purchasing, QSP Manufacturing etc.…).

The date functionality helps us draw the evolution of the average lead-time according the selected scale of time.

The **Figure 51** shows the status of the selected KPIs from a global view (no additional filters).



Figure 52: Lead Time Main Dashboard

Another power thing about the developed application is that it gives you the possibility to monitor cases on a global scale (all Volvo cases, per operational unit, per plant) but more than that, using filters you can progressively go into more reduced scopes and you can even select to monitor a specific case.

The main dashboard presents other features such as displaying:

- The number of cases per HNC category as shown in the **Figure 52:**

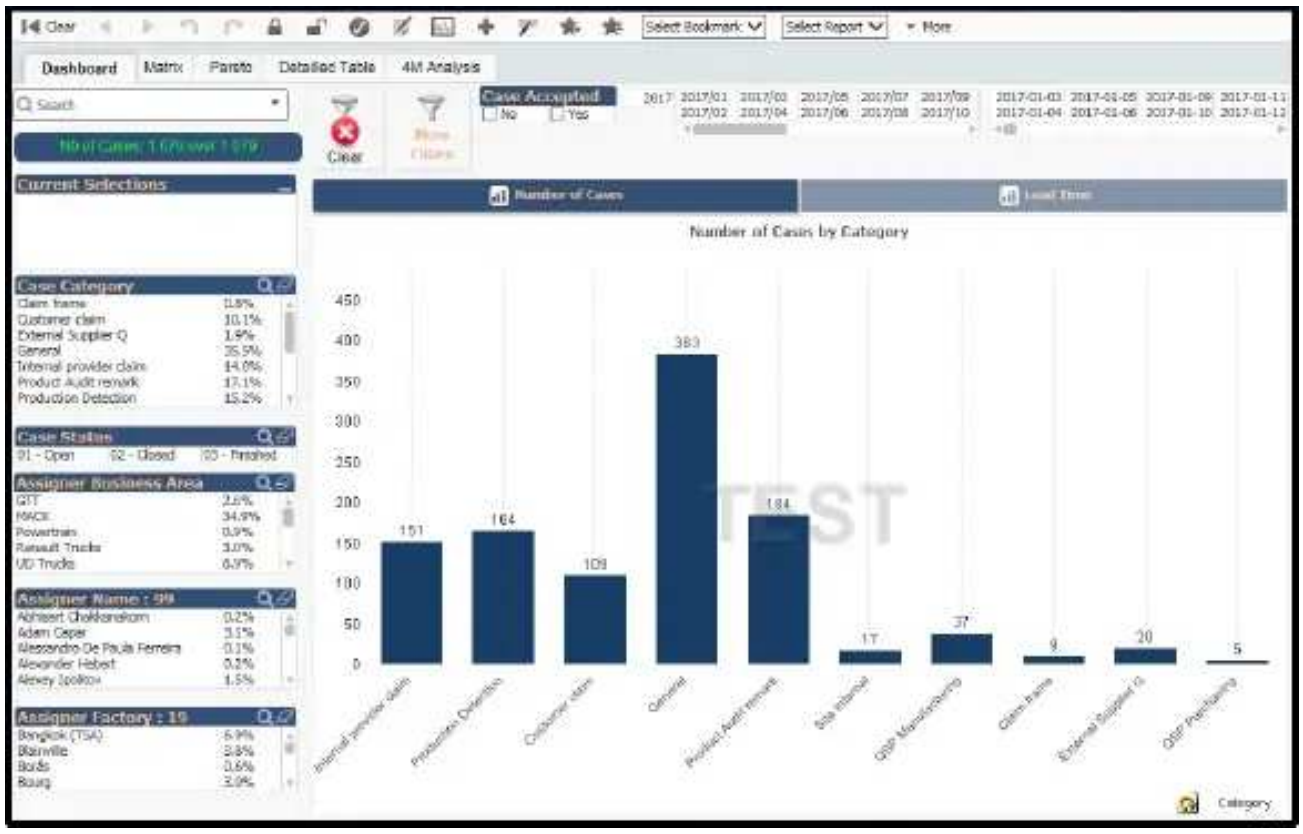


Figure 53: Display of number of QULIS cases per HNC Category

- The number of cases per case status (**Figure 53**), in order to have a view on our progress in handling non-conformities.

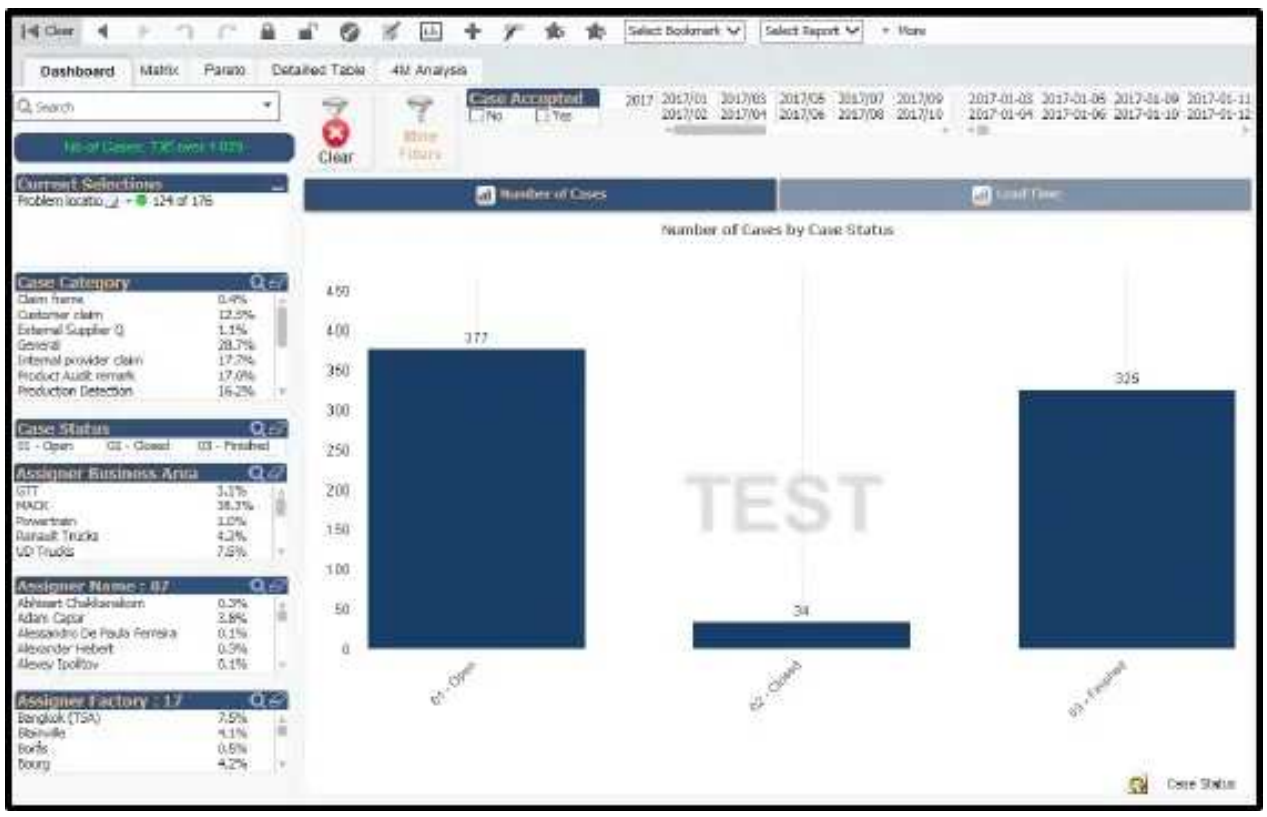


Figure 54 : Number of QULIS cases per case status

To enhance the follow up of the progress of the cases, we have implemented another feature showing the number of cases per QULIS Step, by this we are able to know the number of case that are for example still stuck at the root cause analysis step and act according to that to support the solving team and motivate them.

This feature is presented in the next figure.

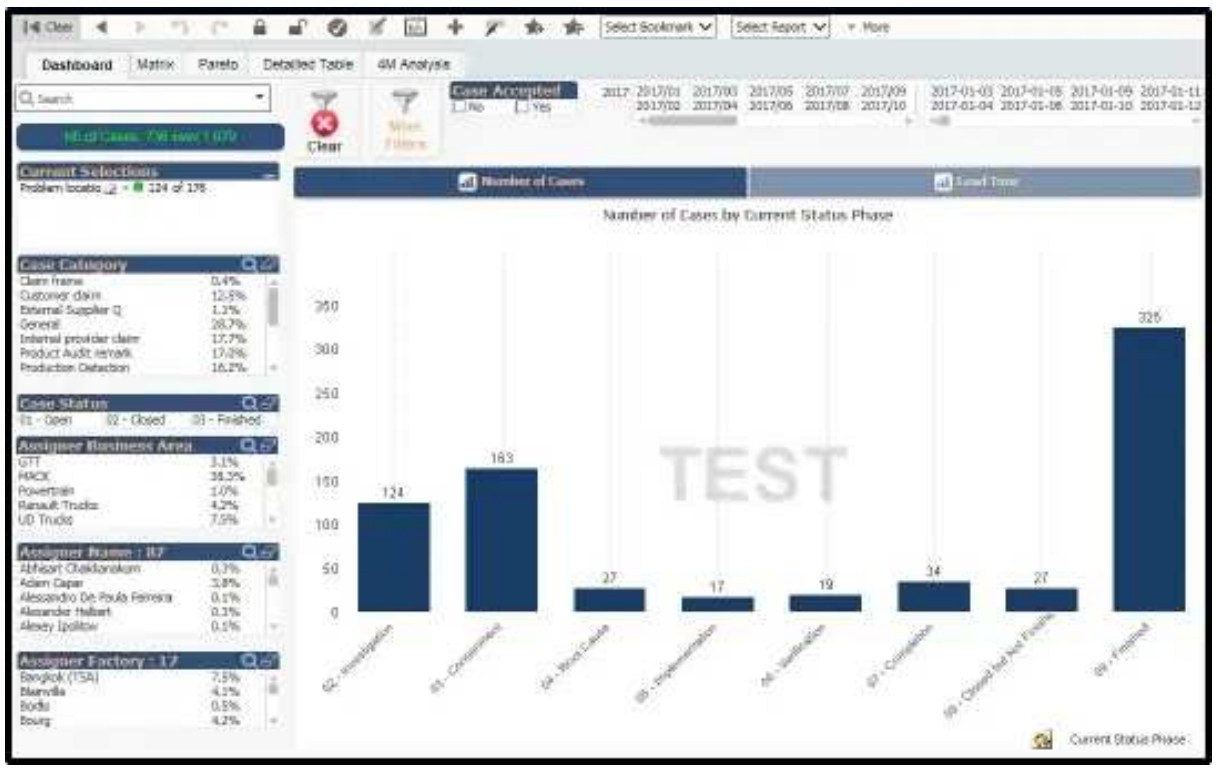


Figure 55: Number of cases per Current Status Phase

In order to support the decision making process, have a clear view on the problem, the contribution of vehicle parts to the non-conformities and by that quickly address the issues whether to the manufacturing or the design side, we have developed more specific features presented in the Figures below:

- The **Figure 55** presents the number of cases per VMS(Vehicle Module Structure / Team)

The Volvo organization divided a vehicle per Vehicle Module Structure that represents a specific part of the vehicle; Engine (50), Transmission (60) etc.

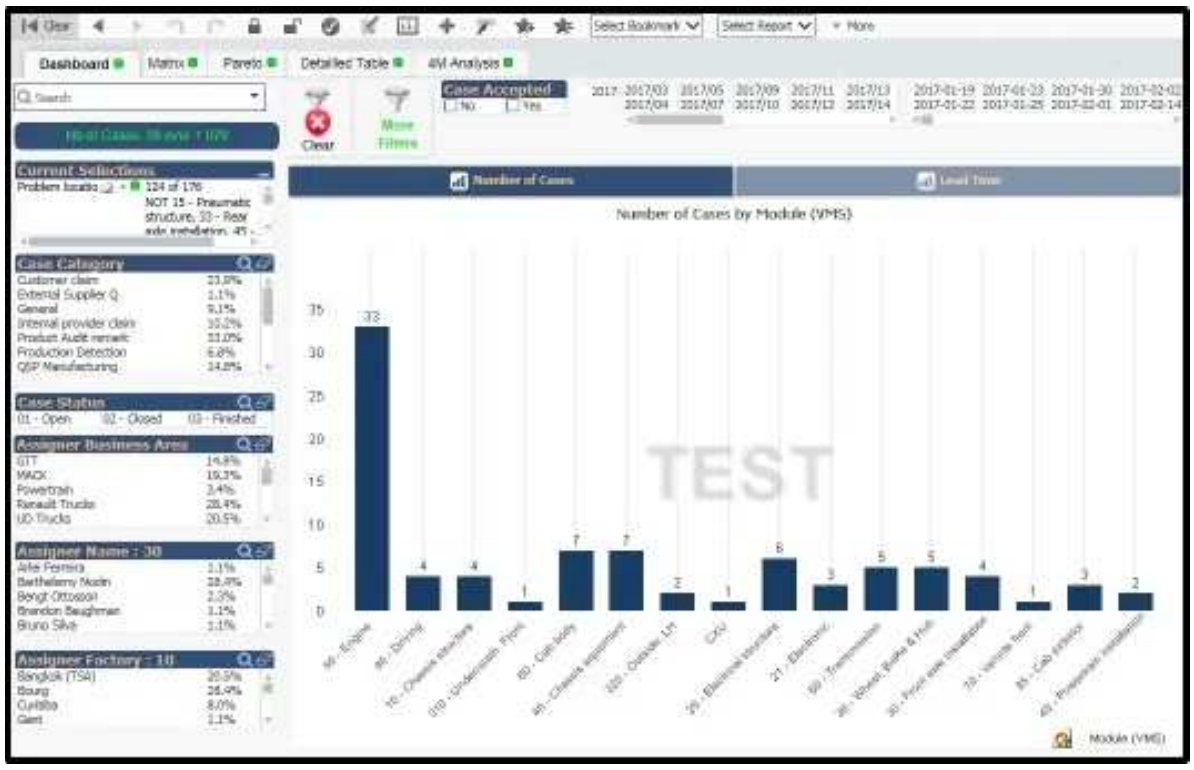


Figure 56: Number of QULIS cases per VMS

- The **Figure 56** displays a more specific count of cases per problem location

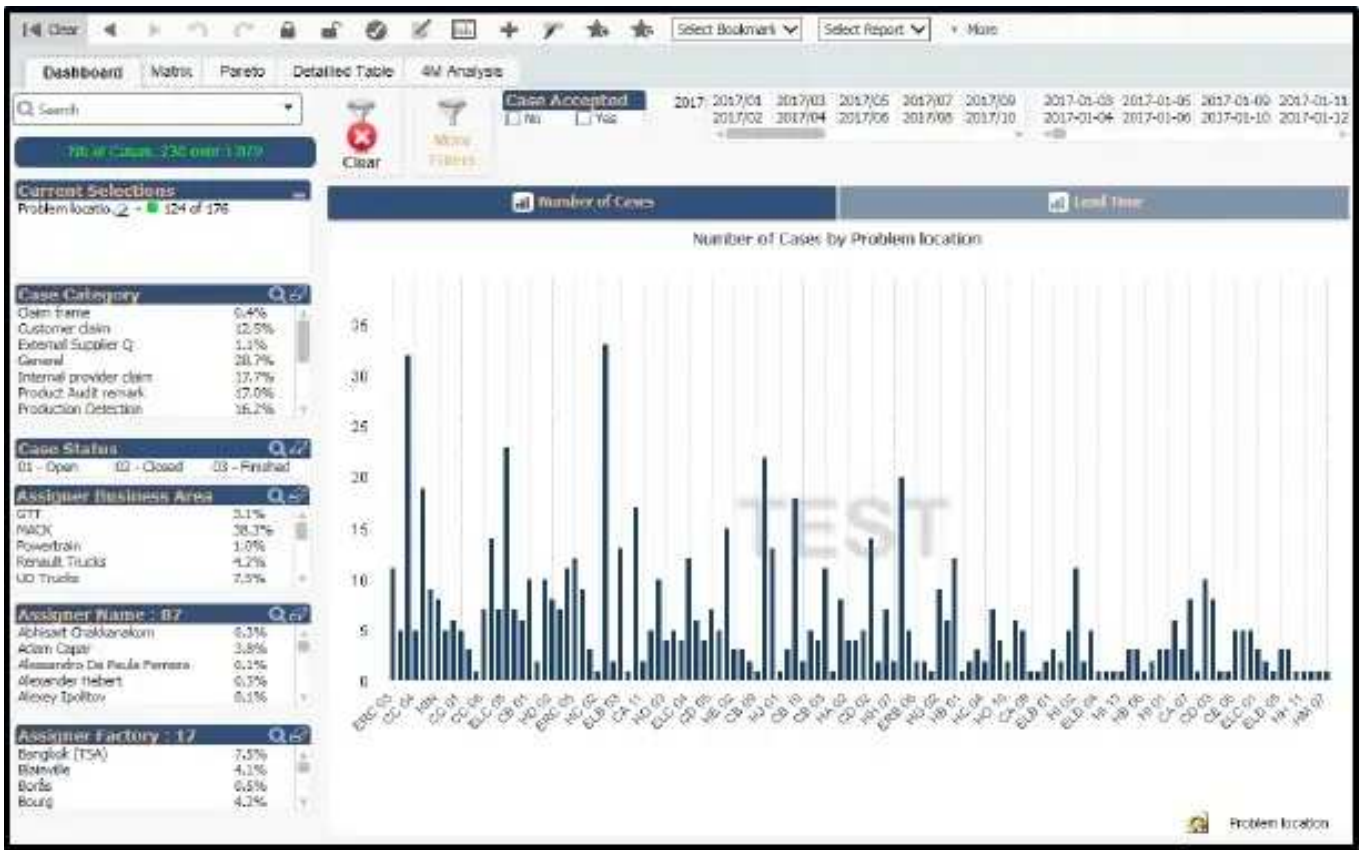


Figure 57: QULIS Cases per Problem Location

- **4M Analysis KPI:**

To have a good assessment of non-conformities root causes, we have implemented this KPI and feature in order monitor the contribution of each M(Machine, Manpower, Method and Material) to the root cause and also to the non-detection of the non-conformities.

We gave the user the possibility to switch between a global 4M root cause analysis pie as shown in the **Figure 57**.

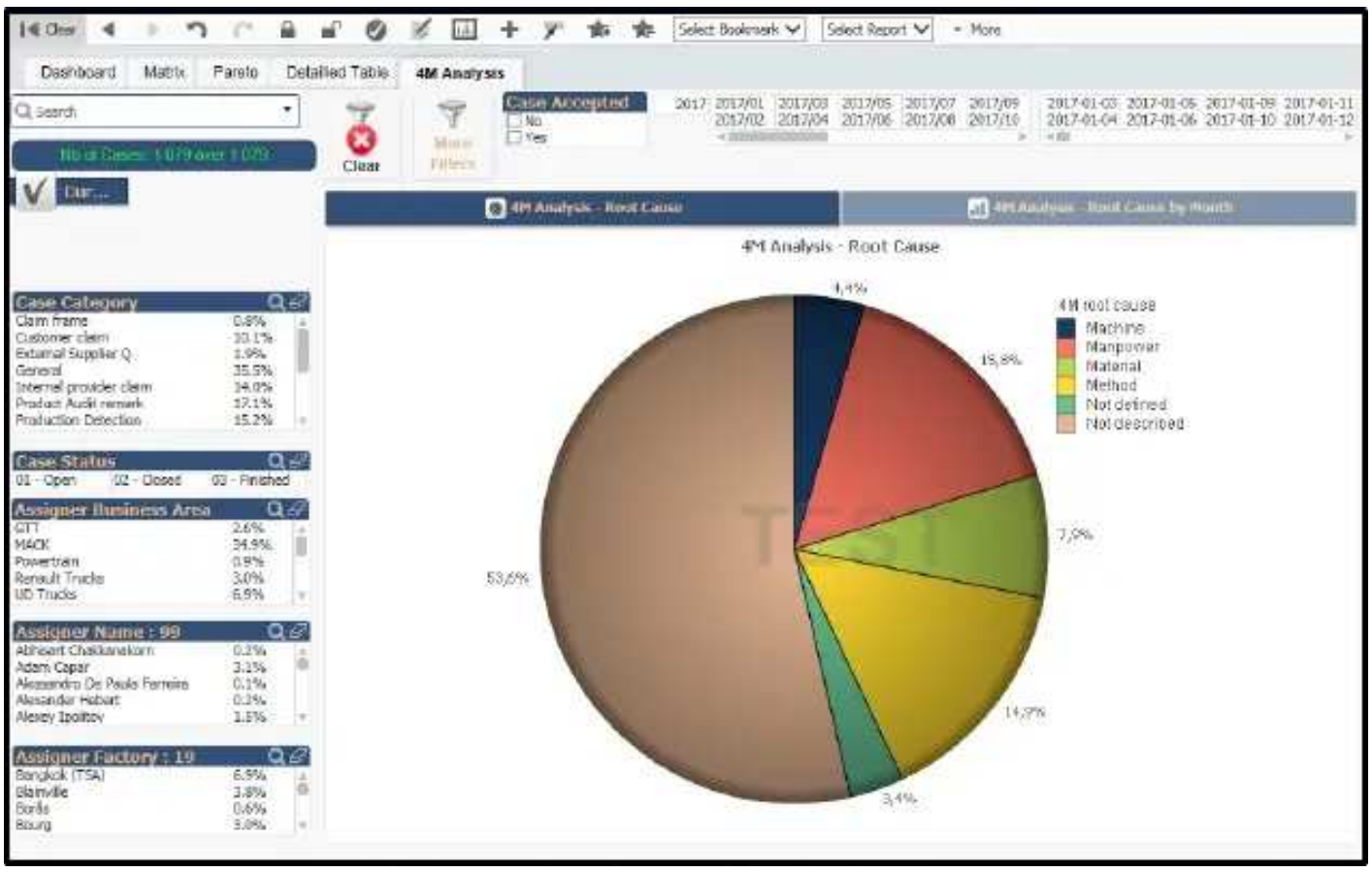


Figure 58: 4M Analysis Root Cause Pie

As for the second possibility it is a monthly follow up of the contribution of each M of the four to the root cause or to the non-detection of the non-conformity.

The **Figure 58** displays, for Curitiba plant (Brazil), the contribution of each M in the last 6 months.

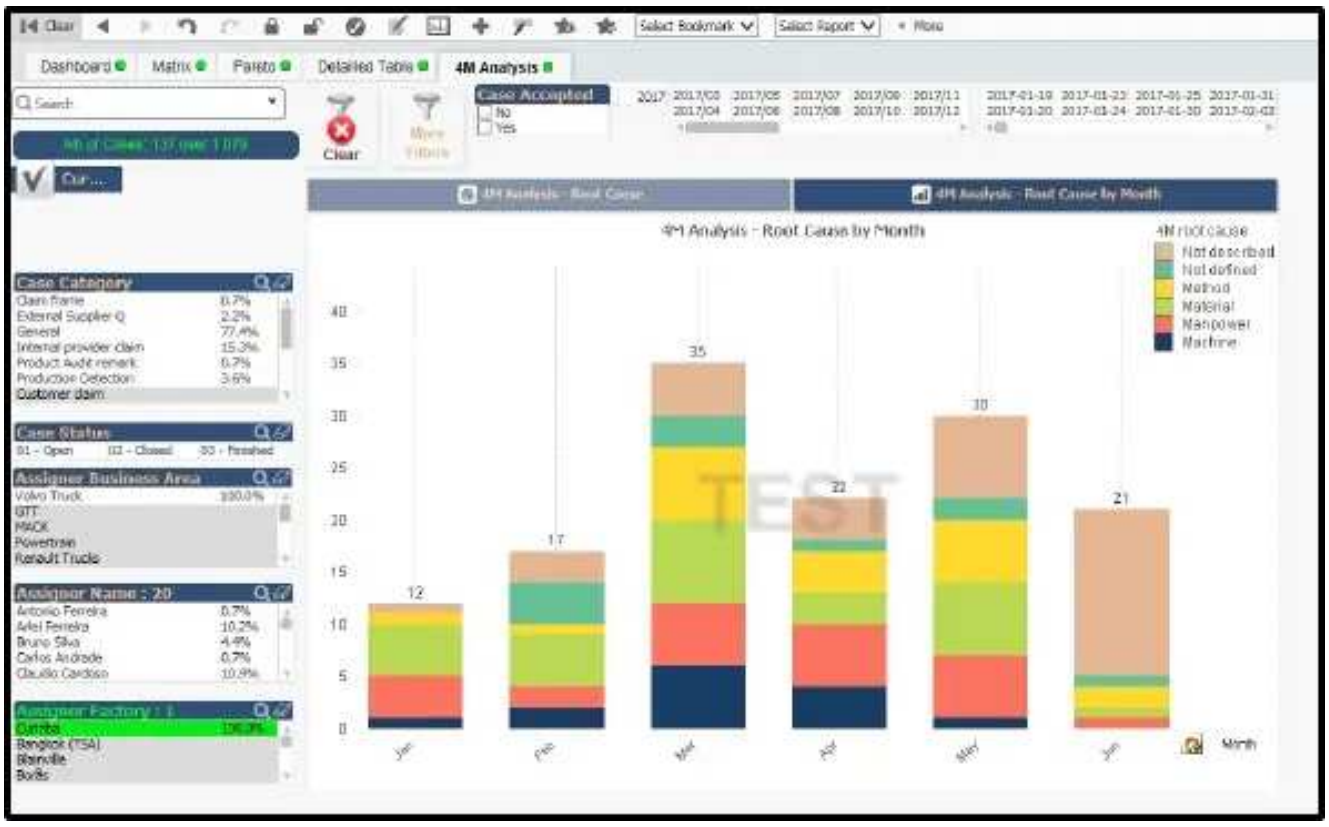


Figure 59: 4M Analysis - Root Cause per Month for Curitiba plant

➢ **Detailed table:**

To easily extract the data related to the selected scope (according to the chosen filters), we have put in place a data extracted portal where the user can select the specific details he needs to process or to communicate around HNC and then by a simple click generate an excel file with the wanted data.

This portal is presented in the **Figure 56**:

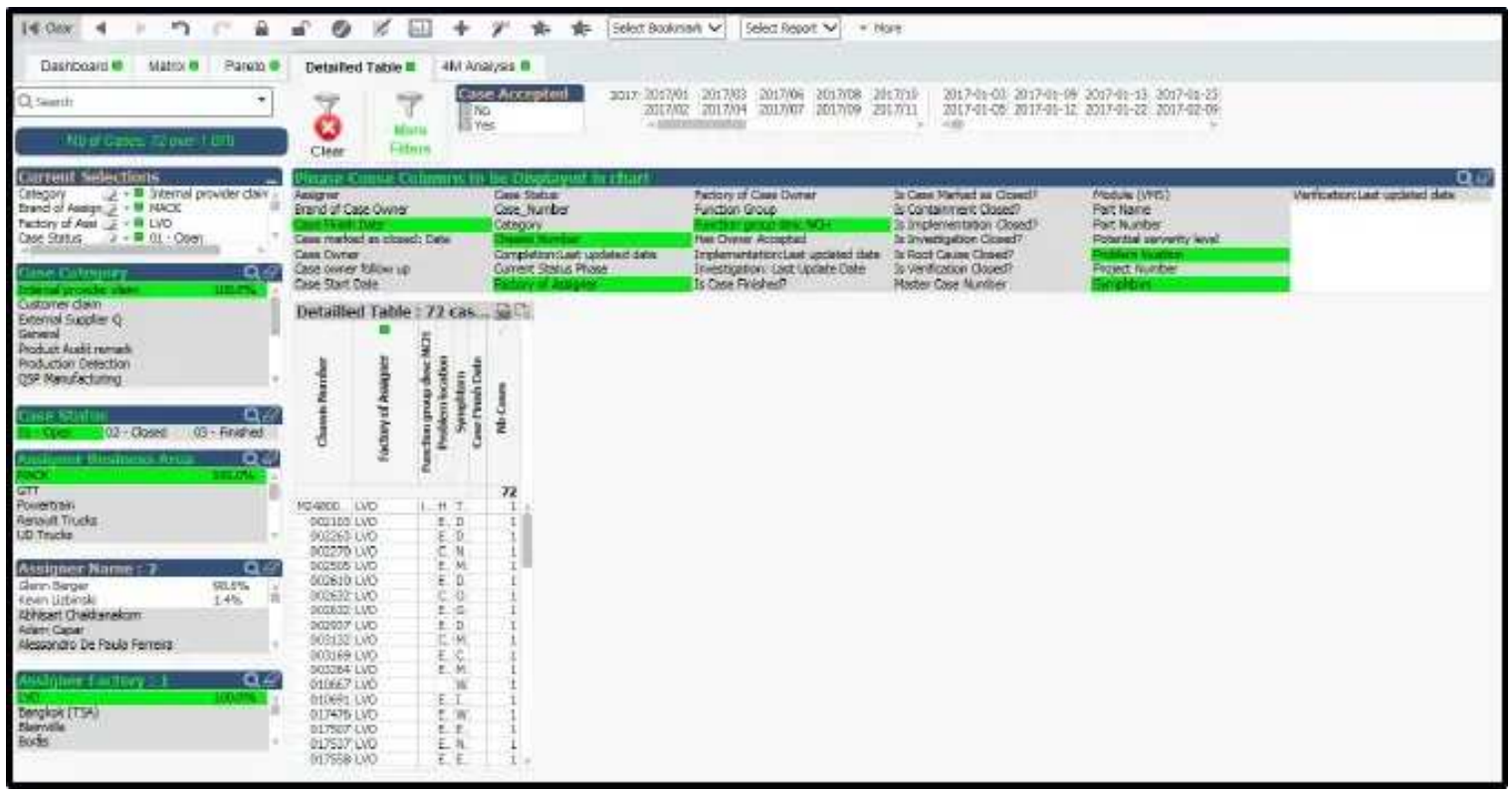


Figure 60: Detailed Table Portal

5. Results

A demo of the developed solution was presented to the different operational units' quality managers during the QULIS working group meeting and received good feedbacks. We have also prepared some training material for the new users.

However, it is now put in the Qlikview QA access point for future users to try it and report possible improvements.

By the end of September the dashboard will be put into production and used by all plants, quality managers, quality engineers and handle non-conformity team as the developed solution will ensure:

- An active monitoring of the handling of quality issues through the display of:
  - Average lead-time of each step of the cases with the possibility to focus on a certain case.
  - Number of cases per :
    - HNC category
    - Actual Step
    - Causal Part
    - Month

- Plant

- An understanding of the main contributors to the non-quality through the 4M analysis KPI.
- Global statistics and a traceability tool for all Volvo AB HNC cases.

These outputs will help the company react actively to any delay or problem with a single case and even support the work on a more global frame (plant, HNC process etc.…) ensuring the delivery and implementation of the final solution within the target deadline.
By that our dashboard would be one of the main contributors to the continuous evolution of the operational excellence within the non-quality solving teams.

IV. Conclusion

The set of actions we have put into place through the software working groups, showed a lot of areas to improve in Volvo's way of working in related issues. We have succeeded through this taskforce to enhance multiple processes touching even the warranty organization and delivering GTT software specialists with the right data to analyze and identify the root causes of the failures causing customer claims.

As for the handling non-conformity processes we were able to define the appropriate performance indicators and develop a flexible and powerful dashboard for the monitoring of these PIs ensuring an up to date view on the situation and an active reaction towards delays and bottlenecks which are slowing down the solving process.

General Conclusion and Perspectives

The Volvo Group goal is to become the world leader in sustainable transport solutions; this can only be achieved by keeping the costumer in focus in whatever it does.
"Changing the game is a mindset" (Volvo Group Concept), the GTO have put "main mindsets and behaviors that will contribute to customer success; Deliver what is committed with a zero defect mindset, communicating in a transparent and open manner and striving for efficiency and effectiveness to quickly solve customer issues.

The main actions that were implemented, during our graduation project, have contributed noticeably in reaching these mindset and behaviors.

At a first place, we have carried out a diagnosis on the current software claims situation and their main causes which allowed us after that to initiate and coordinate two working groups around customer software issues present in our FF1 KPI.

The set of actions that were put in place as output of our WG contributed to improve clearly the current situation as we were able exclude all misleading data from our FF1 that were leading to extra analyzing work for our team when they were not real problem contributing in having a robust performance indicator that reflects the real situation.
We have help in the reinforcement of the communication between the Group's organization and the cross-functional work which led to detect the missing information required by the GTT in order to work properly on solving customer software problem; these missing information were reduced to reach only 21% by August when it was 74% in June.

We were also able to start a deep investigation around software Green OK processes in all plants comparing them to the Global Standard, striving for having on common process in all GTO plants approved by both GTT and GTO.
Our work led to implement monthly software meetings to review the current situation and act upon that.
On a global overview, software claims contribution to customer claims reached only 9,3% by May 2017(12% in March) and a Software FF1 reduced by 45%(in May compared to March 2017).
We have also reported a decrease of 1,4% in the contribution of software claims to total claim payments ,which is almost equivalent to 150K SEK (15K euro).

At the second part, we have worked on the deployment of a global IT tool supporting communication around Handle Non-Conformity cases, where we were able through a change

management strategy to get approval from all GTO plants and GTT Quick Solving Leaders to use it as the only tool for HNC.

After that, we have defined HNC performance indicators and elaborated a global dashboard developed using QlikView through a database connection with QULIS to have a powerful and efficient tool for the monitoring of HNC cases statics and current progress situation helping quality engineers and managers act according the displayed data to reinforce the work on particular cases or on a global view, striving by that for efficiency and effectiveness in solving customer issues and ensuring by that a continuous evolution of the operational excellence.

Nevertheless, there are still work and improvements to be brought to the both treated subjects; as for the Software Product Quality we should ensure the implementation of all the remaining actions and their standardization in all GTO plants and to secure a more efficient and engaging actions from our working groups we should from now on adopting the Visual Performance Management methodology.

For the Handle Non-Conformity dashboard it should be put in production by end of September but it could also be enhanced by introducing other features and performance indictors reflecting the effectiveness of the implemented solution.

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[2] Volvo Group Trucks Operations. *Volvo Operations Concept 1.0*. Goteborg, 2016.

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[9] *Visual Intelligence Web site*. August 2017. 17 August 2017.

[10] *QlikView Developer Course, www.udemy.com*. 2017. 28 August 2017.

Annexes

Annex A: PROSIT+

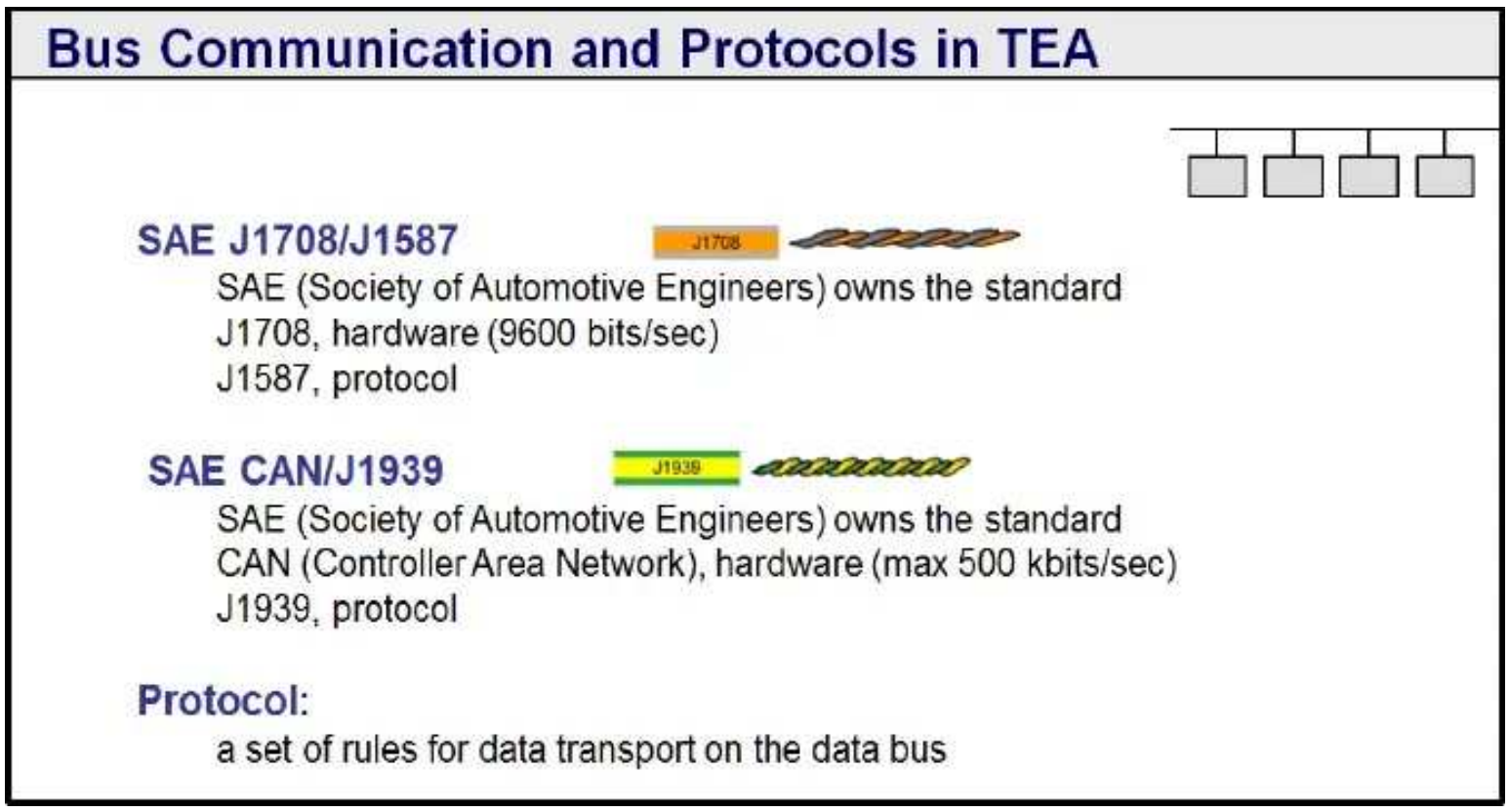


Figure 61: Bus communication and protocols in TEA and PROSIT+

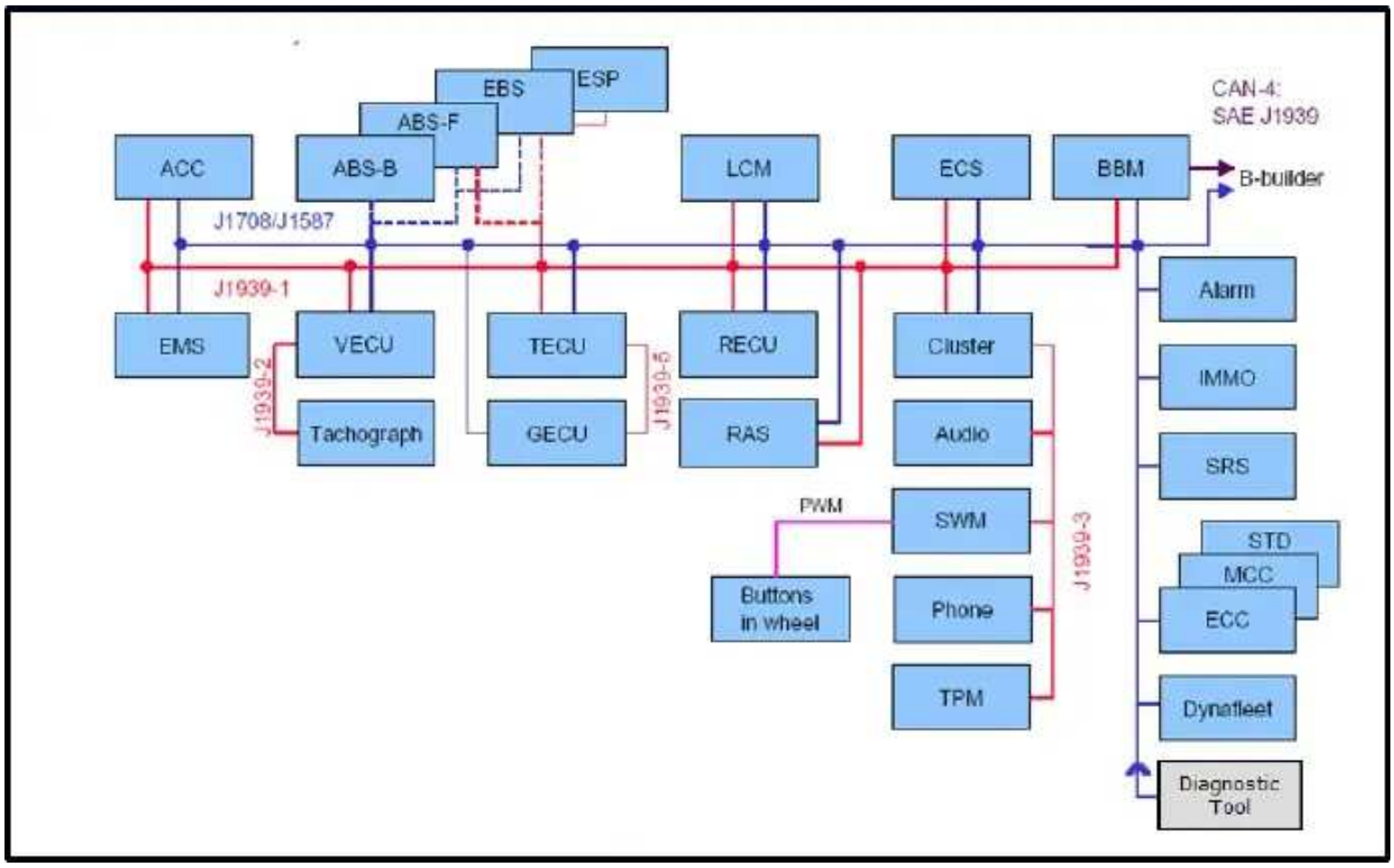


Figure 62: Truck Electronic Architecture

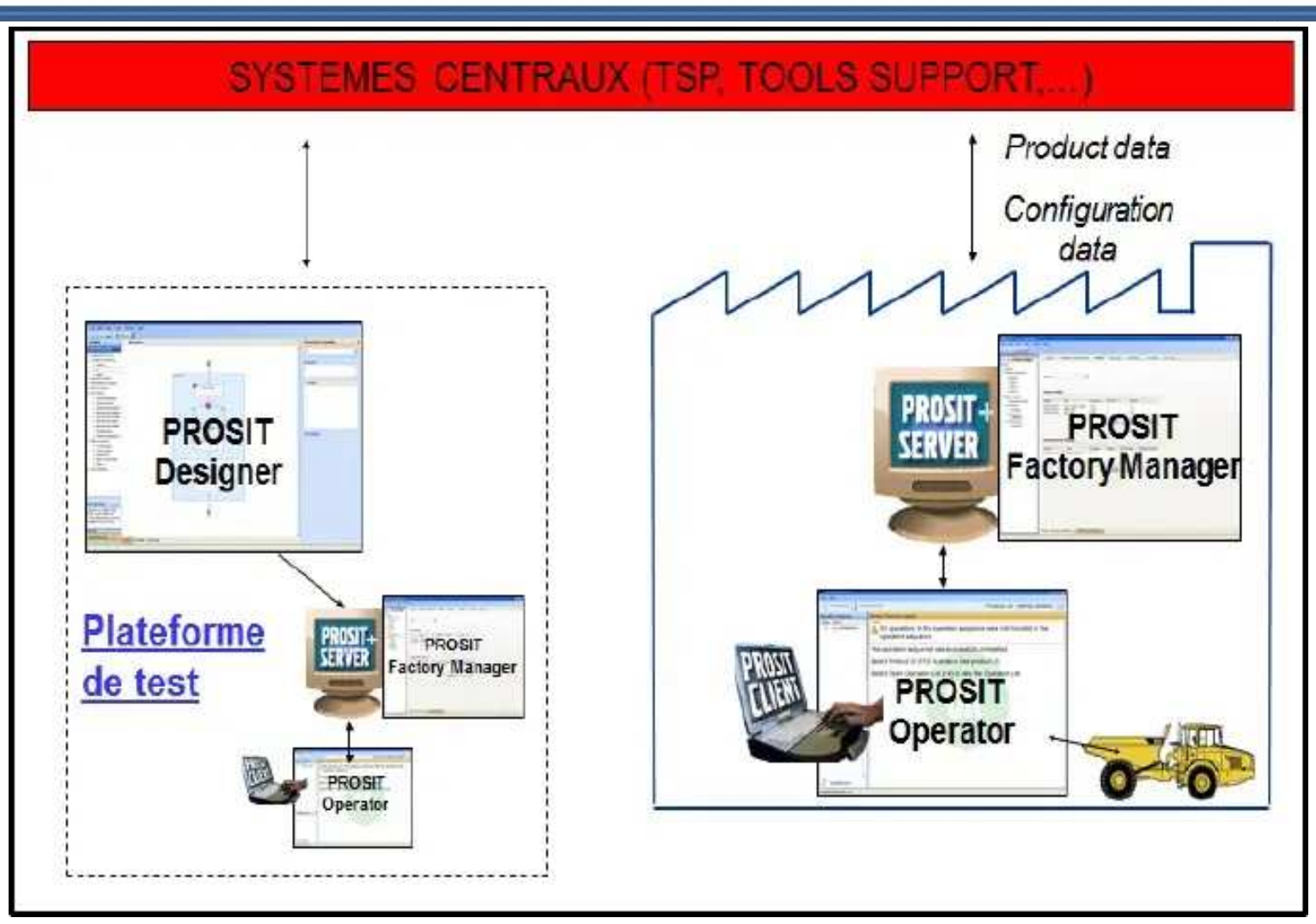


Figure 63: PROSIT+ Architecture

```
[GeneralInfo]
Version = 1.00
MID = 216
ECU_PN = 204227169
MSW_PN = 20425153
MSW_MaxMessageLength = 21
[AreaInfoLevel1]
AreaId = 0,1,2
AreaStartAddress = 058C,05A4,0388
AreaEndAddress = 05A3,07FB,03B5
AreaRWPWAddress = 0590,0592,0592
. . .
. . .
[DSWContent]
:00000001FF
[MSWContent]
:020000020000FC
:108000001BF1DC19F016CE982CC60718023171044A
:1080100031F94AB25E0504411253182717CD040105
:108020000A53182717ED53182617FD4A853402F610
:1080300030A5C50118260E63C76B838759C3320B61
. . .
[DSTContent_1]
. . .
[DSTContent_2]
. . .
```

Figure 64: Example of a programming file

Annex B: Green OK Process Analysis

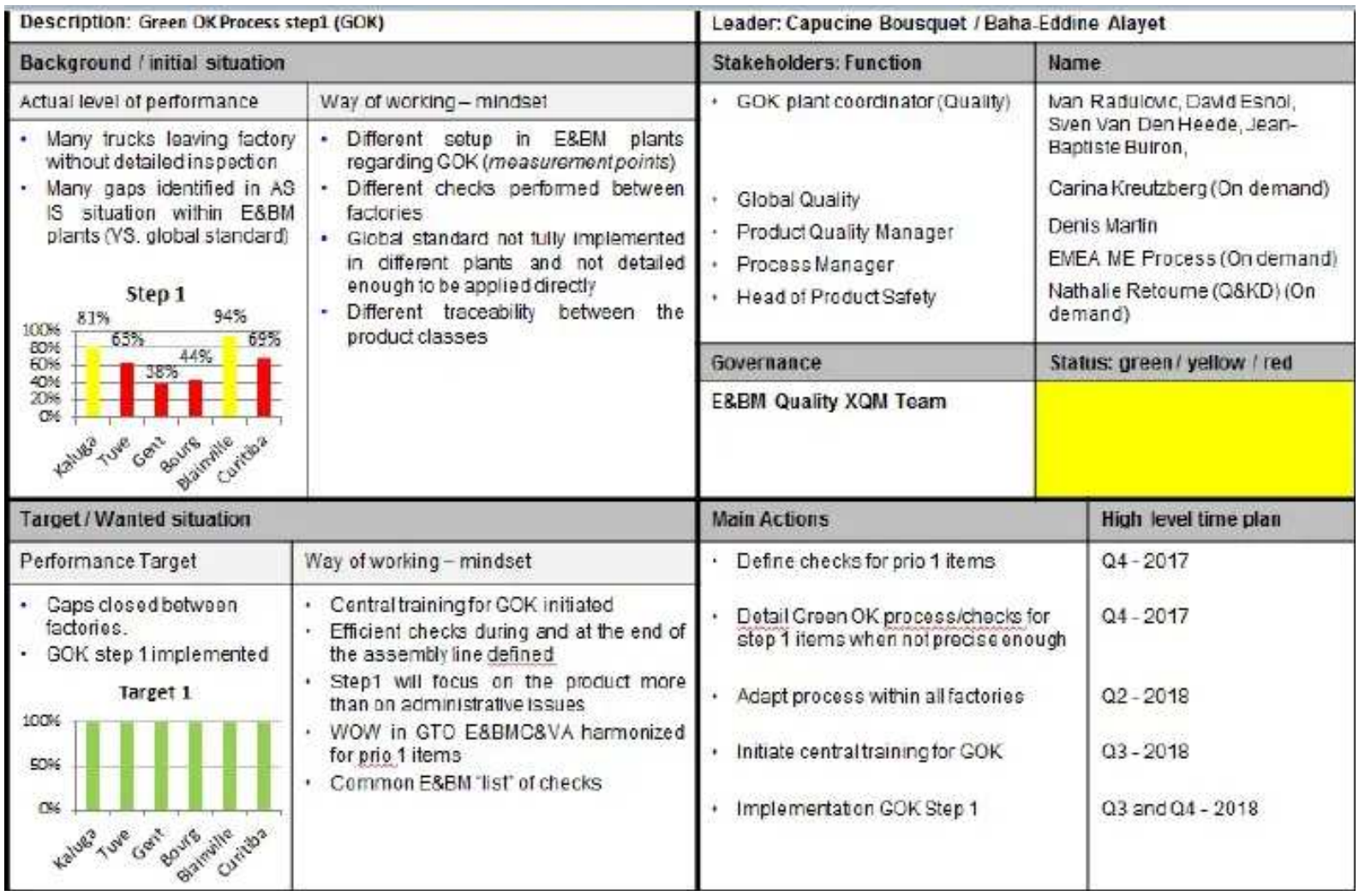


Figure 65: Green OK Process Status

Annex C: QULIS Deployment Status

		QULIS NEW CASE HANDLING				
		QSF Handling	claiming between supplying plant and receiving plant	handling of internal deviations	remarks recording	Legacy case handling
CDE	Tuve	ready but start when GSL will initiate in QULIS	With PTP plants			internal deviation, with Umea
	Gent	ready but start when GSL will initiate in QULIS	With PTP plants			with Umea
	Kaluga	ready but start when GSL will initiate in QULIS	With PTP plants			internal deviation
	Berug	ready but start when GSL will initiate in QULIS	With PTP plants			
	Blainville	ready but start when GSL will initiate in QULIS	With PTP plants			
	Umea Cab	ready but start when GSL will initiate in QULIS	supplying plant			internal deviation, external deviation
	Blainville cab	ready but start when GSL will initiate in QULIS	CDE decision: will not be used			
	Curitiba	ready but start when GSL will initiate in QULIS	implemented	communication with material quality dpt (implemented)		
	Curitiba cab	ready but start when GSL will initiate in QULIS	implemented	communication with material quality dpt (implemented)		with Umea
Cons	LVO	cases initiated by GTO			45 stations on the floor	
	NRV	cases initiated by GTO				used for
	LVO Cab	cases initiated by GTO				
	NRV cab					CSRs
SD & GTA	Wuxi	cases initiated by GTO	With PTP plants & cab	with New Case Handling	local system used (SPCS)	closing phase
	TSA (cab + VA)	cases initiated by GTO	With PTP plants & cab	with New Case Handling		closing phase
	Bangalore (cab + VA)	cases initiated by GTO	With PTP plants & cab	with New Case Handling		closing phase
PTP	Ageo	ready but start when GSL will initiate in QULIS				
	Curitiba	ready but start when GSL will initiate in QULIS				
	Hagerstown	ready but start when GSL will initiate in QULIS	use has just started			
	Skovde	ready but start when GSL will initiate in QULIS				
	Vannassaasa	ready but start when GSL will initiate in QULIS				
	Koping	ready but start when GSL will initiate in QULIS				
JV	Powertrain JV	QULIS not implemented yet	QULIS not implemented yet	No decision yet	QULIS not implemented yet	
UD	Ageo (cab + VA)	ready to use as soon as 1 new case		By old case handling		internal deviation handling
	Ageo (engine & Transm)	no decision yet	implemented			

implemented

decided & not implemented

not decided

decided & implementation ongoing

Figure 66: QULIS Deployment Status